

- definition of force – slide 1**
- revisiting vectors: components of forces – slide 7**
- adding vectors in 2D:
Tail to head method – Algebraic method – slide 10**
- types of forces: weight, normal force, tension – slide 27**
- static equilibrium – slide 46**
- frictional force – slide 51**
- coefficient of friction – slide 64**
- rotational equilibrium – static equilibrium slide 66**

A force is a push or pull. A force causes an object to change its state of motion:

- get it moving
- stopping it
- slowing down or speeding it up
- changing its direction (making it turn)

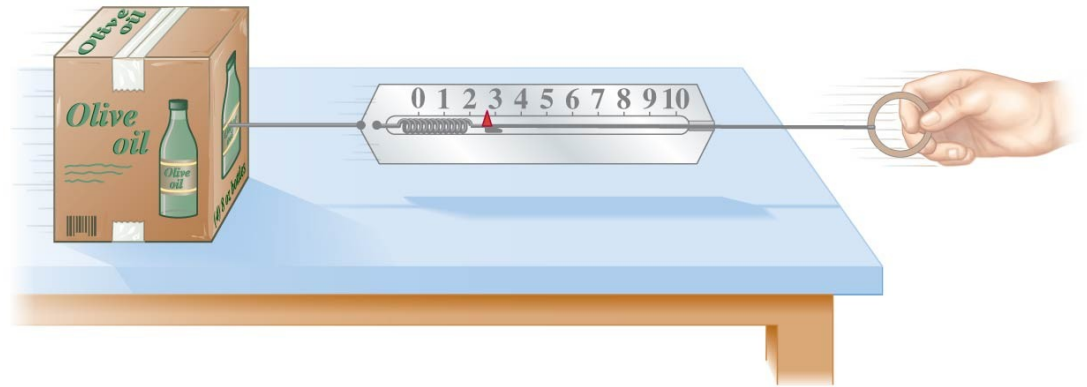


Force → change of velocity → acceleration

(Newton's 2nd law : $F = m a$ or acceleration is the force per unit mass)

We will see that once the object is moving, it will keep moving (with the same velocity) without push/pull . Unless there is friction which is a force.

The magnitude of a force can be measured using a spring scale.



Units of Force

- SI unit of force is a Newton (N)

$$1 \text{ N} \equiv 1 \frac{\text{kg m}}{\text{s}^2}$$

- US Customary unit of force is a pound (lb)

- 1 N = 0.225 lb

10 N is about 2 pounds which is about

The force of gravity acting on a mass of 1 kg

(to go from newtons to pounds divide by 5)

- In Europe, the bathroom scale reads kg (mass)

In the US, it reads pounds (weight). Divide pounds by 2.2 to get kg

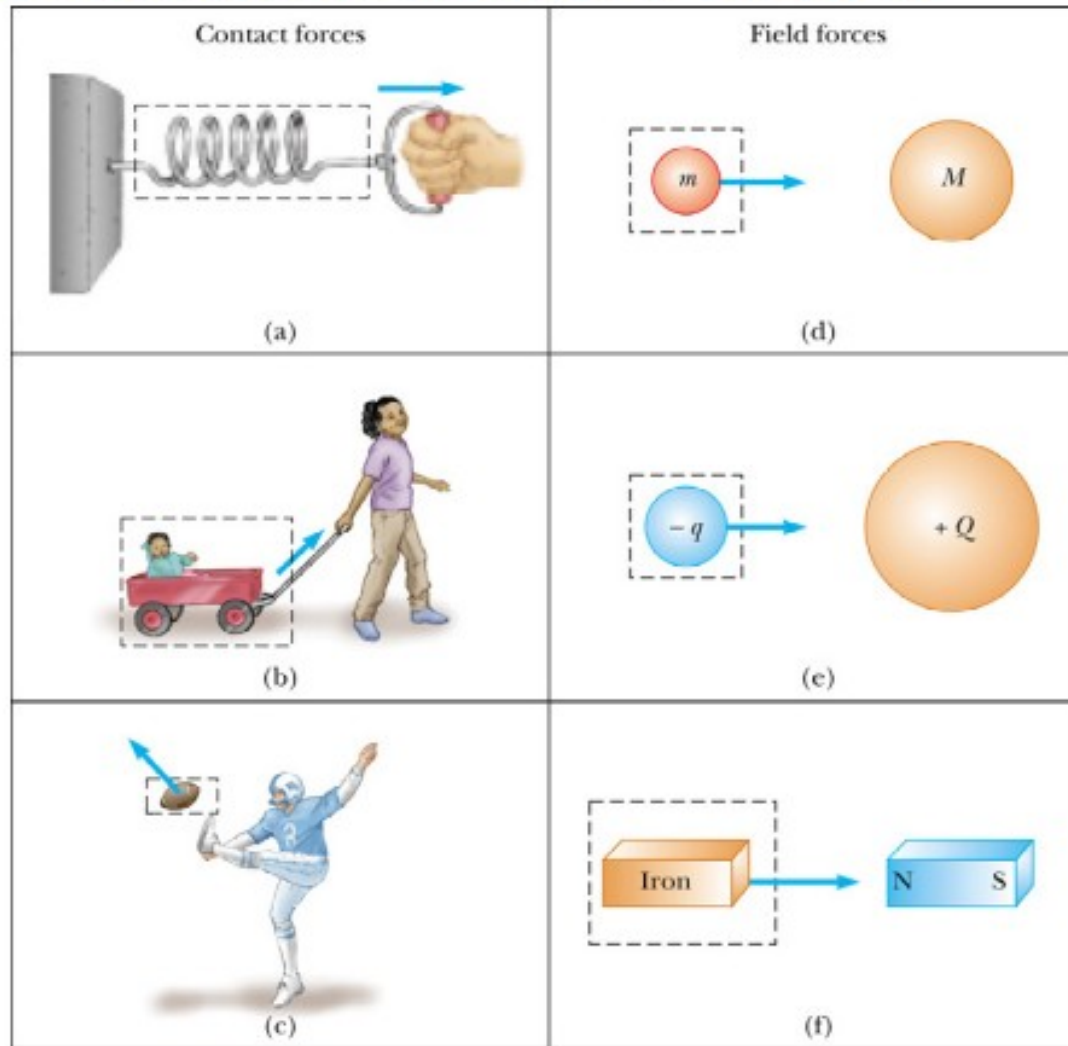
Example 180 pounds in _____ kg ?

A ***force*** is a push or a pull.

Contact forces arise from physical contact .

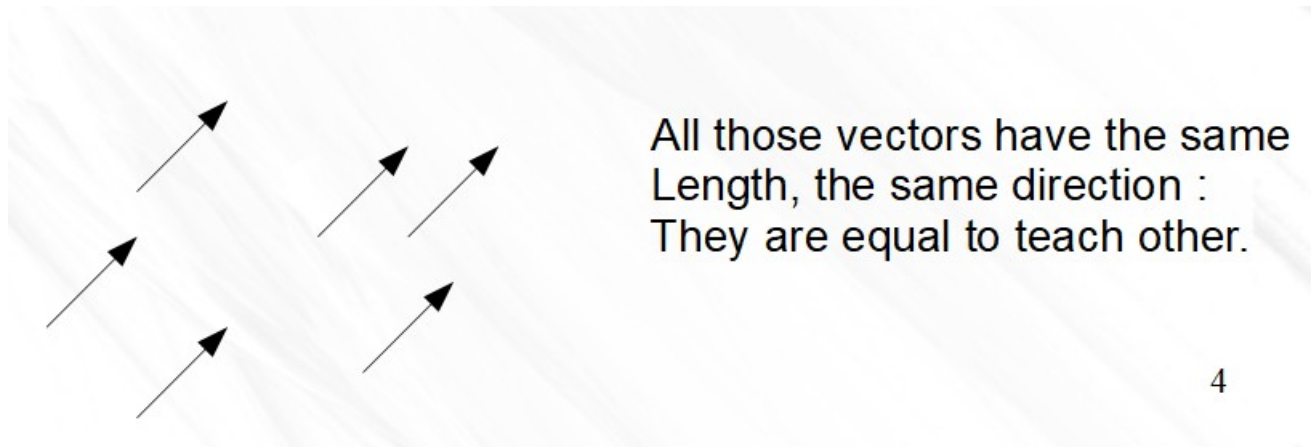
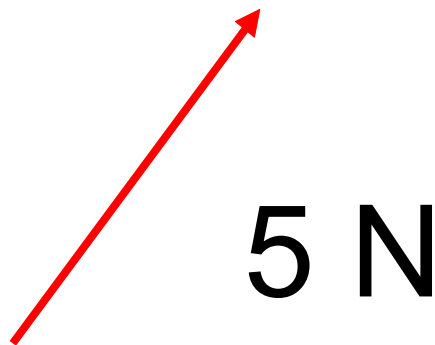
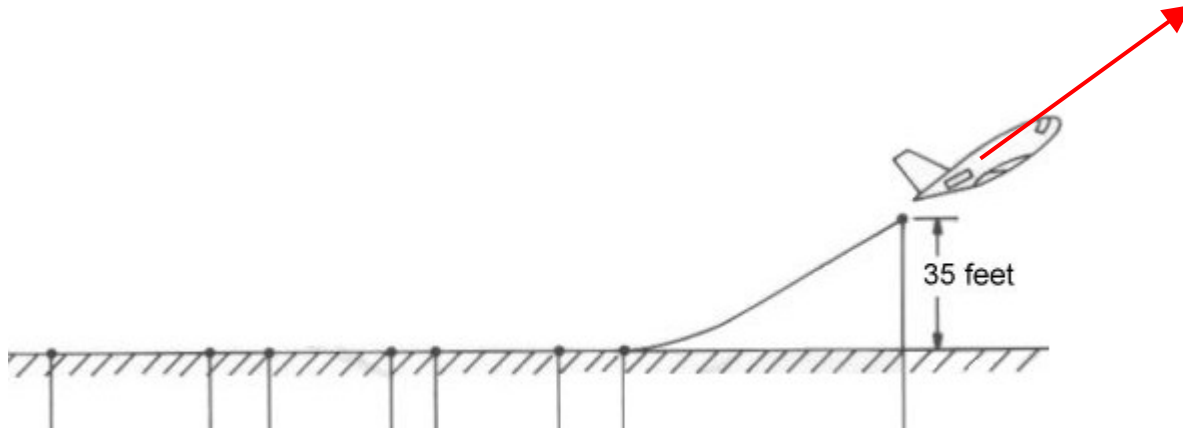
Action-at-a-distance forces do not require contact and include gravity and electrical forces.

Contact and Field Forces



Forces are vectors.

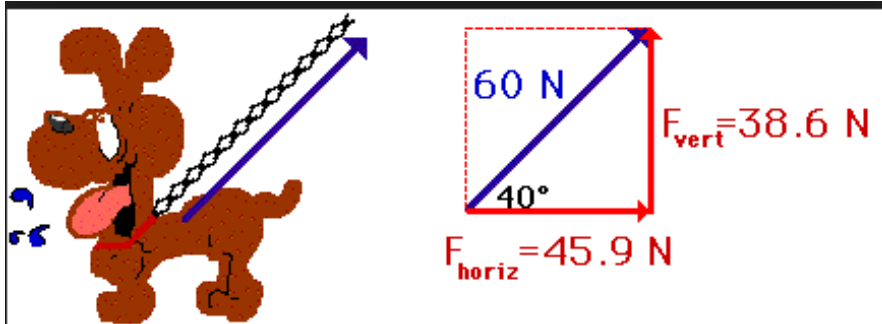
Forces have a magnitude (how much), a unit (of what) and direction (which way): **Boeing737 150 mph @ 15 degrees**



REMEMBER VECTORS COME WITH COMPONENTS !!

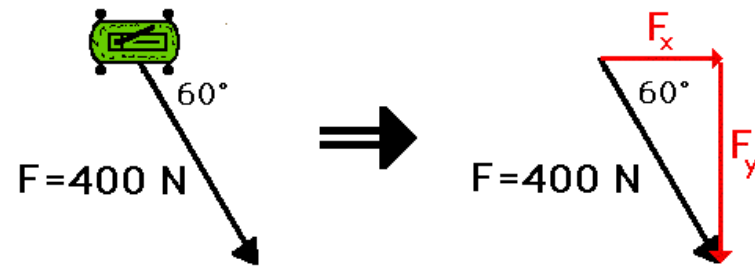
Components are independent from each other and have “different” job.

Source: <http://www.physicsclassroom.com/class/vectors/Lesson-3/Resolution-of-Forces>



Find the components of the force
Pulling the dog through the leash.

What is the “job” of each component



Find the components of the force
ff the wind pushing the sail.

What is the “job” of each component
($F_x = 200 \text{ N}$, $F_y = -173 \text{ N}$)

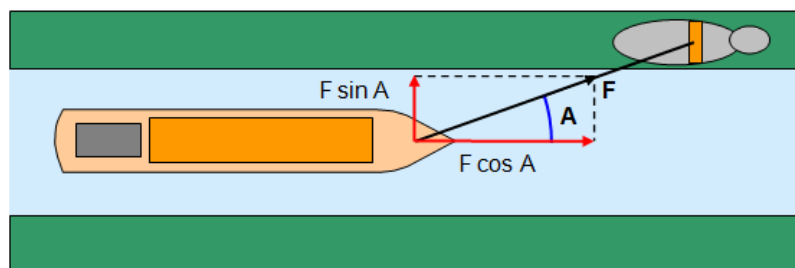
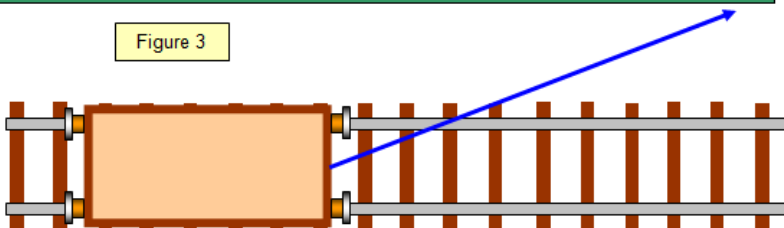
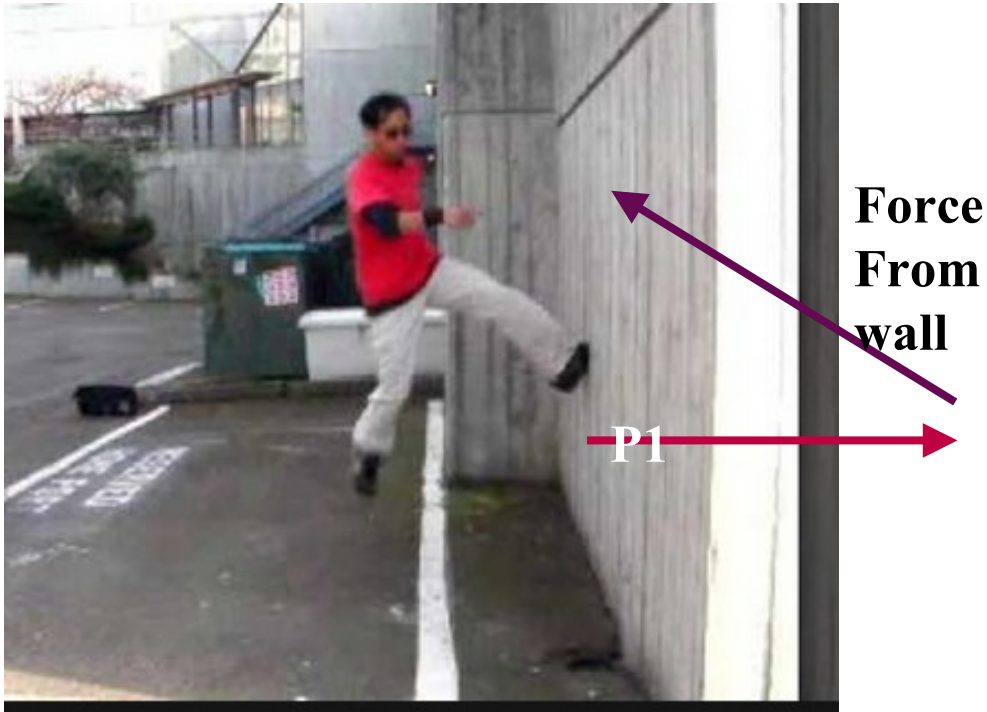


Figure 3



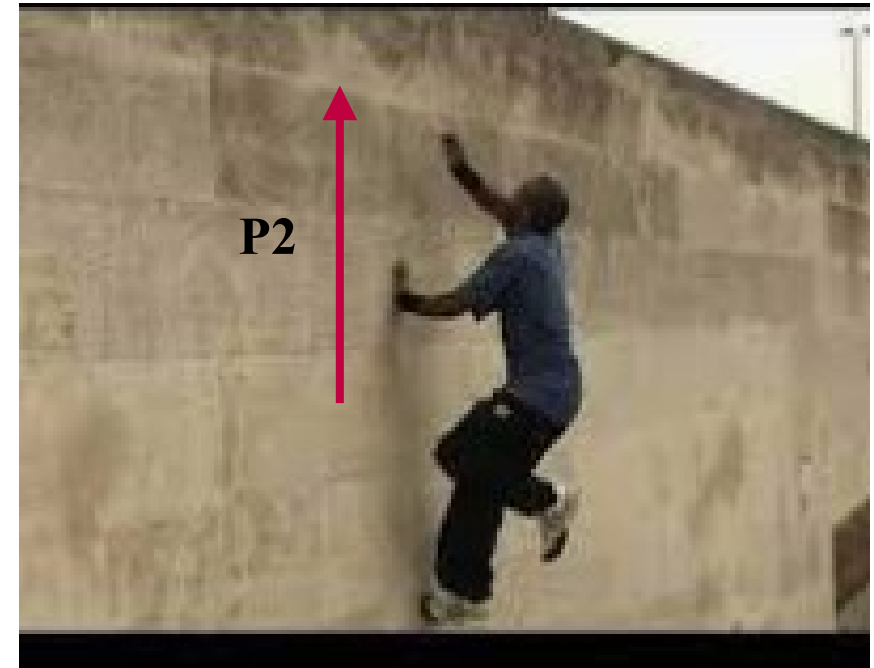
Describe in each
Case, what is the “job”
Of each component.





The force from the wall has 2 components.

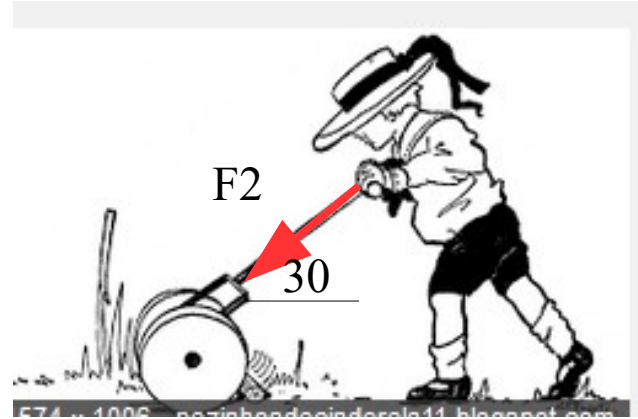
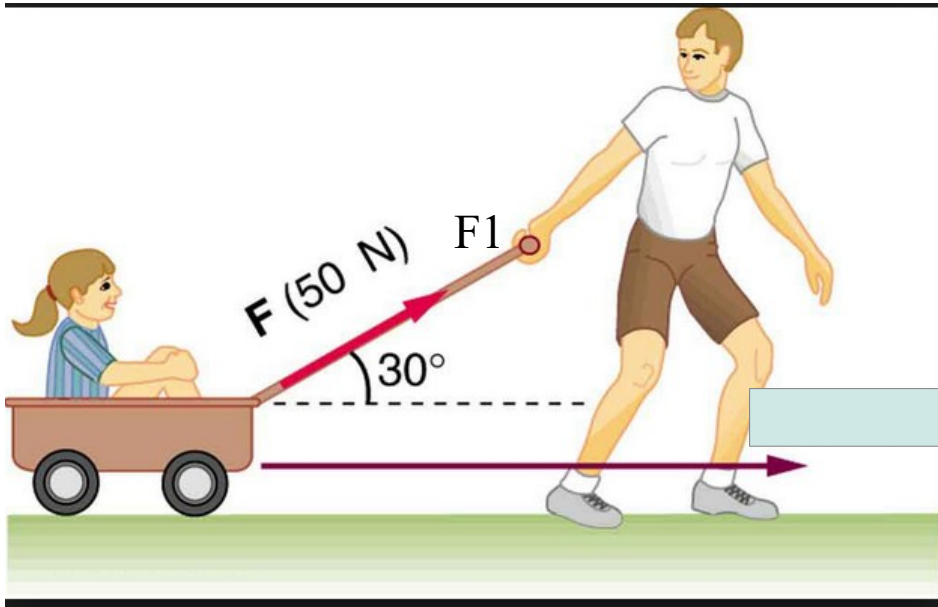
One component stops the person and one component
Propels the person up



The vertical momentum →
Huge force over small time
Force is friction and propels
upward

The effect that a force has in any direction can be found by calculating its component in that direction

If the vector (force) is in standard notation → cosine gives the x-component
→ sine gives the y-component



Draw each vector in a free-body diagram (x-y coordinate system). The tail is attached to the origin. Compute the components of the vectors. The magnitude of the force is the same but not its direction.

$F_1 = 50\text{ N @ } \underline{\hspace{2cm}}$ but $F_2 = 50\text{ N @ } \underline{\hspace{2cm}}$ (use standard notation)

The components are not the same : $F_1 (\quad , \quad)$ $F_2 (\quad , \quad)$

Describe what the x-component and the y-component are doing to each object.

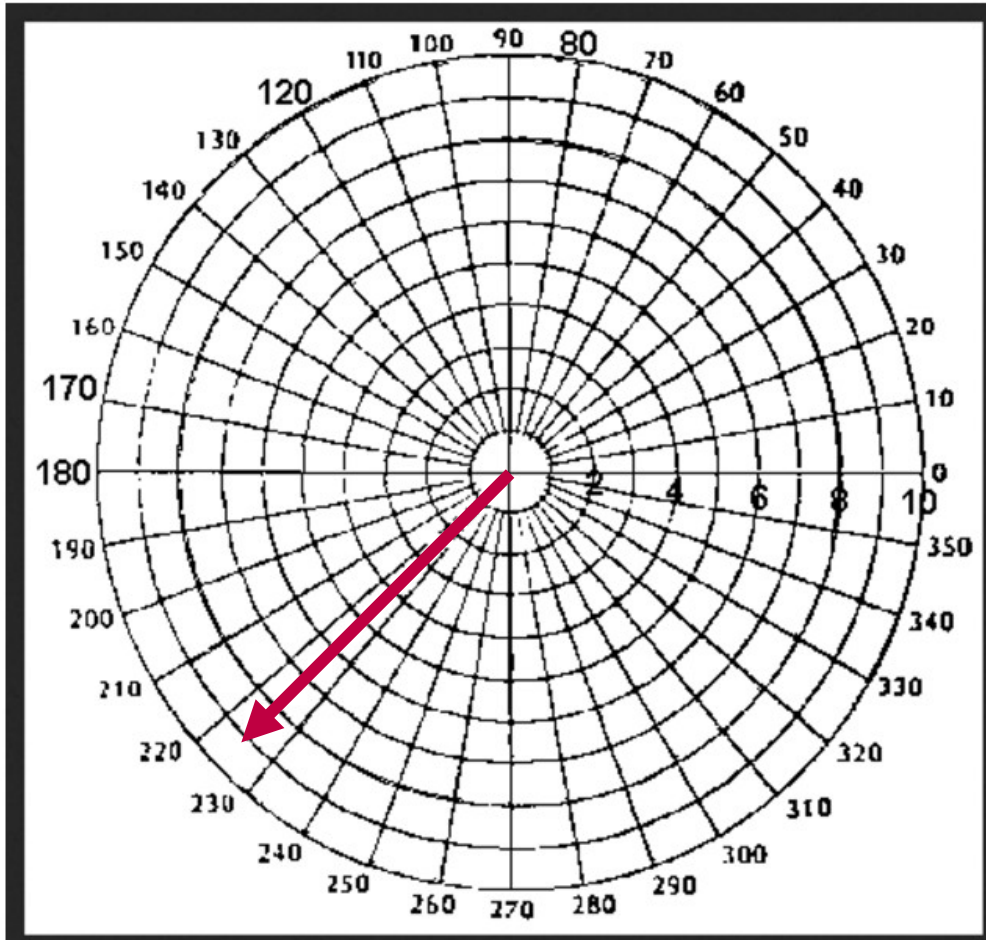
The y-component of F_1 is the wagon with a force of .

The y-component of F_2 is

If there is friction. Which way is the best to pull with less effort ?

$F_1(43,25)$ $F_2(-43,-25)$

To find the components of vectors. You can use a polar graph if you don't get trig (you should get it).

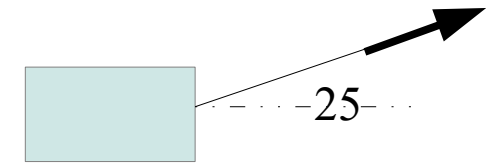


In a free-body diagram you find the components of the forces.
The x-axis is independent from the y-component

1) you are dragging a crate and the rope makes a 25 degrees angle with horizontal. The tension in the rope is 250N. Make a free-body diagram and show the force.

A) How much force is being used to drag the crate ?
(find the horizontal component)

B) How much force (from you) is pulling the crate upward ?

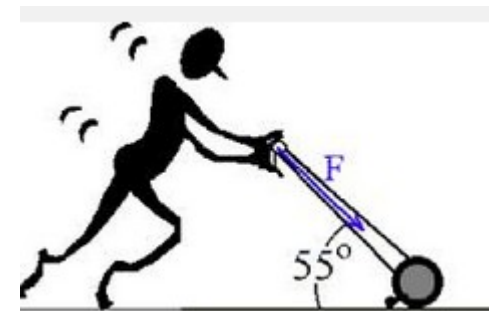


Solution: $F(226.6, 105.6)$

3) You push a lawn mower with a force of 160N, exerted directly along its shaft. The shaft makes an angle of 55 degrees with the ground. Make a free body-diagram

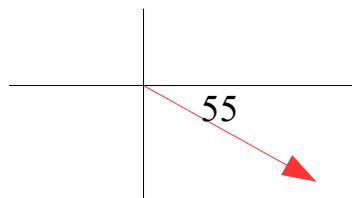
A) How much force is moving the lawn mower (along horizontal)

B) how much force is pushing the lawn mower toward the ground.



Solution:

$F(92, -131)$

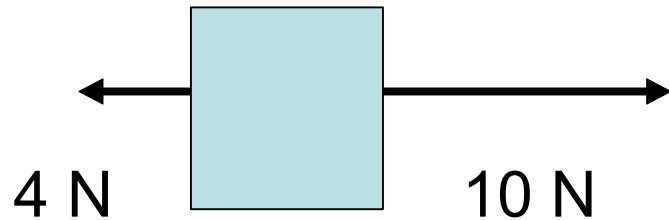


Adding vectors

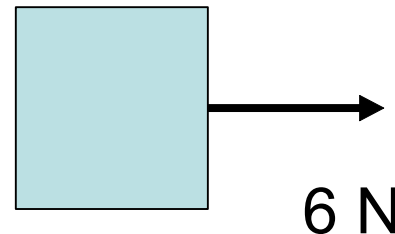
The net force on an object is the vector sum of all forces acting on that object.

The SI unit of force is the Newton (N).

Individual Forces

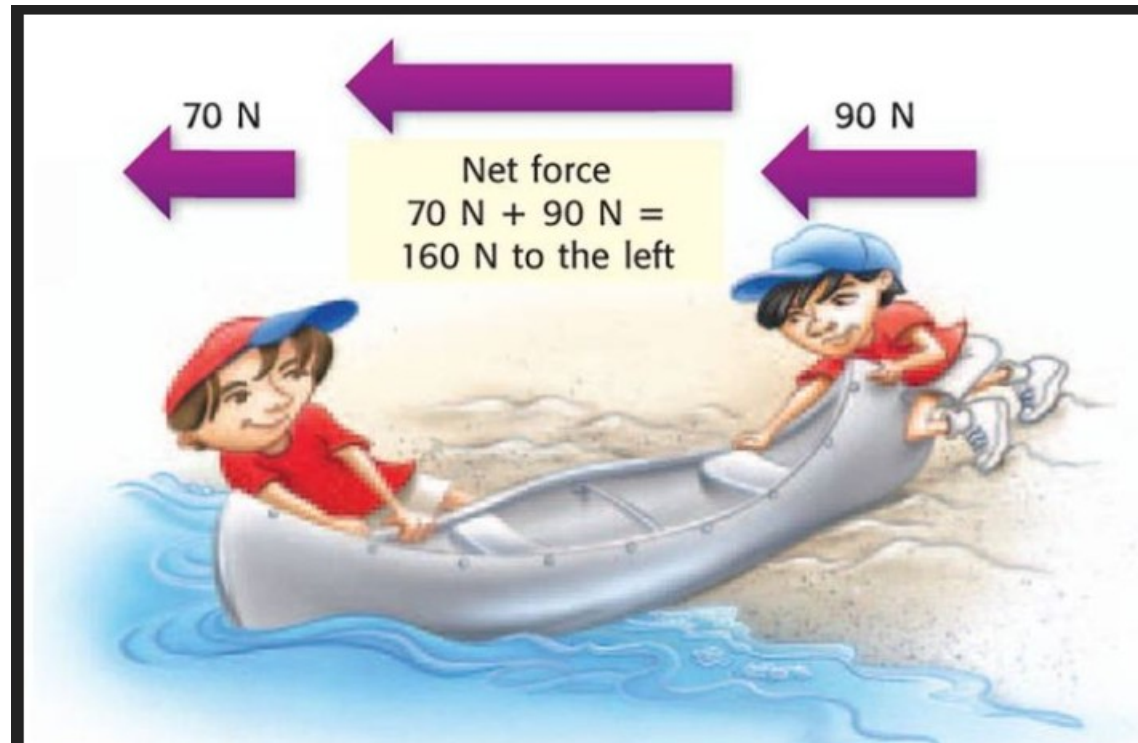


Net Force



Adding vectors:

Finding the net force acting on an object → which way the object will move because of Those forces and what is its acceleration ($F_{\text{net}} = \text{mass} \times \text{acceleration}$)



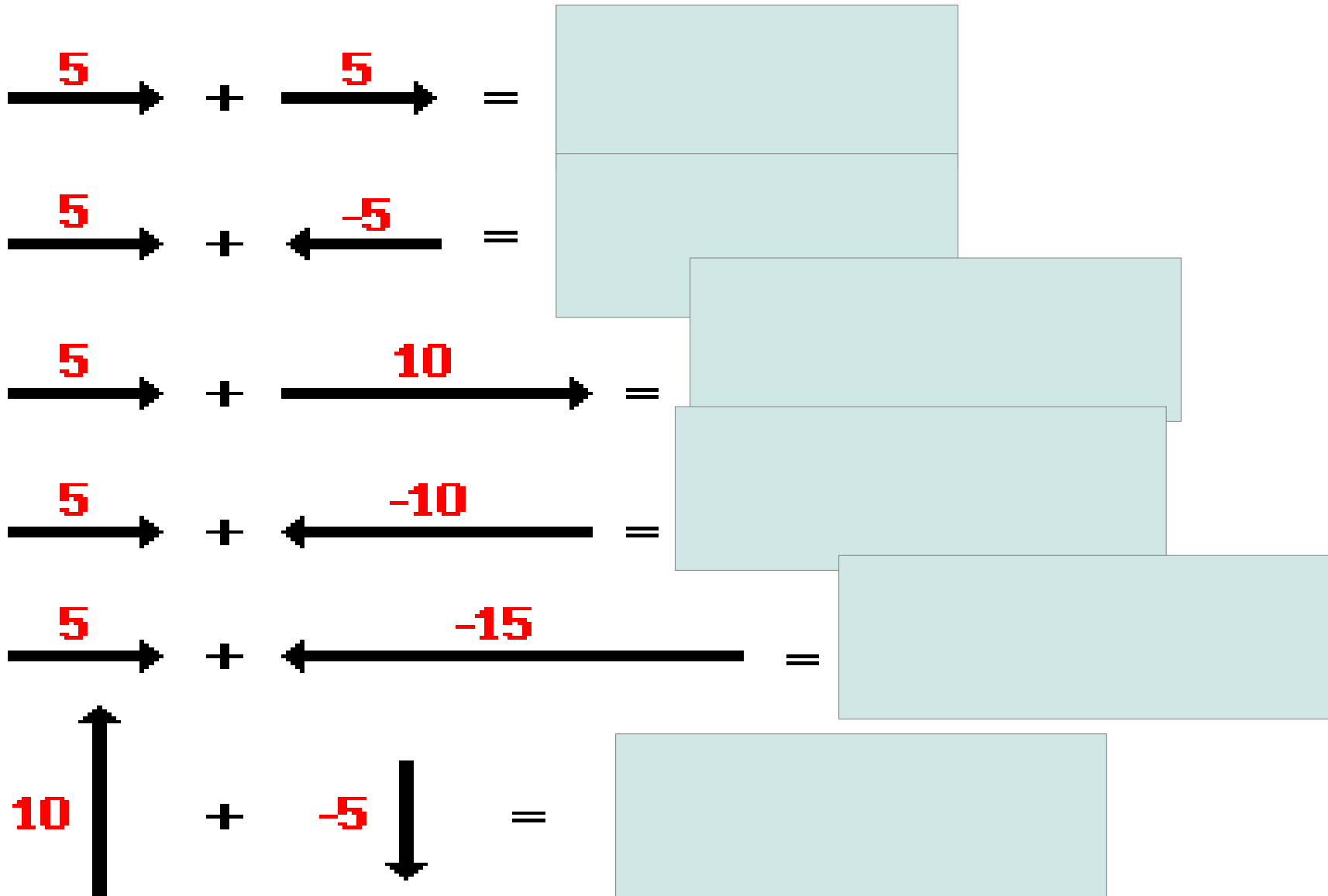
Mathematically, the net force is written as

$$\sum \vec{\mathbf{F}}$$

where the Greek letter sigma denotes the vector sum.

Adding forces in 1D is easy:

The *net force (sum)* is the vector sum of all of the forces acting on an object.

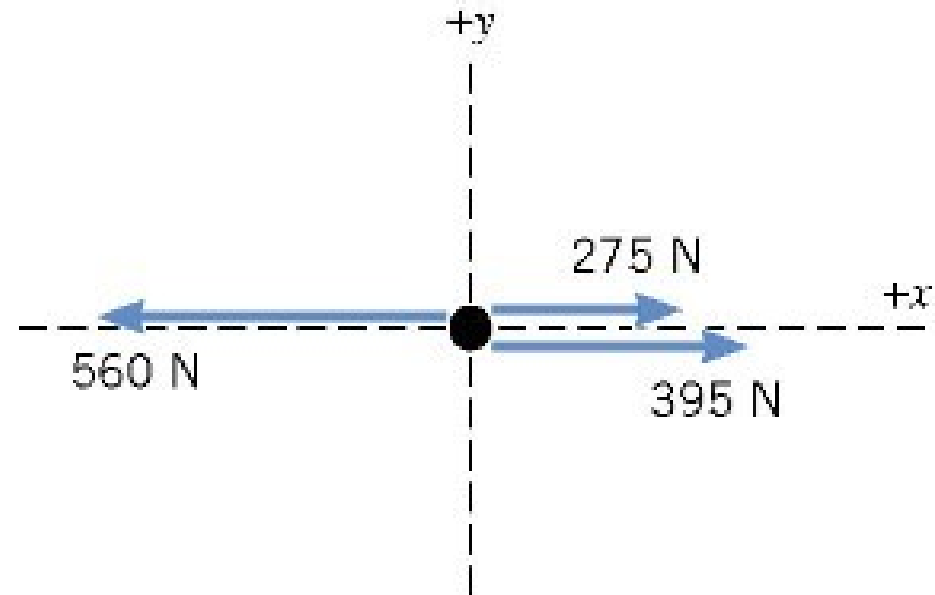


Source:

4.3 Newton's Second Law of Motion



(a)



(b) Free-body diagram of the car

The net force in this case is:

and is directed along the $+x$ axis of the coordinate system.

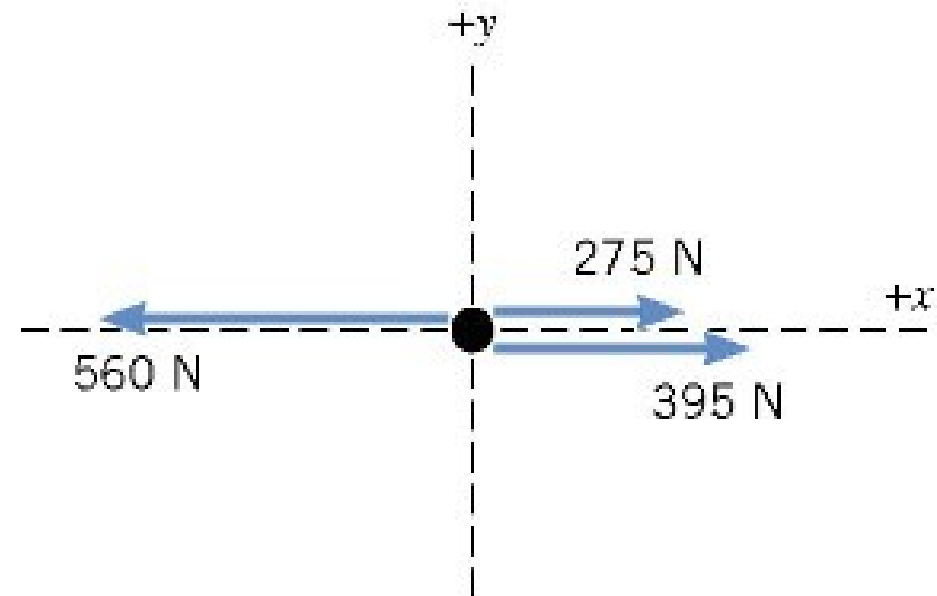
A ***free-body-diagram*** is a diagram that represents the object and the forces that act on it.

Find the net force

$$275 \text{ N} + 395 \text{ N} - 560 \text{ N} = +110 \text{ N}$$



(a)



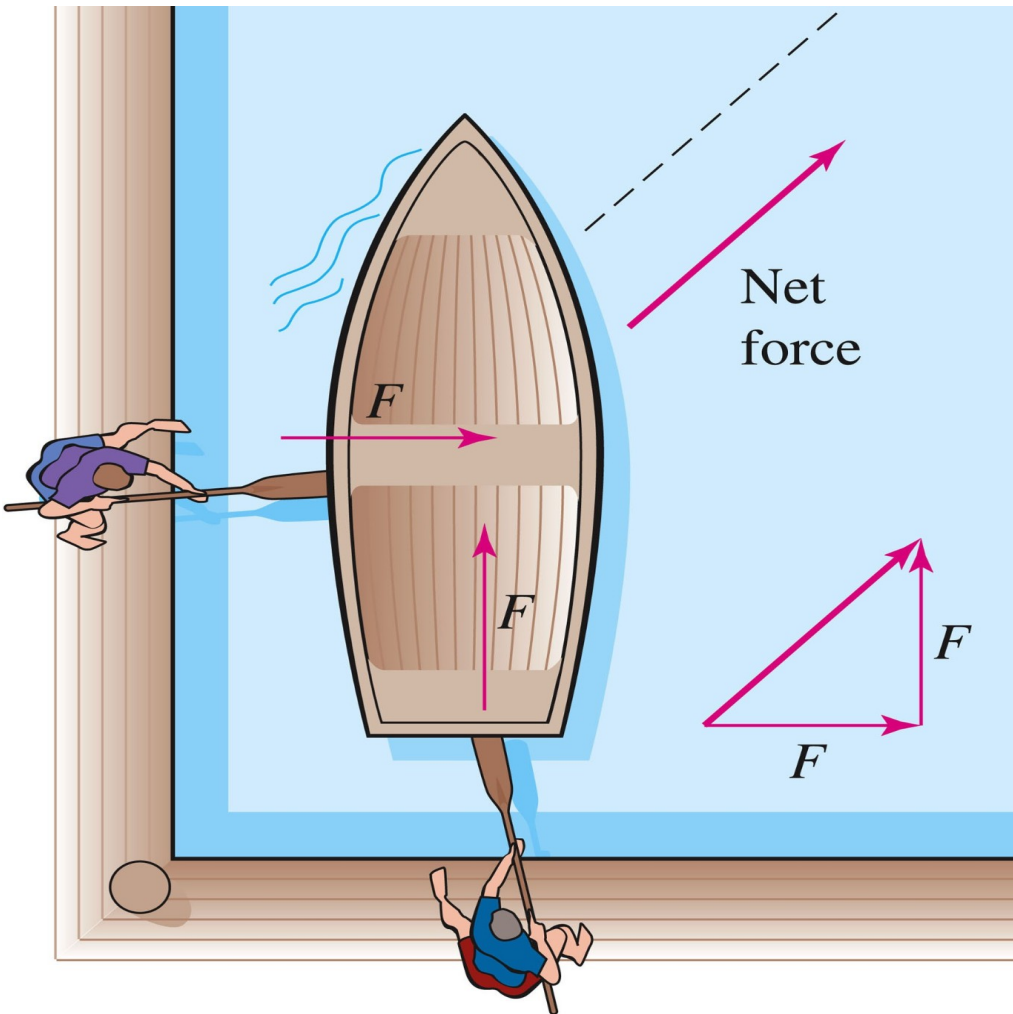
(b) Free-body diagram of the car

Which one of the following tools is useful in representing the forces acting on an object and simplifies problem solving?

- a) free-body diagram
- b) scalar drawing
- c) vector analyzer
- d) Newton's ladder
- e) force monitor

HOW to add VECTORS in 2D → tail to head method.
Easy if the vectors are perpendicular.

In which direction do you think the boat will move ? =
Draw the resultant F_{net} or the net force applied to the boat (a vector).
The boat will move in the same
Direction as F_{net} . (Newton's second law → $F_{\text{net}} = \text{mass} \times \text{acceleration}$
→ the boat will accelerate in that direction.



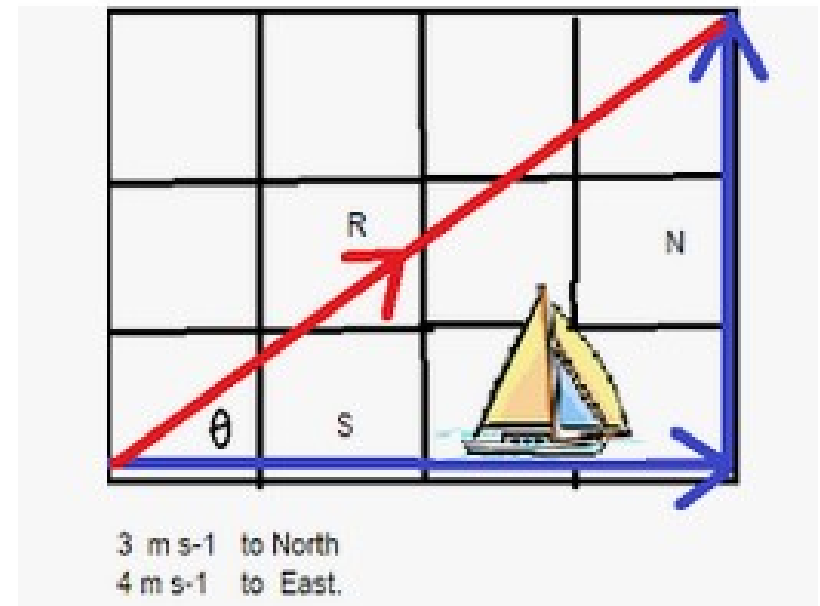
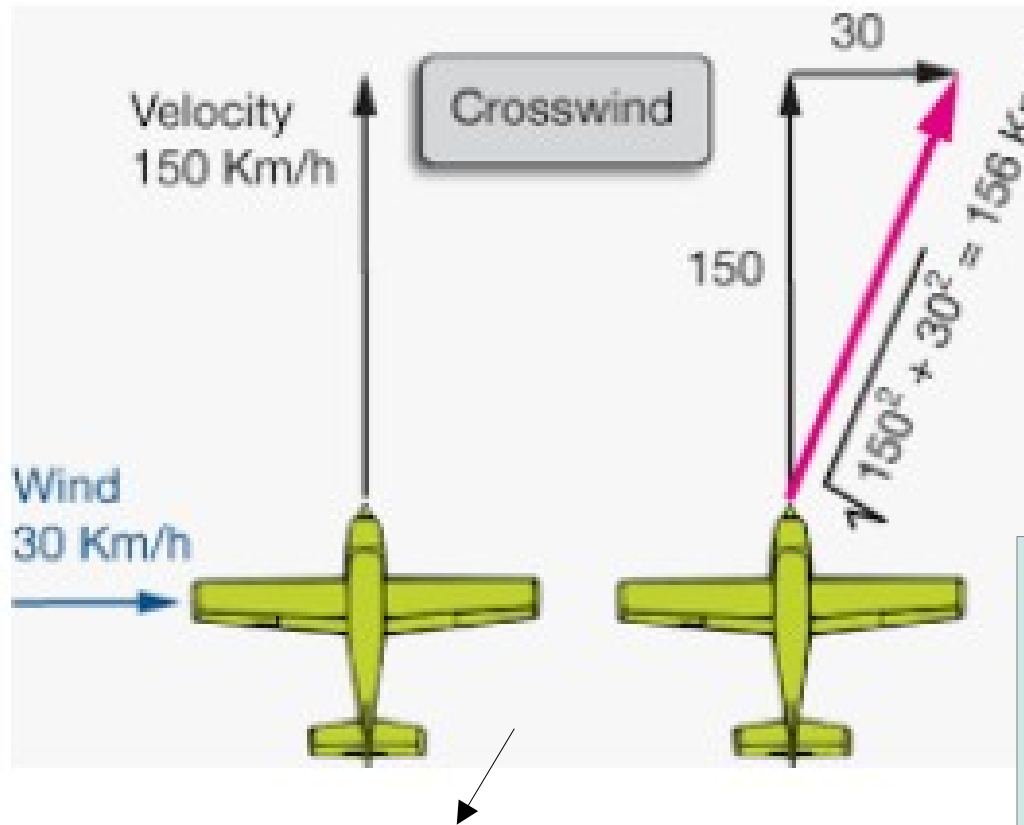
1) If $|F| = 100\text{N}$
Find R or F_{net}
Use the tail to head method.

2) draw a free-body
attach the tails
Of the forces at the origin.
The boat is reduced to a point.

Adding vectors , you can also add vector velocities

Again → tail to head method.

If the vectors are perpendicular, its easy to find the vector sum



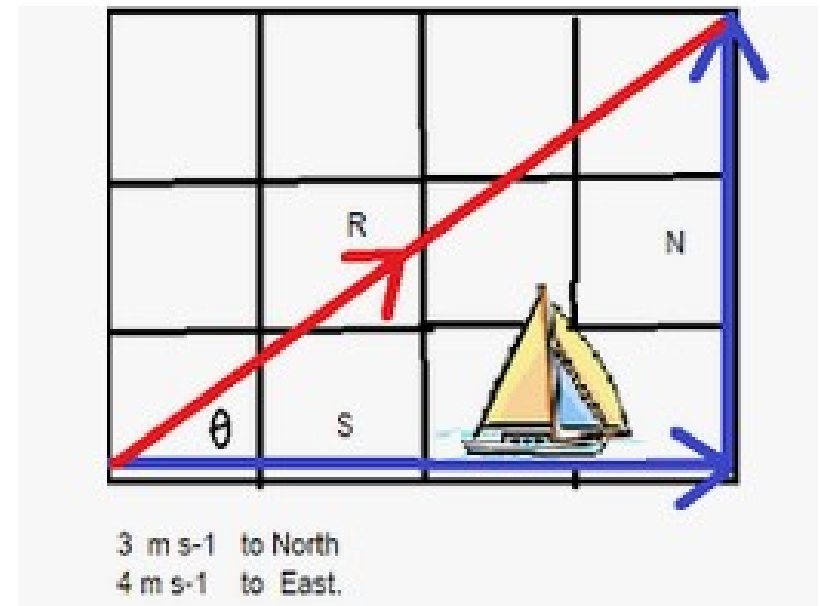
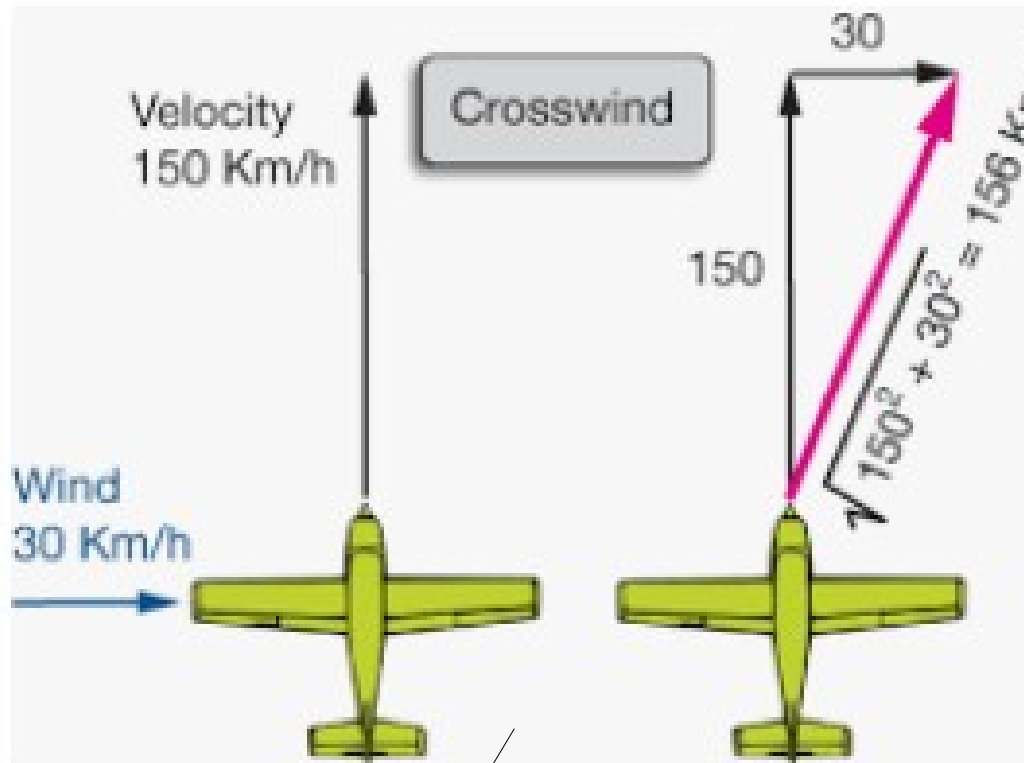
1 square is 1 m/s
Find the vector sum

Find the velocity of the plane relative to ground.

Adding vectors , you can also add vector velocities

Again → tail to head method.

If the vectors are perpendicular, its easy to find the vector sum



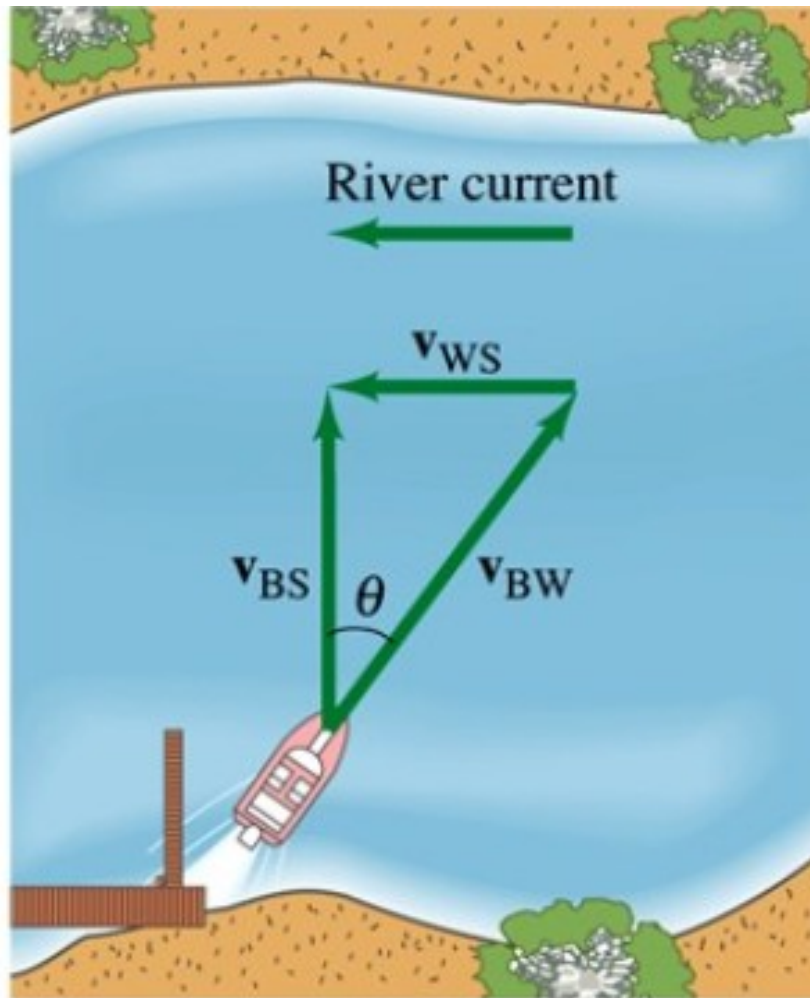
Angle = $\tan^{-1} (3/4) = 37^\circ$
Vector sum = 5 m/s @ 37° N of E

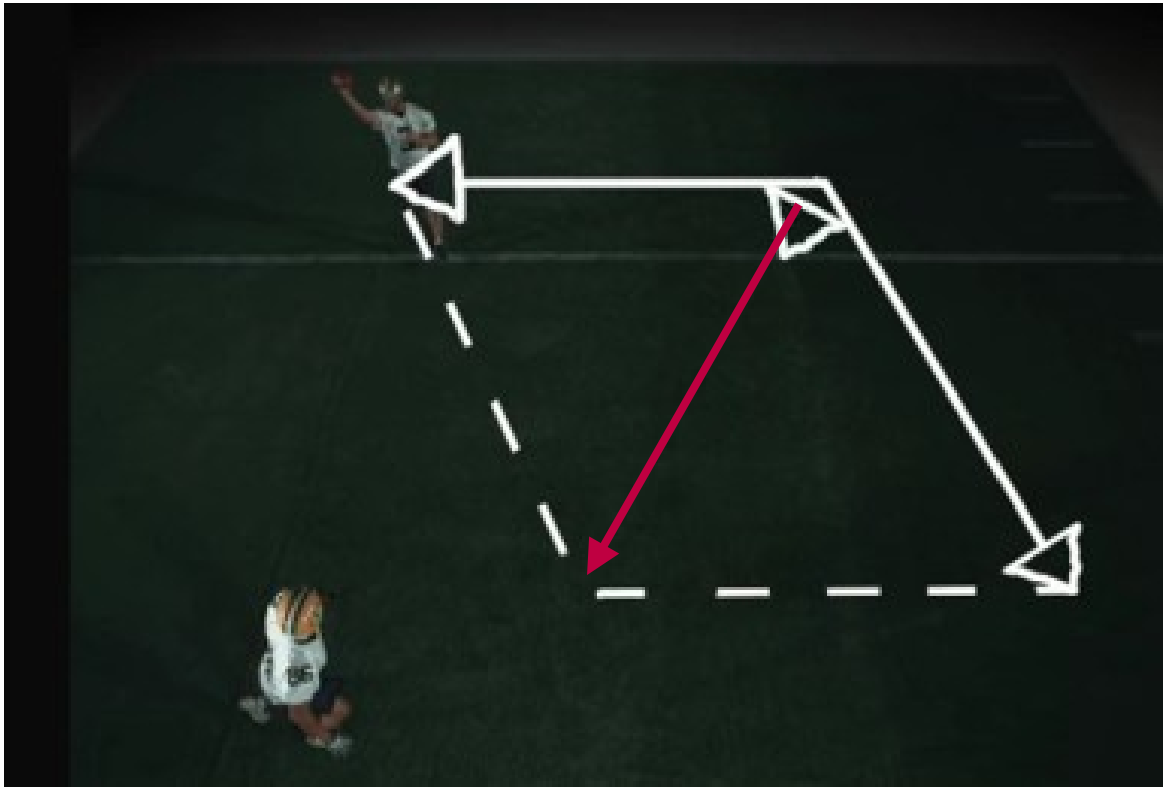
Angle = $\tan^{-1} (30/150) = 11^\circ$

Vector sum = 156 km/h @ 11° E of N

When finding relative velocity $\rightarrow$ you are Adding vectors
(tail to head method)

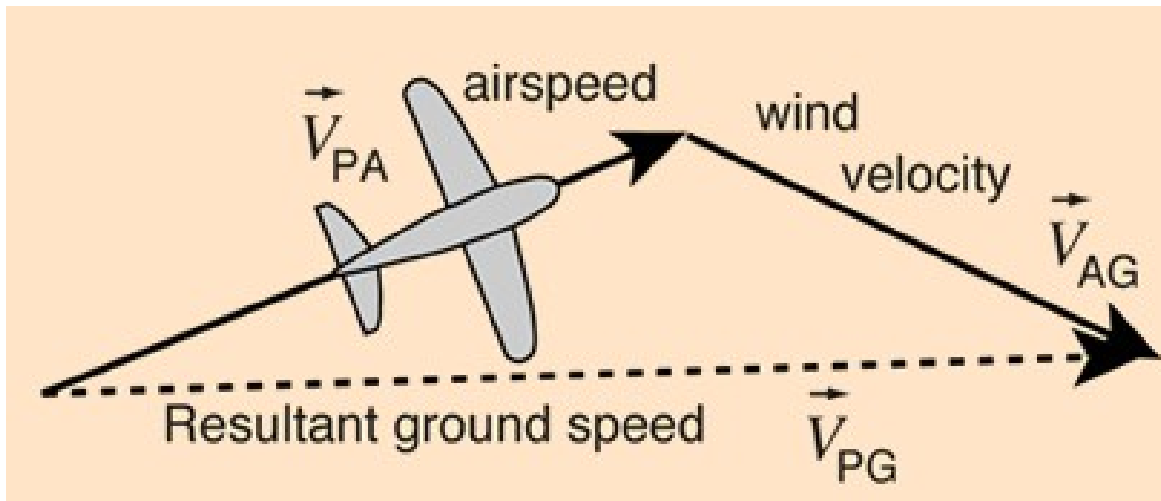
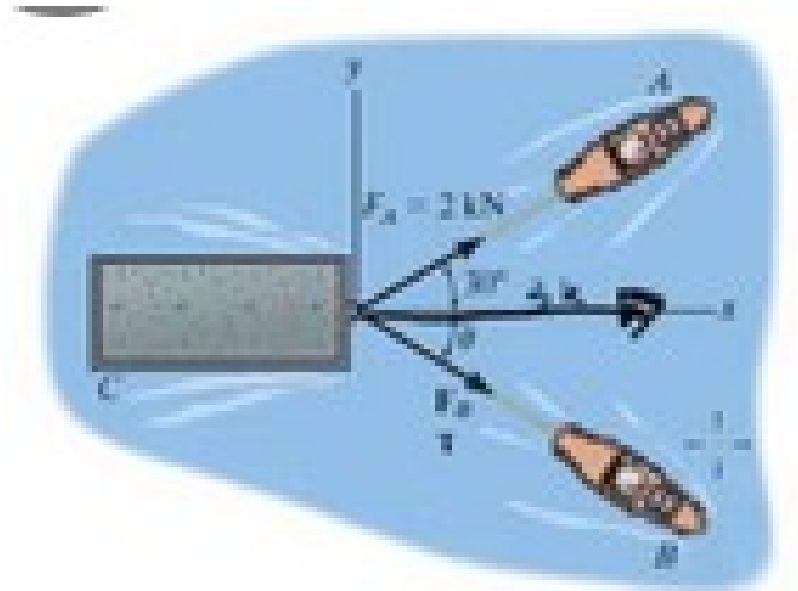
$$\mathbf{V}_{B/S} = \mathbf{V}_{B/W} + \mathbf{V}_{W/S}$$





The vector velocity of the ball has 2 components (perpendicular)
The motion of the ball is defined by the final velocity
= the vector sum of the 2 components

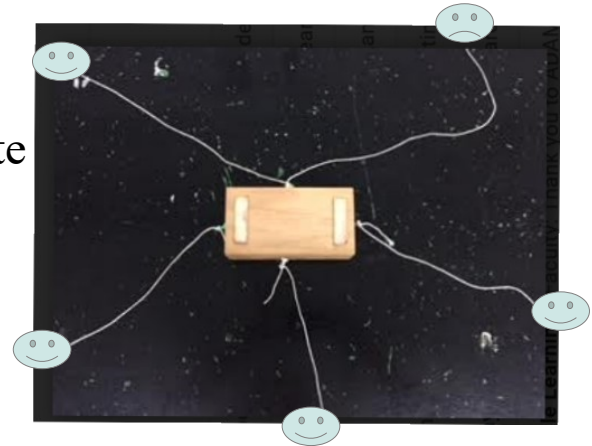
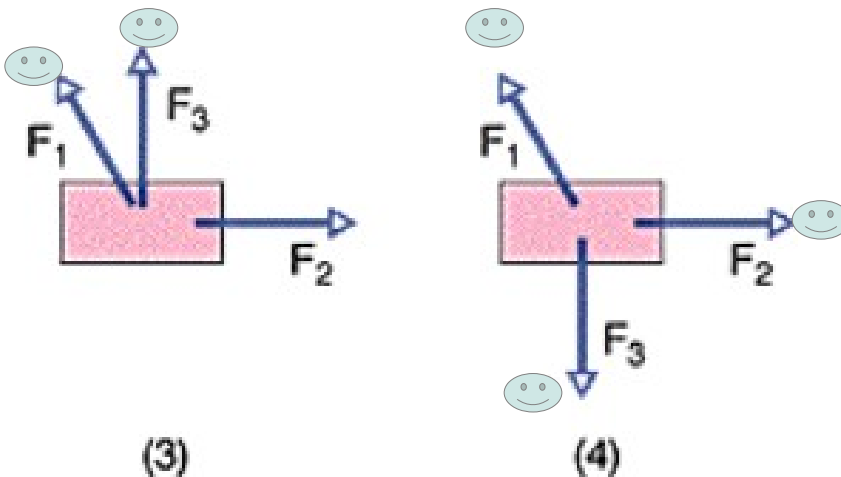
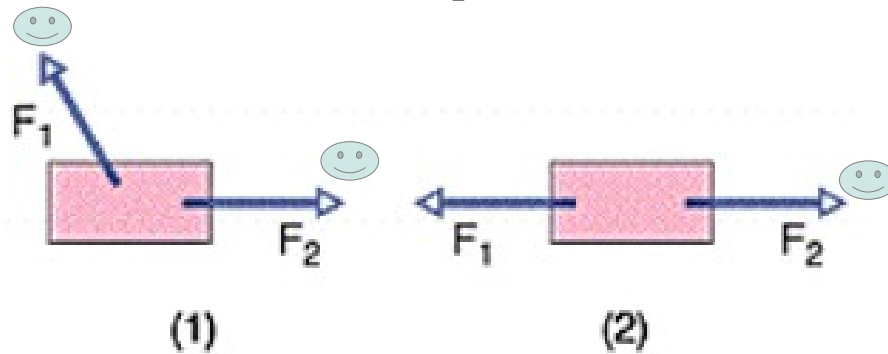
Adding forces in 2D when the forces are not perpendicular to each other → geometrically use head to tail method.



A block is placed on a frictionless table and students are pulling with strings in different direction.
A) In which situations you think the block is not moving (intuition) ? (in equilibrium)

B) If the block is moving which way (about it is moving)

C) for each case add the vectors F_1 and F_2 (geometrically)
To trace the net force. Attach the net force to a x-y coordinate
System. And trace the components.

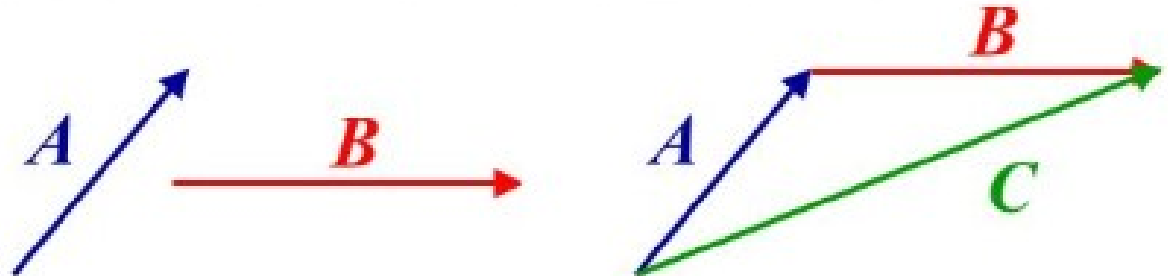


You can find the net force using
This geometric method if
You know each vector 's
Magnitude and direction but you
Need a protractor and a ruler and
To trace vectors with a scale.
(like 1 in = 10N)

Let's do it using Algebra.

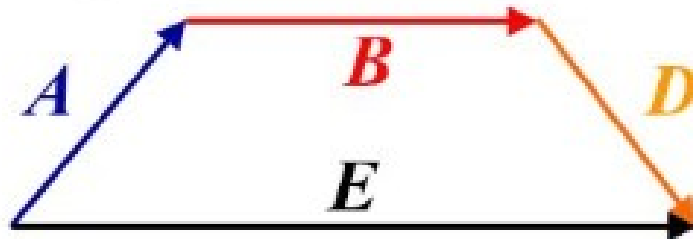
Vector Addition and Subtraction

- For $A + B = C$:



– Vector addition is commutative (order doesn't matter) – try it!

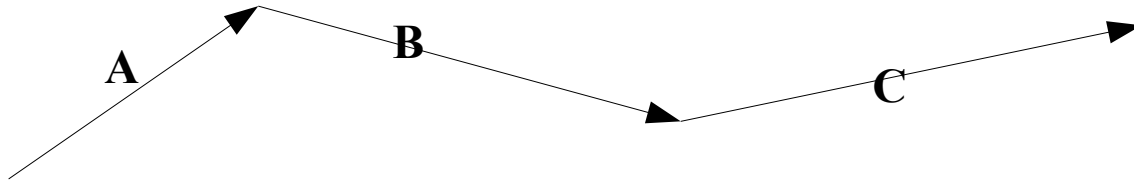
- For multiple vectors $A + B + D = E$:



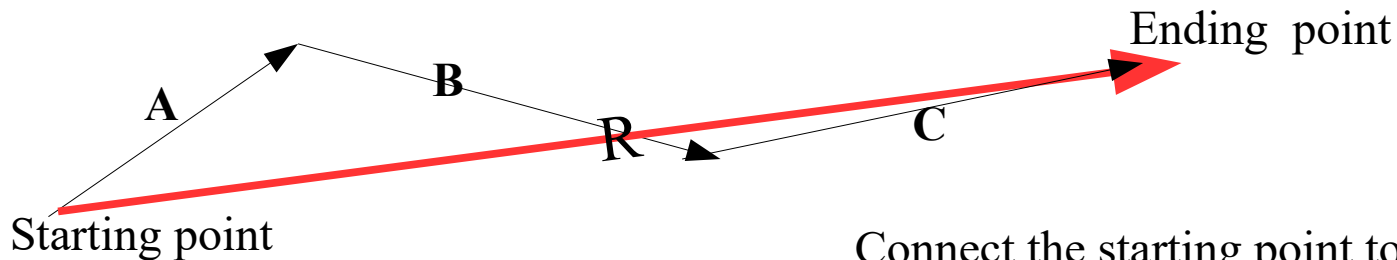
Vector Addition and
Subtraction
Interactive

How to add vectors: method head to tail (tail to head)

Example: add the vectors A, B, C

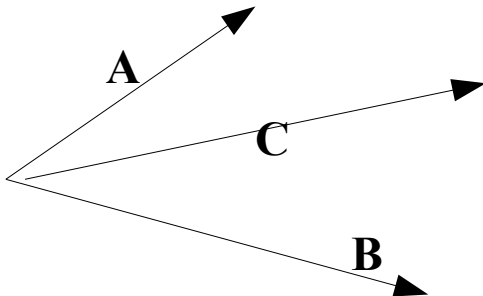


$A+B+C = R$ R is the vector sum (or resultant)

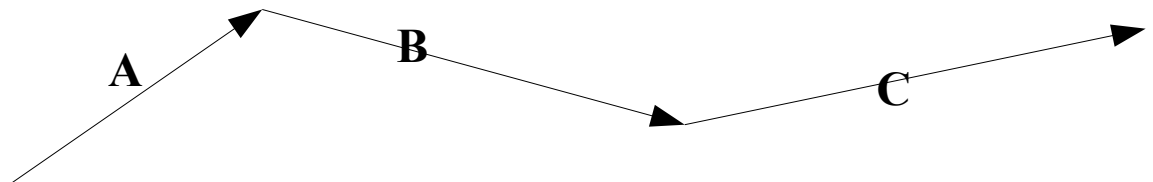


Connect the starting point to ending point
The vectors are tail to head (head to tail)

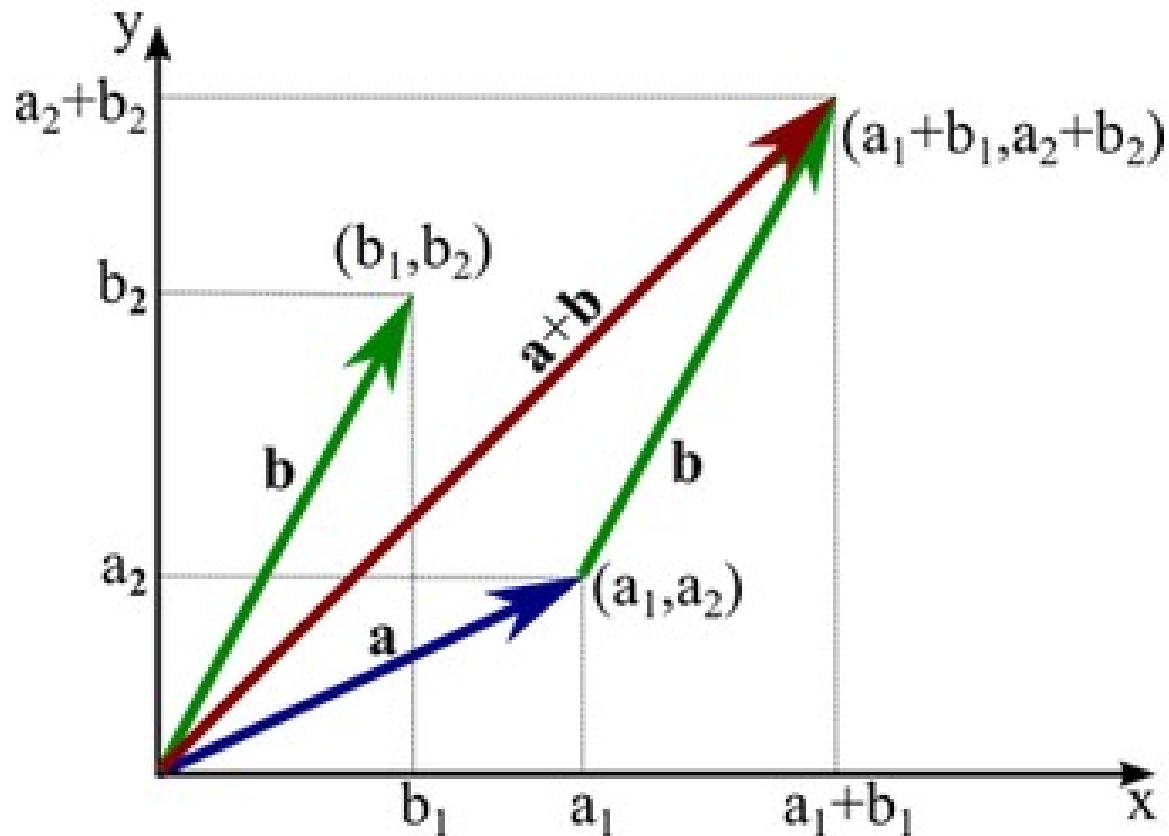
If the vectors are not tail to head then move them tail to head



Move the tail of B on head of A and the tail of C on head of B



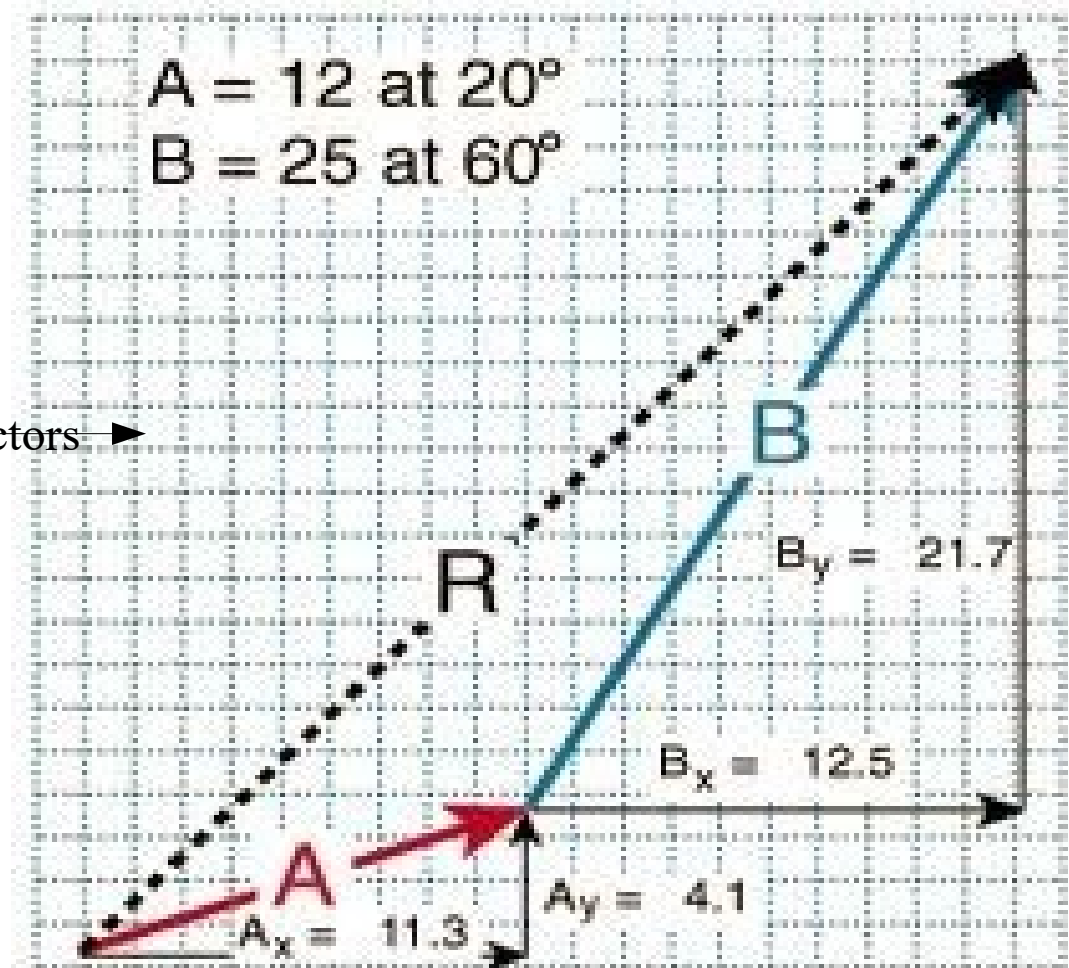
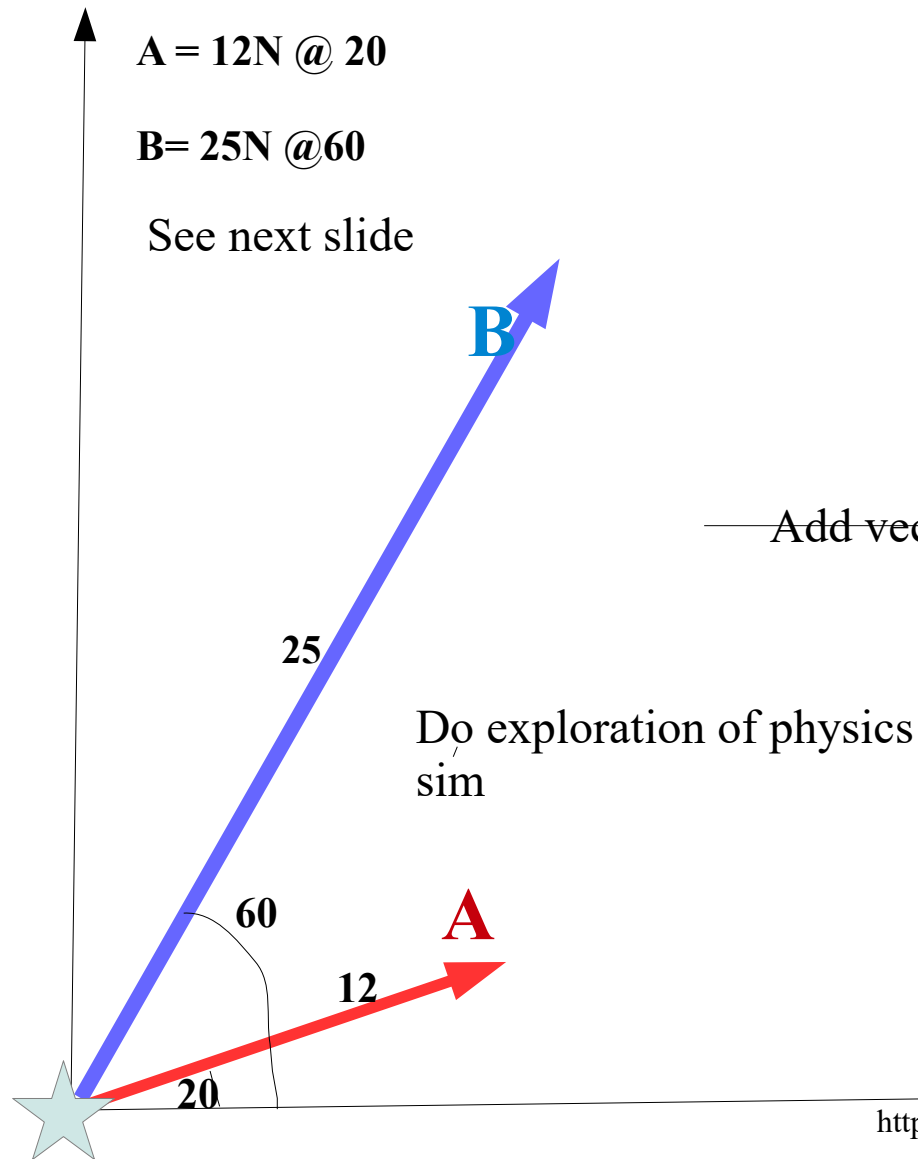
Instead of using geometry, we can use algebra. No need for graph paper, protractor, ruler, scaling.



Adding vectors see exploration of physics

Adding vector algebraically See exploration of physics.

A star is being pulled on a table by 2 students. Trace the net force.

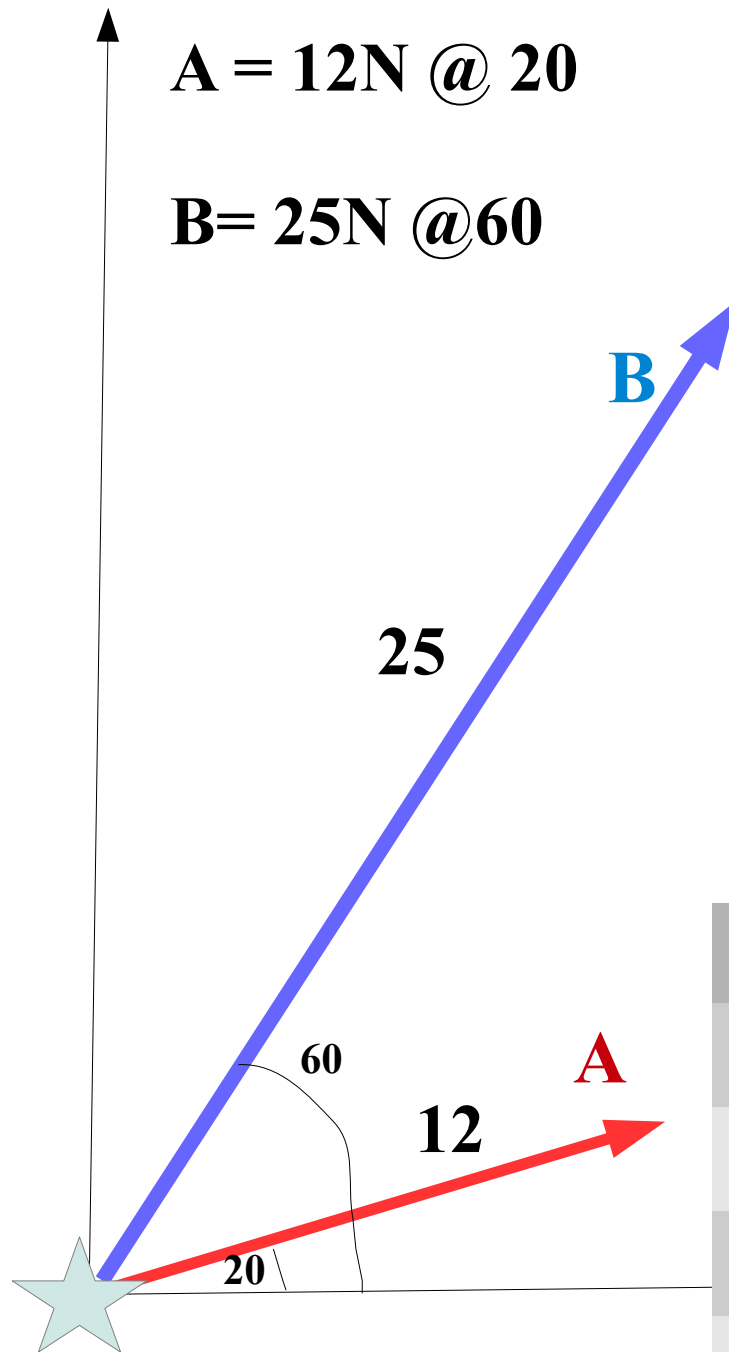


https://phet.colorado.edu/sims/vector-addition/vector-addition_en.html

$$R_x = A_x + B_x = 12.5 + 11.3 = \underline{\hspace{2cm}} \quad R_y = A_y + B_y = \underline{\hspace{2cm}} \quad \text{so magnitude of } R = \underline{\hspace{2cm}}$$

$$\text{Direction} = \arctan(R_y/R_x) \underline{\hspace{2cm}} \quad \text{Do the assignment in class.}$$

When you add vectors you add the components



How to add vector ?

1) trace the vectors in a x-y system = free body diagram

Attach tails to -y coordinate system

2) find the standard notation for each vector

(12N@20 and 25 N @ 60)

3) find the components

$A_x = 25\cos(60)$ $A_y = 25\sin(60)$

$B_x = 12\cos(12)$ $B_y = 12\sin(12)$

4) add the components to find the components of R

$R_x = A_x + B_x = 24$ N

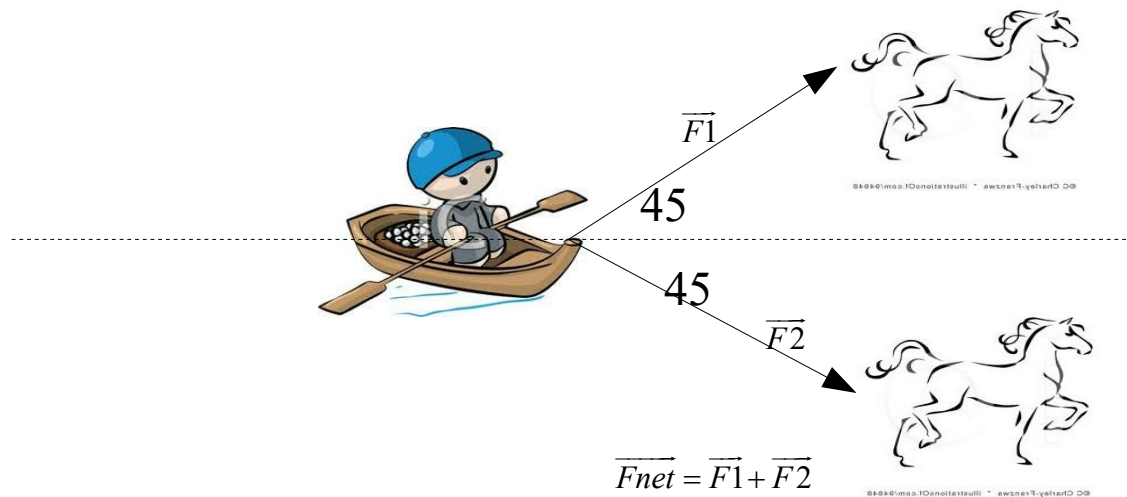
$R_y = A_y + B_y = 26$ N

5) with a different color draw R (R_x , R_y and R) in a free Body diagram)

6) find R in standard notation

$R = 35\text{N @ } 47$ degrees

	X	Y
B		
A		
R		
Direction = $\arctan(\text{opp/adj}) =$		
Magnitude =		

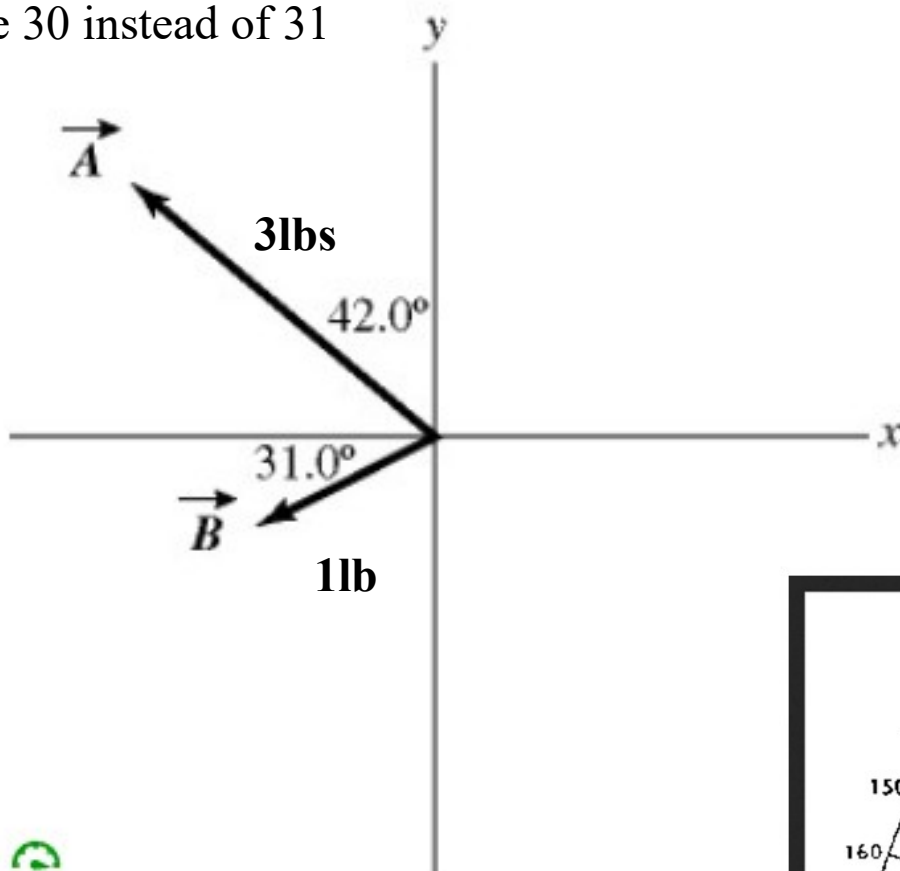


Each horse applies a force of 1000N. What is the net force F_{net} ?

Note horses are capable of pulling up to 15 times their body weight



Take 40 instead of 42
Take 30 instead of 31



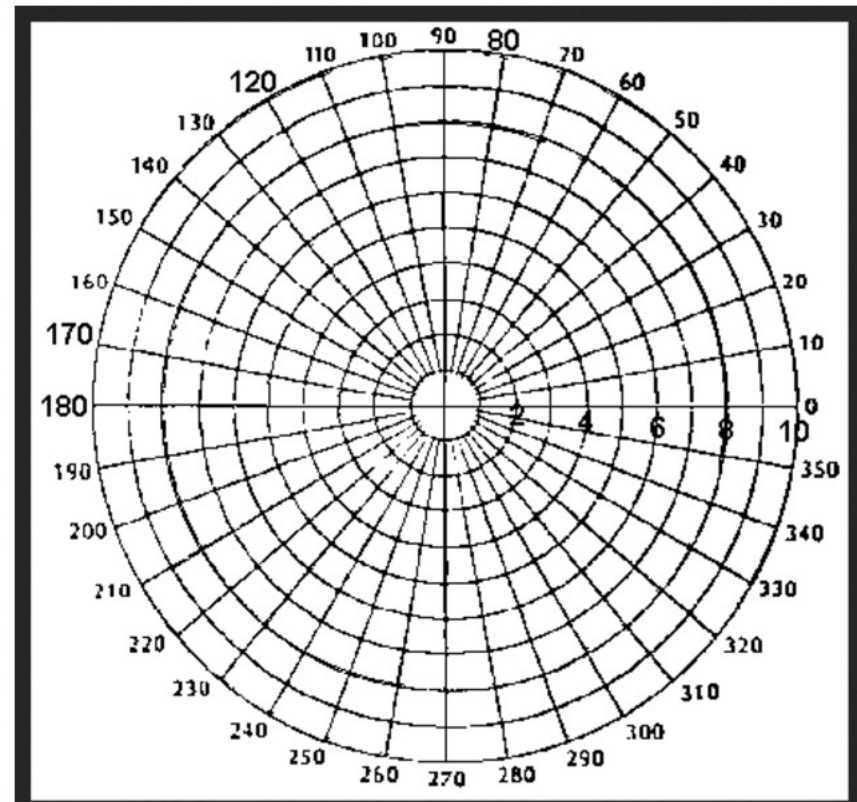
Find the vector sum $R = A + B$

A(., .)

B(., .)

R(., .)

**Draw R in a new X-Y system.
Find angle and mag**



KINDS of FORCES:

Tension in a rope

Normal force (also called support force)

Friction (static or kinetic)

Weight (fundamental force)

WEIGHT: Don't confuse weight and mass.

Mass is a measure of the amount of “stuff” contained in an object.

- Mass does not depend on the Planet the object is placed.

Unit is kilogram (kg)

WEIGHT IS THE PULL DUE TO GRAVITY.

WEIGHT is a FORCE but MASS is STUFF(INERTIA)
WEIGHT is measured in Newtons (or pounds).

**1 kg has a weight of 9.8N (about 10N)
or 2.2 pounds.**

Pound is the unit for force in the British system. The unit for Mass in the British system is the slug !

WEIGHT IS THE PULL DUE TO GRAVITY.
WEIGHT is a FORCE but MASS is STUFF(INERTIA)
WEIGHT is measured in Newtons (or pounds).

weight = mass x g **g is the acceleration due to gravity (m/s/s)**

Mass is in kg

Weight in Newton

On Earth g = 9.8 m/s/s about 10m/s/s

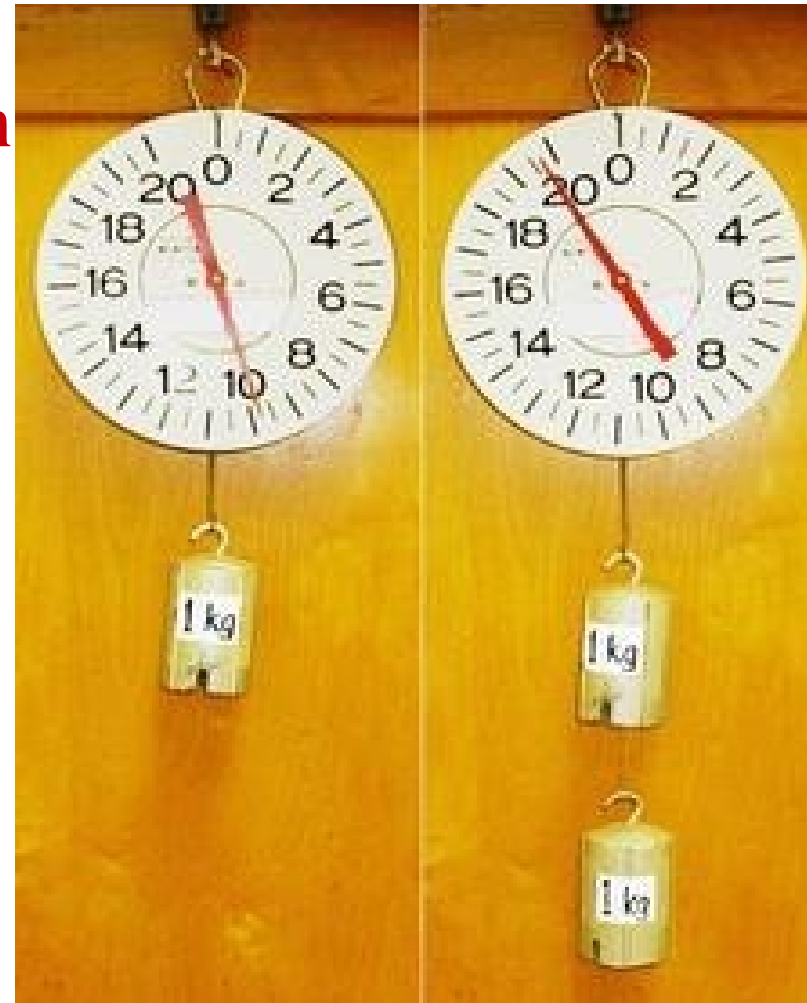
so weight = mass x 10

2.2 Pounds on Earth

1 kg has a weight of 10N

So 10N ↔ 2pounds

Divide by 5

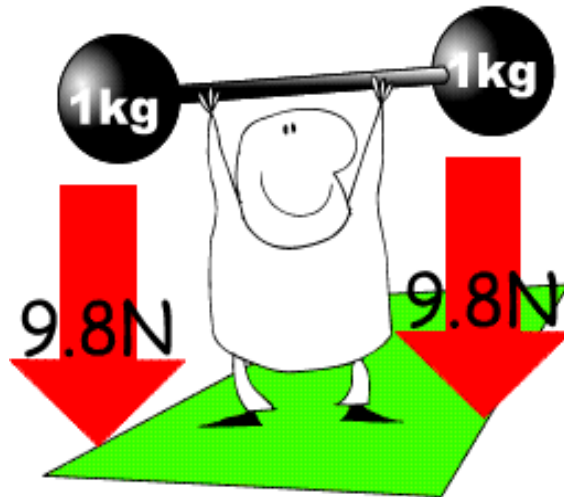


WEIGHT is a FORCE = pull due to gravity

It depends on the planet, star, object.. you are standing on.

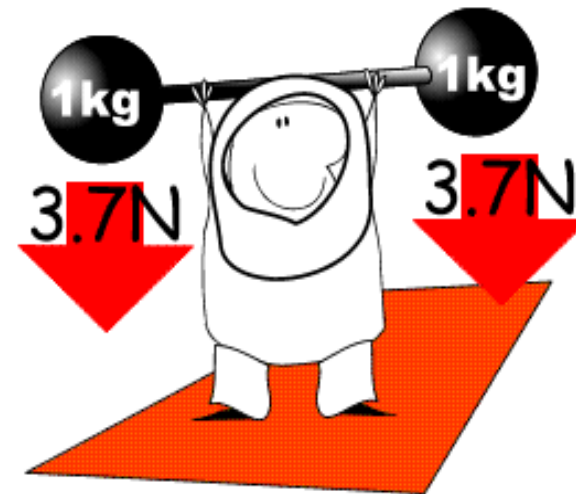
Earth

$$g = 9.8 \text{ N/kg}$$



Mars

$$g = 3.7 \text{ N/kg}$$

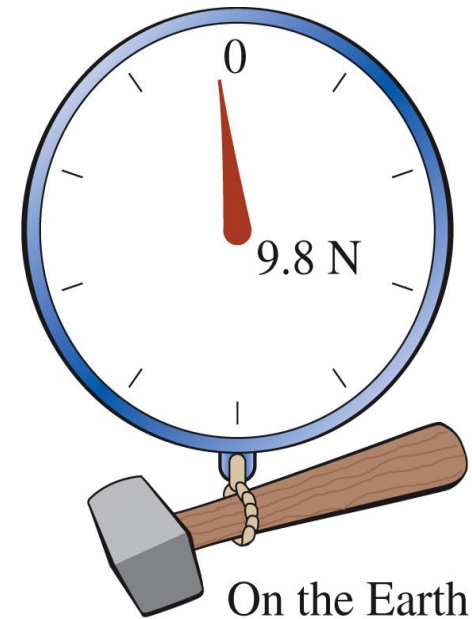
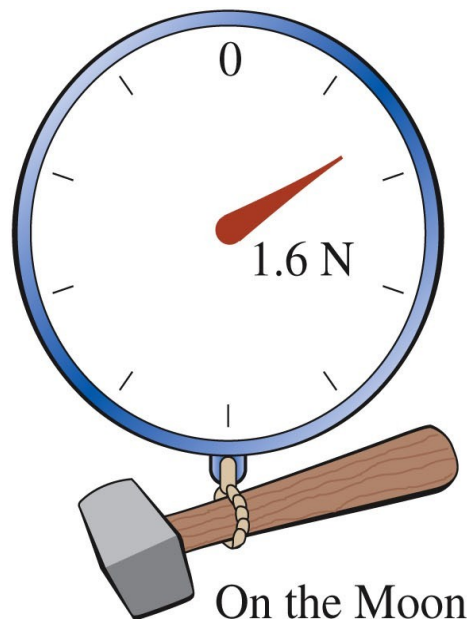
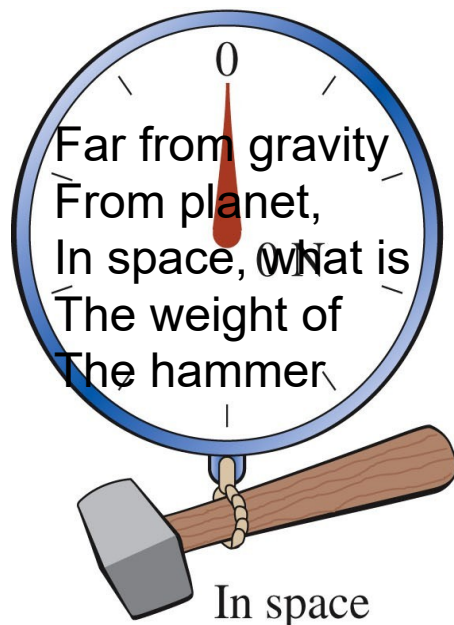


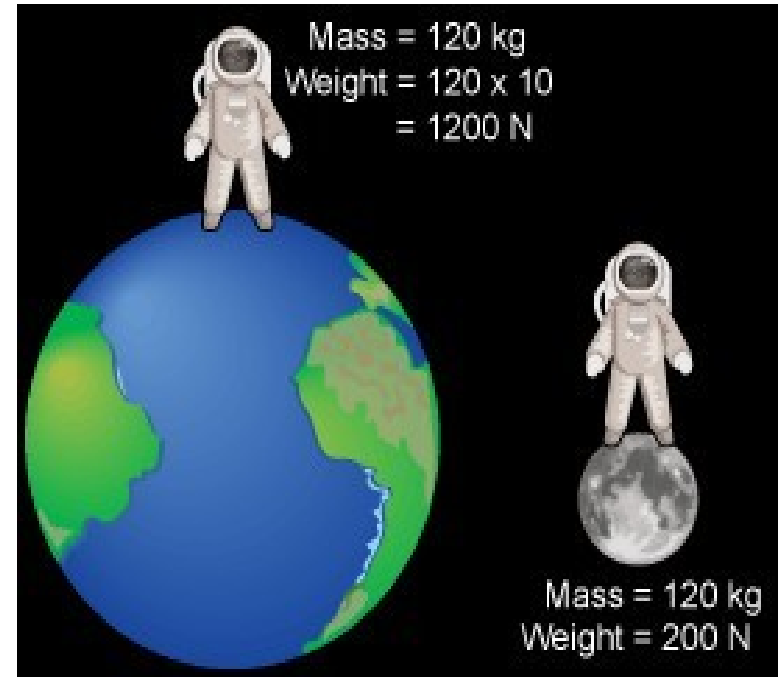
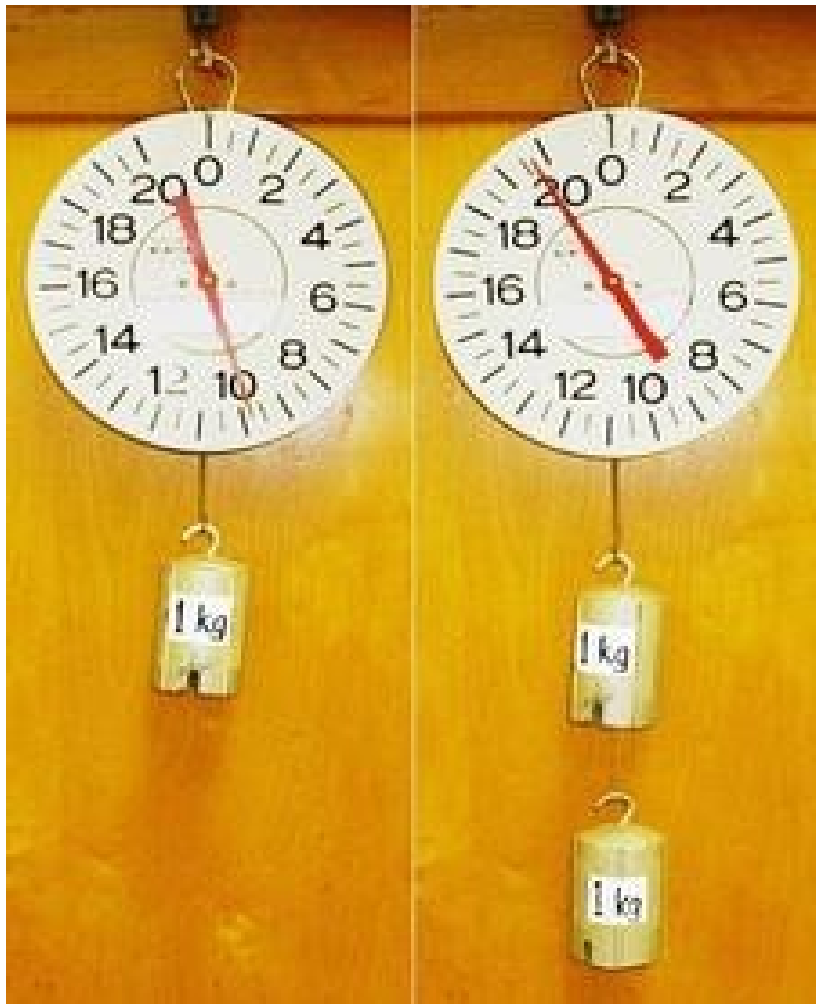
So if your mass is 80kg , your mass on the Moon
Will be : the same ? Greater ? Larger

What about your weight ?

Mass

- The weight of an object depends on which planet you measure the weight.
- The mass of the object is independent of the planet.



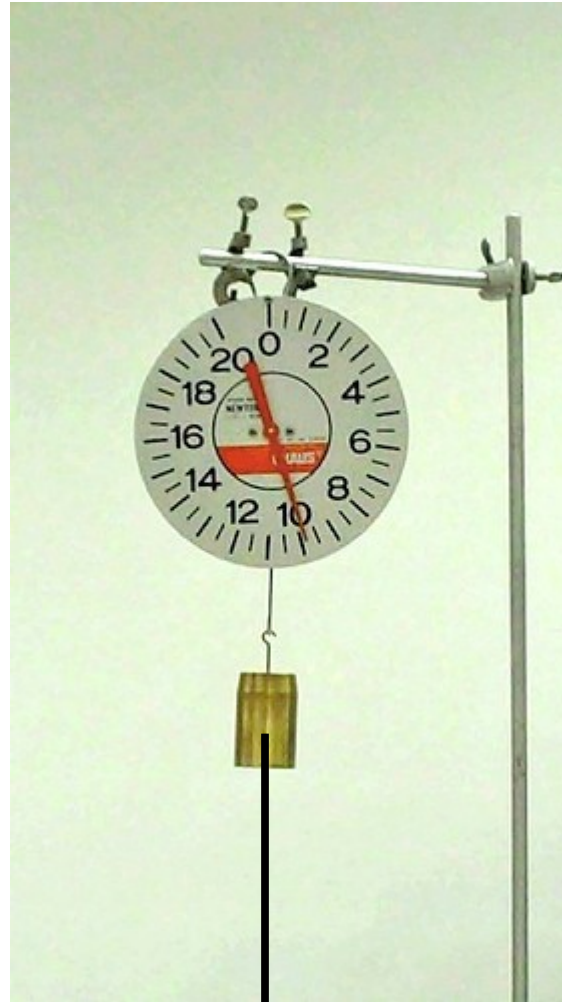


For a 70kg person :
 150 pounds on Earth
 27 pounds on Moon
 400 pounds on Jupiter
 2 tons on the Sun
 ½ ounce on an asteroid (5 pennies)
 (6mph to escape, versus 7 miles per second for Earth = 25 200 mph)



- 1) If a hammer has a mass of 2.5kg, how much does it weigh on Earth ?
On the Moon/ (gravity is divided by 6 on the Moon)
- 2) What is the mass of a girl who weighs 340N on earth ?
- 3) what is the mass of a dog that weighs 75N ?
- 4) convert 10 pounds in newtons.
- 5) On Earth 10 pounds corresponds to what mass in kg ?
- 6) convert 10 newtons in pounds ? What is the mass ?

Weight W is straight down to center of Earth



Mass = 1kg

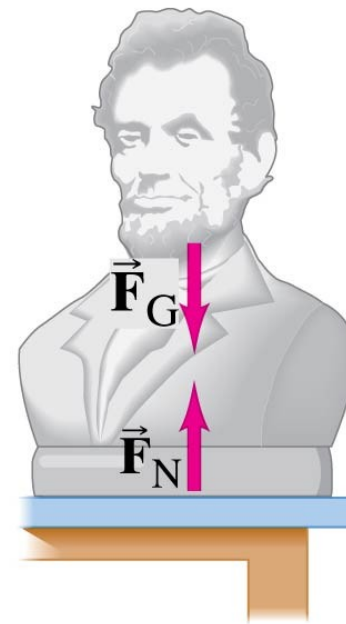
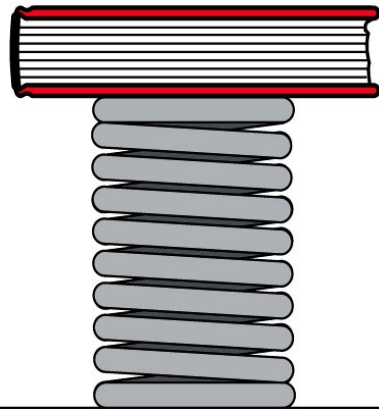
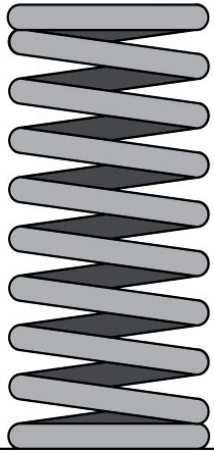
Weight is 10N@down

Definition of the Normal Force

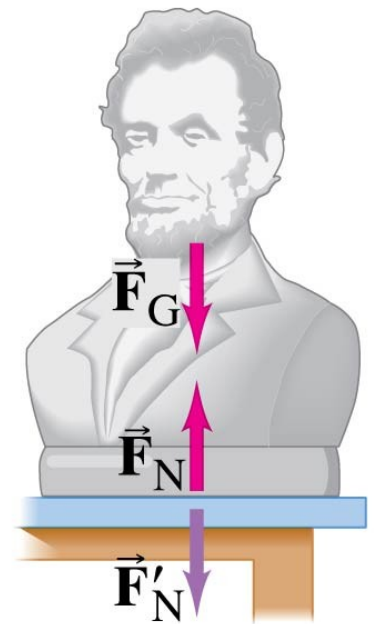
Support force

The normal force is one component of the force that a surface exerts on an object with which it is in contact – namely, the component that is perpendicular to the surface.

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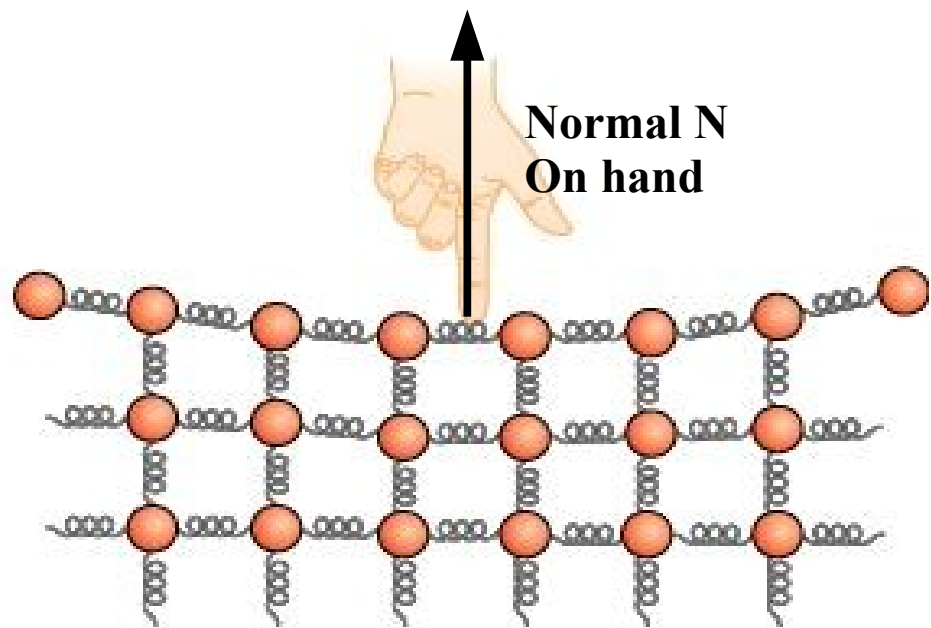


(a)

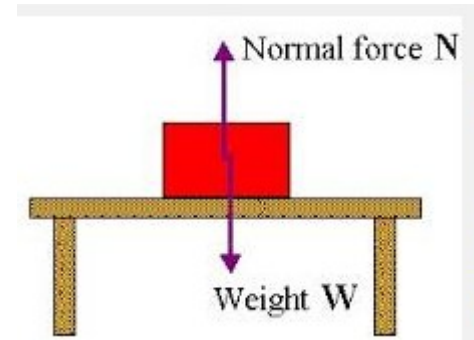


(b)

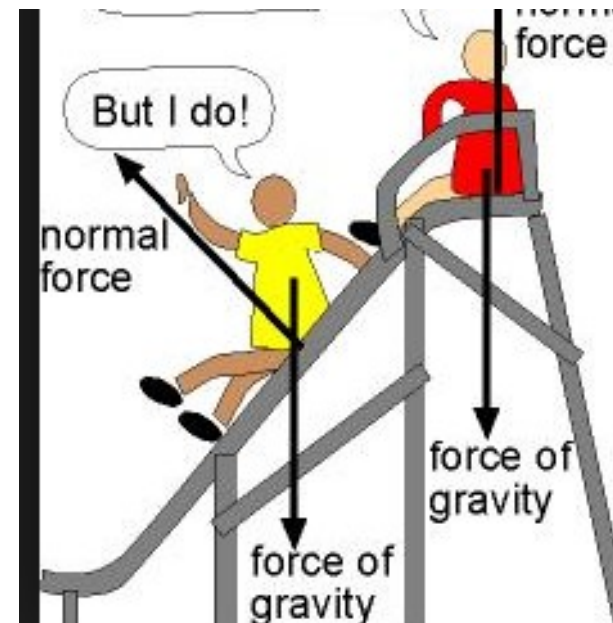
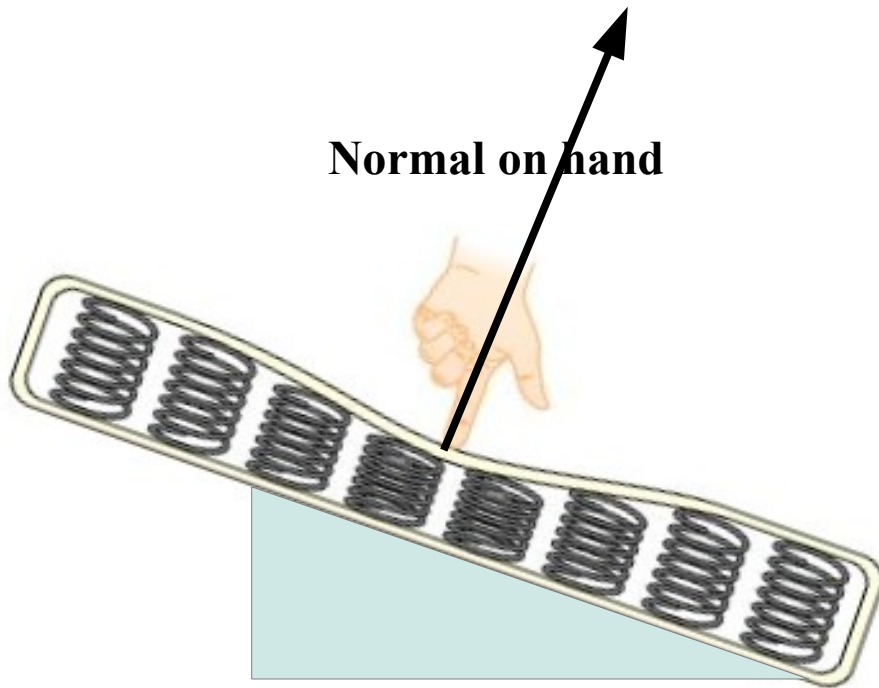
The normal force is a reaction (recoil) force from a surface.

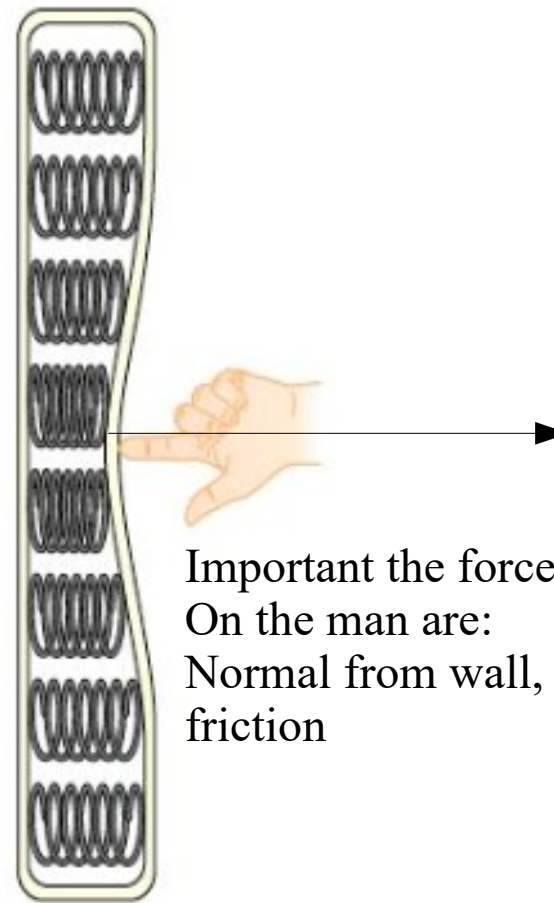
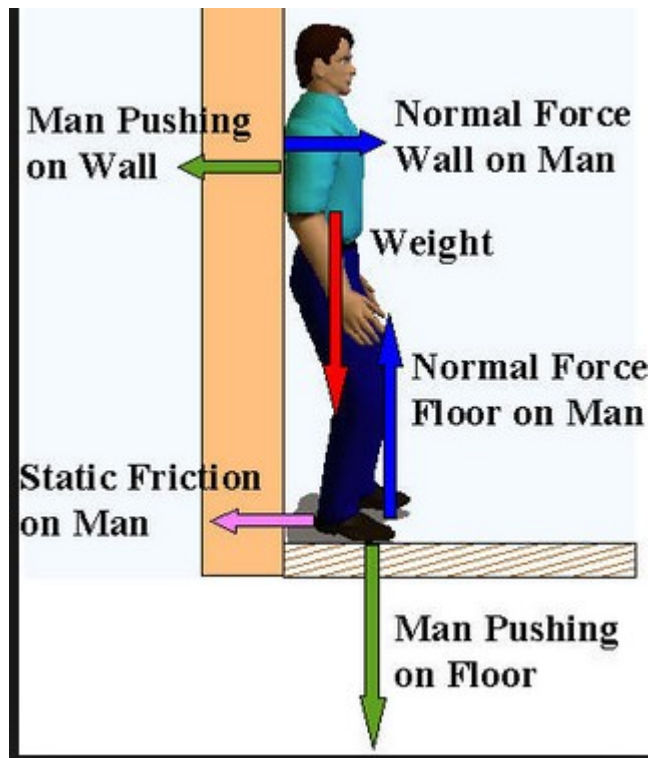


Normal force (also called support force)



Normal on hand





Important the forces acting
On the man are:
Normal from wall, normal from floor,
friction

What is the meaning of the word “normal” in the term “normal force?”

- a) that it is in magnitude and opposite in direction to the weight of the object
- b) that it is one that is encountered in everyday life
- c) that it is directed perpendicular to a surface
- d) that it is measurable
- e) that it has a magnitude of 1 newton

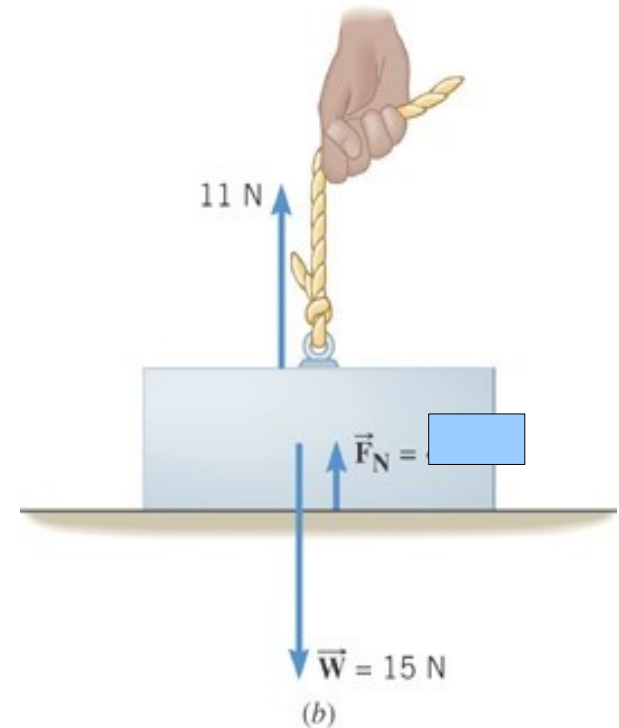
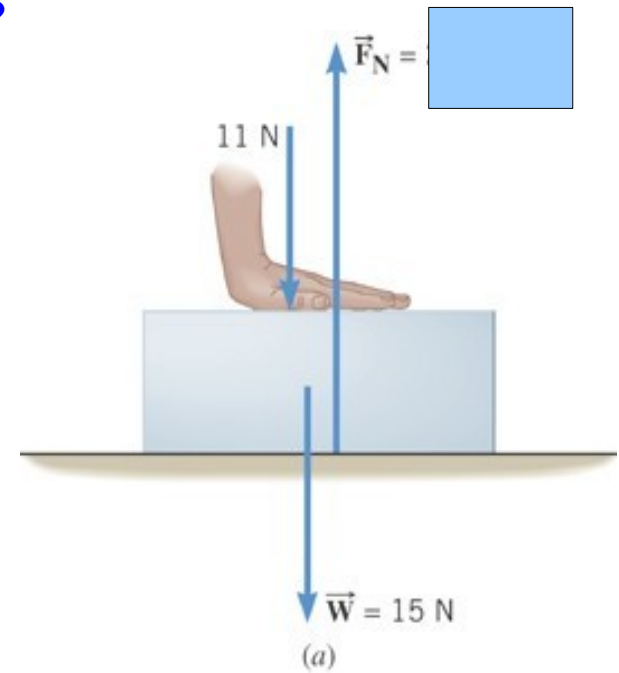
FIND The Normal Force ?

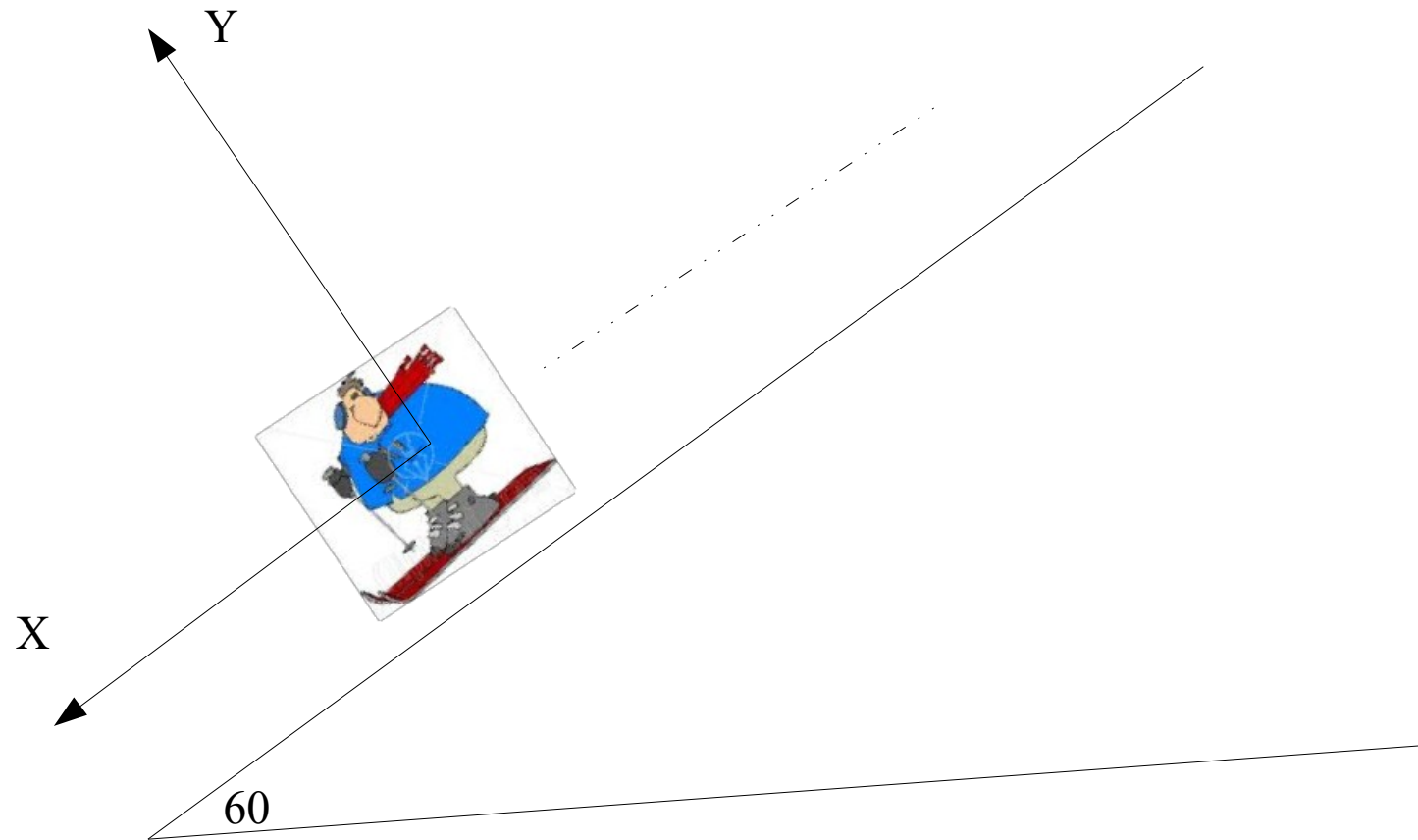
$$F_N - 11 \text{ N} - 15 \text{ N} = 0$$

$$F_N = \boxed{}$$

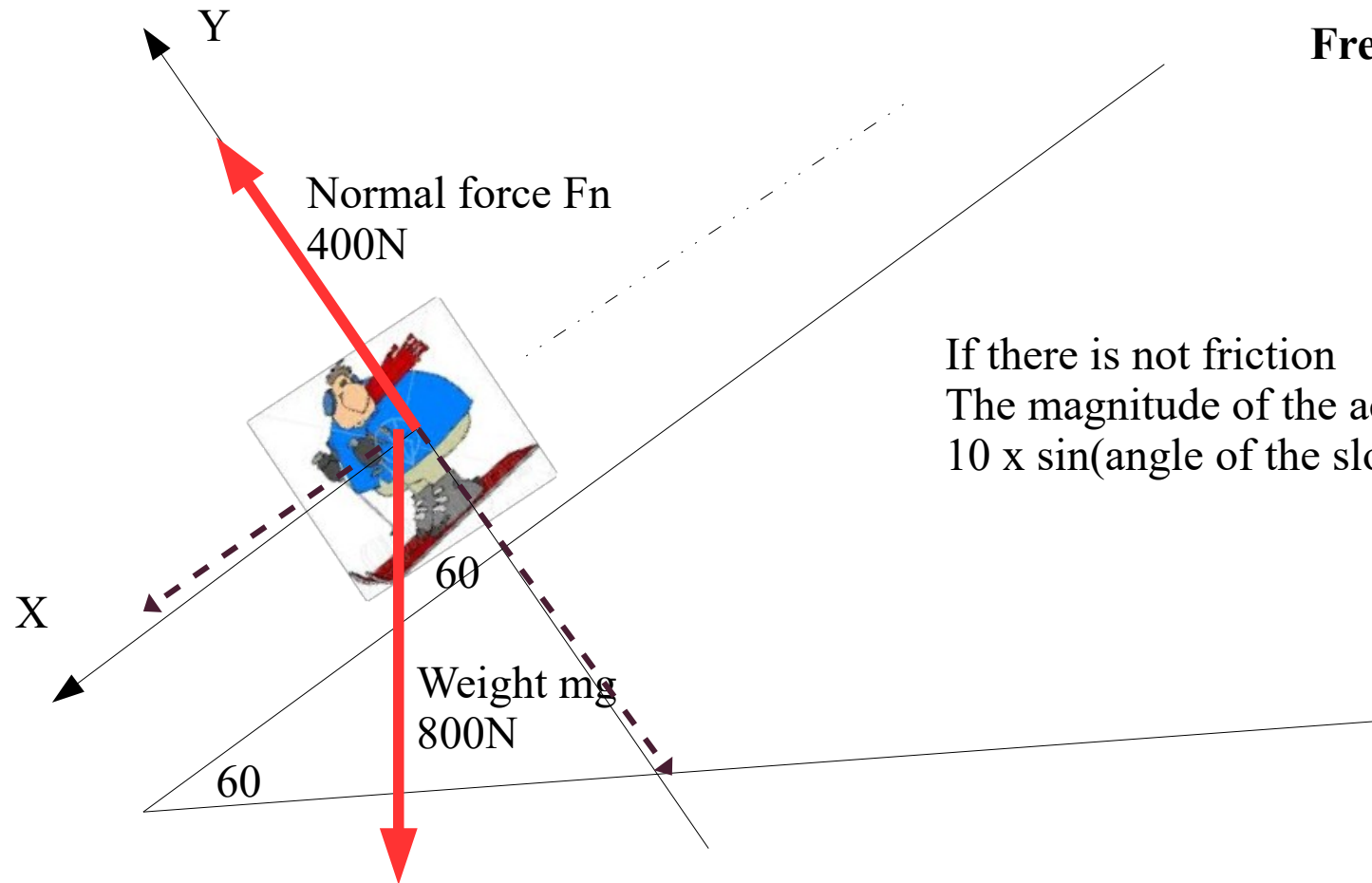
$$F_N + 11 \text{ N} - 15 \text{ N} = 0$$

$$F_N = \boxed{}$$





- 1) Draw the forces applied to the skier. No friction. Just the normal and the weight
- 2) Trace the components of the weight. Discuss their “jobs”
- 3) Trace the resultant. Tail to head (adding the vectors geometrically)
- 4) Find the resultant using algebra. The weight is 800N (about 80kg)



Free-body diagram

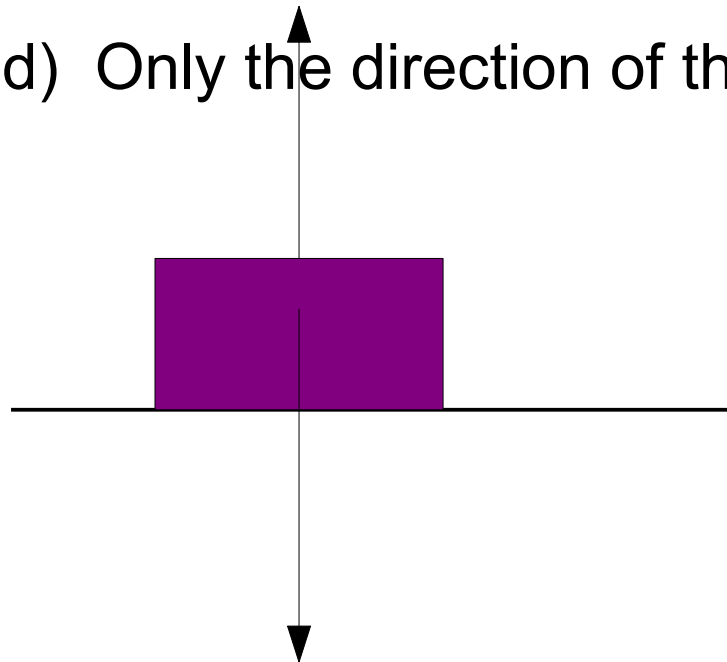
If there is not friction
The magnitude of the acceleration is
 $10 \times \sin(\text{angle of the slope})$

We find $R = 693\text{N}$ @ 180

A brick (like skier) is resting on the surface of a flat board. As one end of the board is slowly raised, what change, if any, is there in the normal force exerted on the brick?

- a) The normal force increases.
- b) The normal force decreases.
- c) The normal force remains constant.

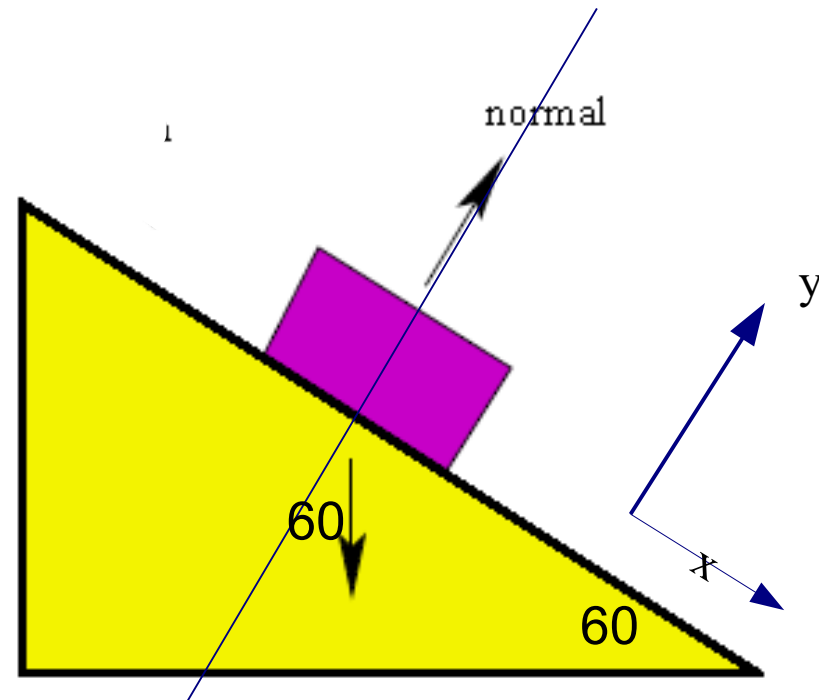
d) Only the direction of the normal force changes.



the apparent weight

What about the scale?
Its called the apparent weight

Same reading ?
Do demo

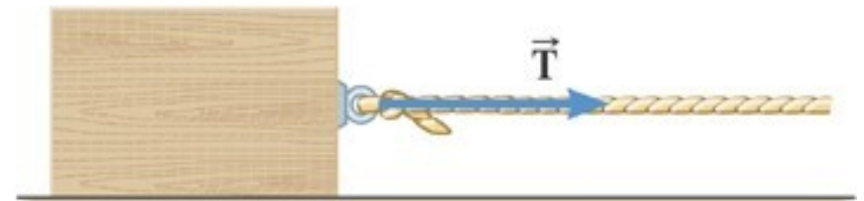


The Tension Force

Cables and ropes transmit forces through ***tension***.



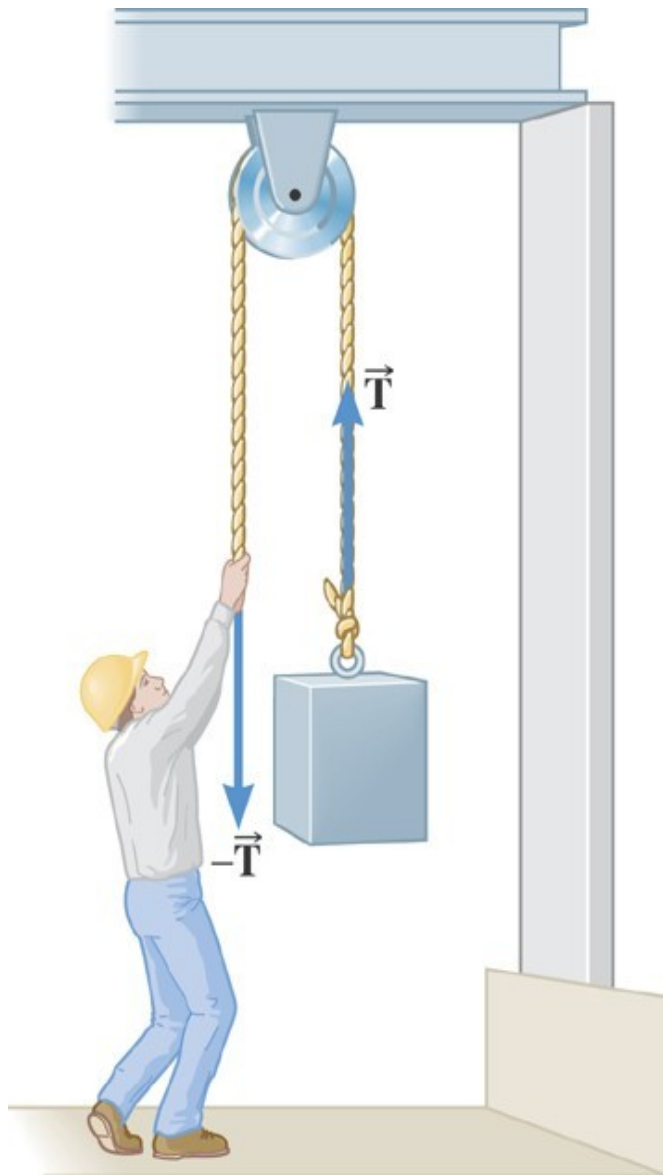
(a)



(b)



(c)



A massless rope will transmit tension undiminished from one end to the other.

If the rope passes around a massless, frictionless pulley, the tension will be transmitted to the other end of the rope undiminished.

demo

https://javalab.org/en/pulley_1_en/

https://sites.ualberta.ca/~dnobes/Teaching_Section/NOBES_SIM_Pulley.html

Mechanical equilibrium



Check it first. Show how the forces “eat each other out”

Static EQUILIBRIUM MEANS : no acceleration

MEANS : The net force is 0

$$\sum \vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 0$$

Algebraically (we use):

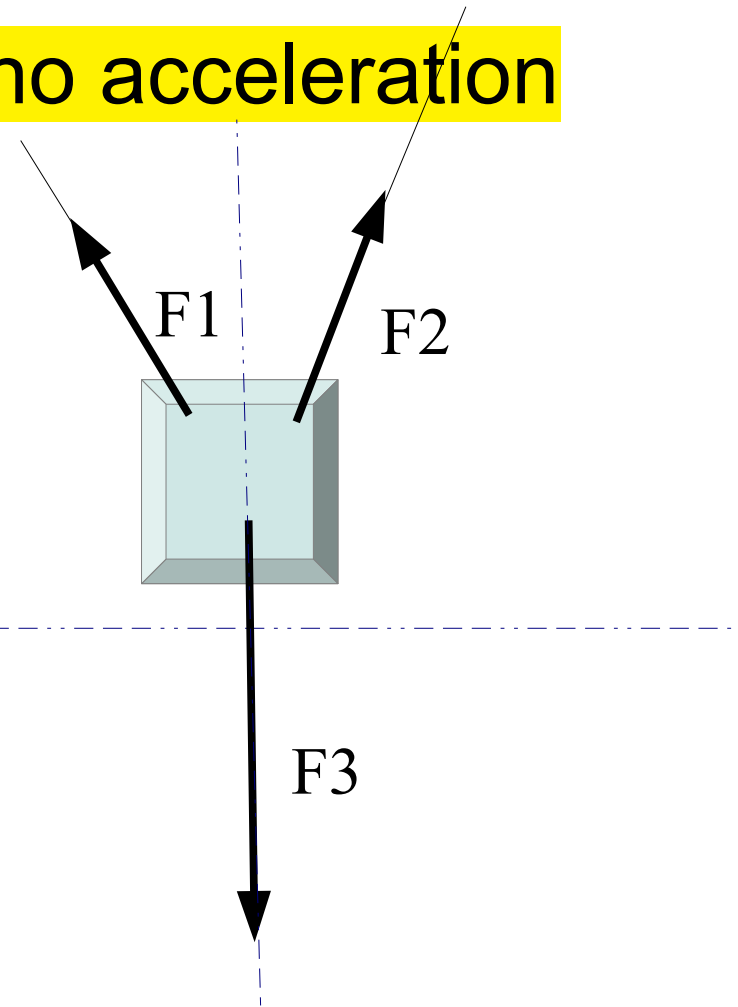
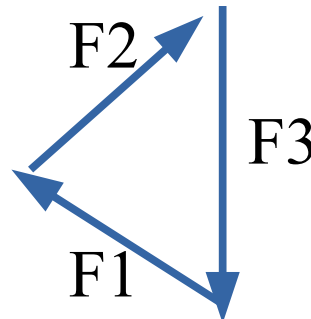
Sum of x-component = 0

Sum of y-components = 0

$$F_1x + F_2x + F_3x = 0$$

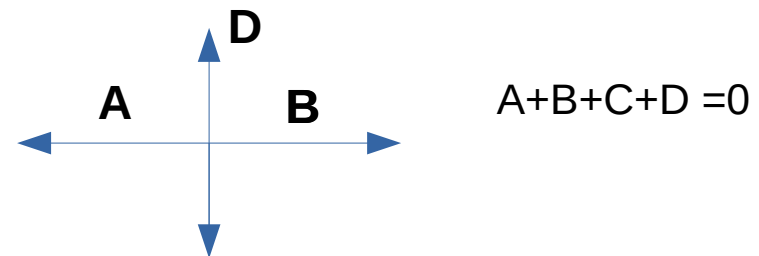
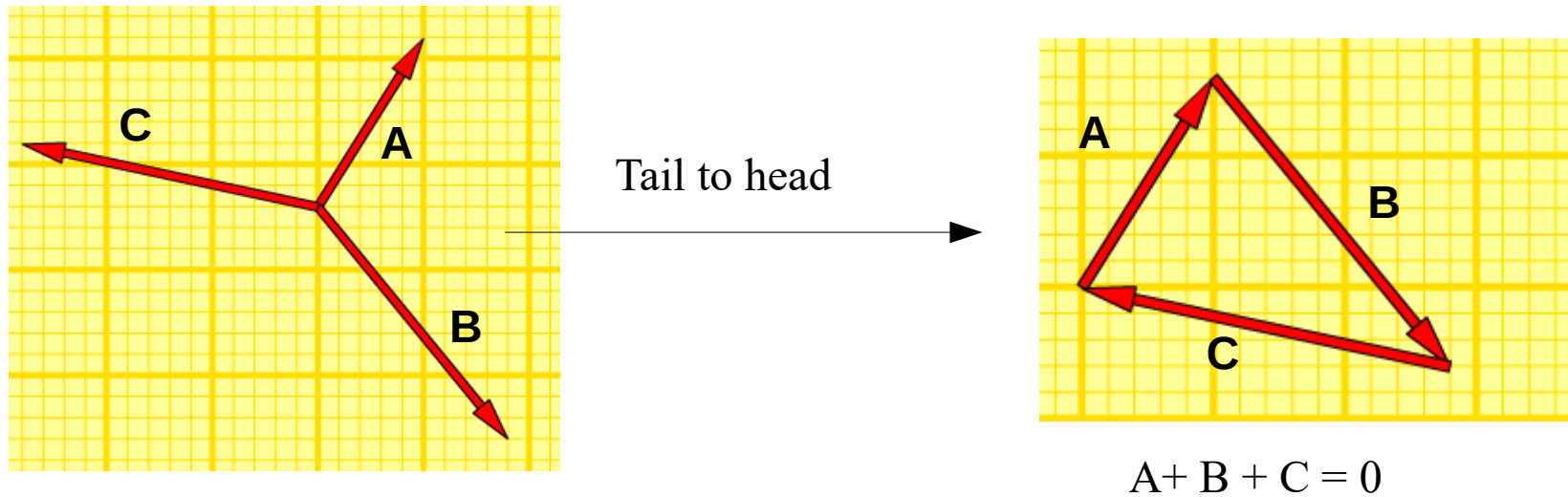
$$F_1y + F_2y + F_3y = 0$$

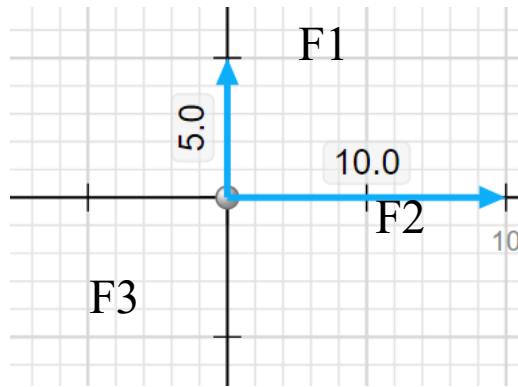
Geometrically:



This is adding vectors. Its like Displacement = 0
What happens if not 0?
There is a net force.

Note: from previous slide. (2) (4)
The net force is 0. The block is in equilibrium.
The vectors add up to 0. .





The crate placed at the origin is at rest. 3 forces are acting on it.

$$F1 = 5\text{ N @ } 90^\circ \quad F2 = 10\text{ N @ } 0^\circ \quad F3 = ?$$

1st trace using the parallelogram method. Can you see the components Of F3?

	x-axis	y-axis
F1		
F2		
F3	F3x	F3y
SUM=FNET	0	0

1) Fill the table

2) Use $F1x + F2x + F3x = 0$ and $F1y + F2y + F3y = 0$ to find F3x and F3y

3) Compute the magnitude of F3 and its direction

$$F3 = \underline{\hspace{2cm}} \text{ N @ } \underline{\hspace{2cm}}$$

1) An object is being acted upon by three forces and $\square$ as a result moves with a constant velocity. One force is 60.0 N along the +x-axis, and the second is 75.0 N along the +y-axis. What is the magnitude of the third force?

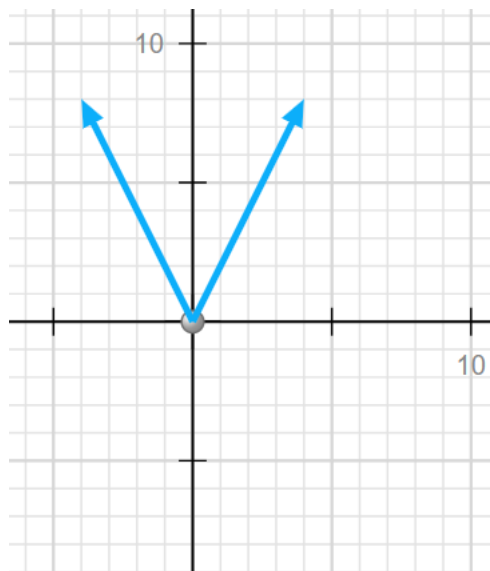
A) 67.5 N

B) 96.0 N

C) 135 N

D) 48.0 N

E) 15.0 N



II) In this situation F_1 and F_2 have a 10N magnitude and make an angle of 30 degrees With vertical. Find the third vector to achieve equilibrium

	x-axis	y-axis
F1		
F2		
F3	F3x	F3y
SUM=FNET	0	0

$F_1(-6.2, 4.5)$ $F_2(9.1, 4)$

2) A 400-kg box is lifted vertically upward with constant velocity by means of two cables pulling at 40.0° on either side of the vertical direction. What is the tension in each cable?

A) 2560 N

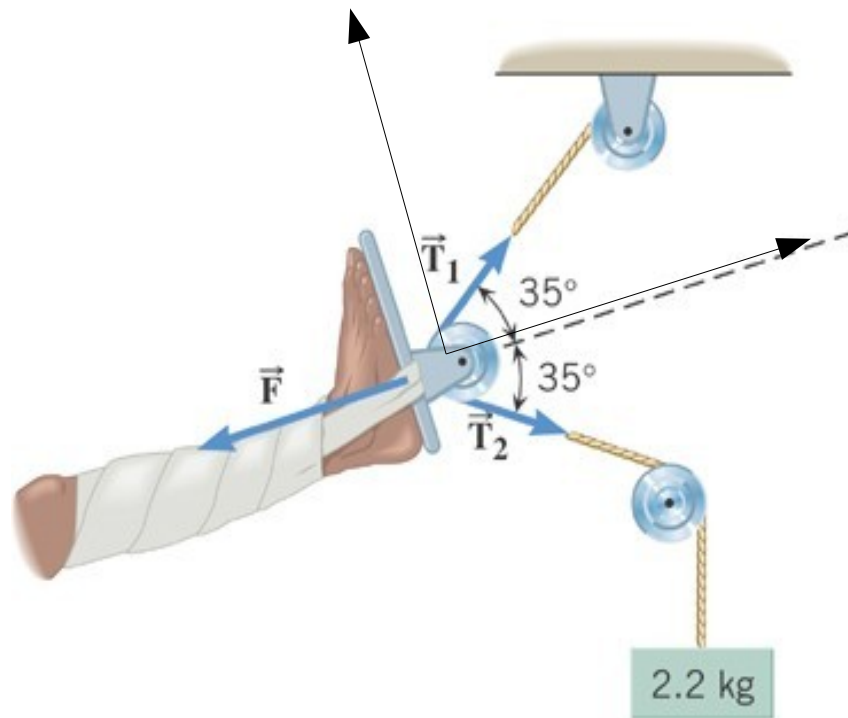
B) 3920 N

C) 231 N

D) 400 N

E) 800 N

4.11 Equilibrium Application of Newton's Laws of Motion



(a)

First draw a free-body diagram
Write each force in standard notation
Use: $T_1 = T_2 = \text{weight} = 22\text{N}$
(we will see that $\text{weight} = \text{mass} \times 10$)

Then find the magnitude of F
The force stretching the leg.
It's an equilibrium so $\text{sum} = 0$

(b) Free-body diagram for the foot pulley

1) $|T_2|$ is a tension so $|T_2| = 2.2 \times 10 = \text{weight of the mass} = |T_1|$

	x-axis	y-axis
F	F_x	F_y
T1		
T2		
SUM=FNET	0	0

Source: Paul Hewitt – Conceptual Physics

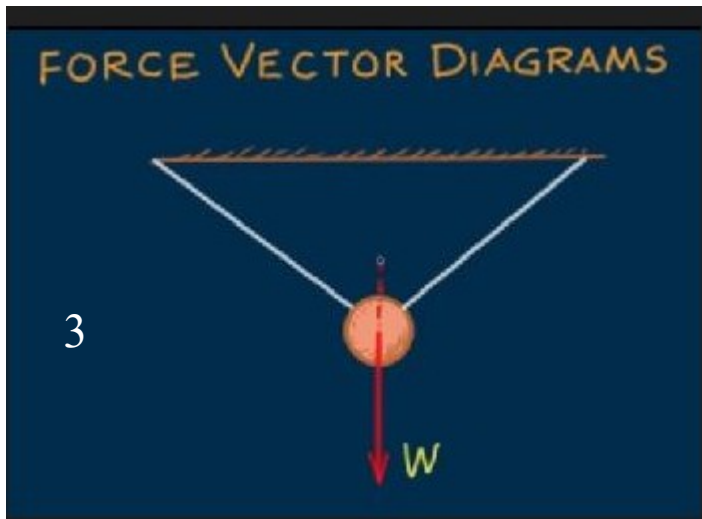
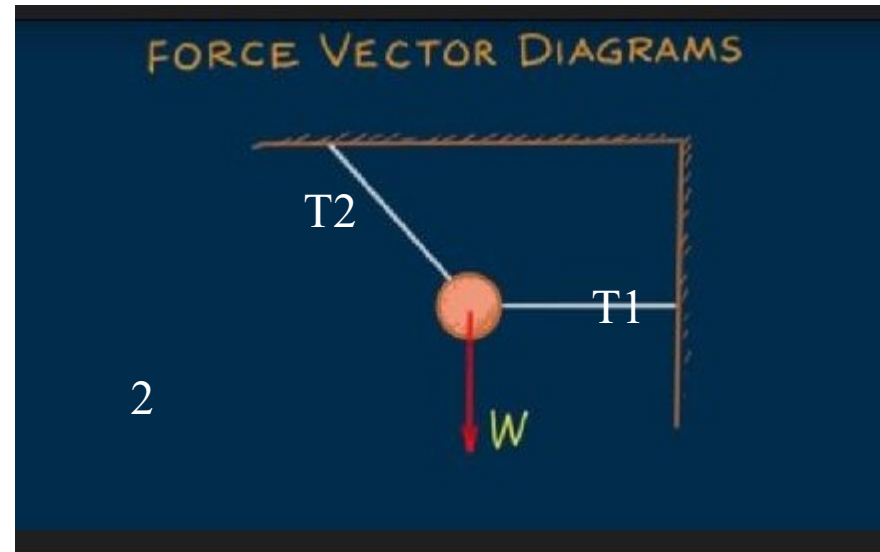
Do demo. Trace the vector sum of the 2 tension for each case.

1st picture what is the tension in each string? Do demo in class.

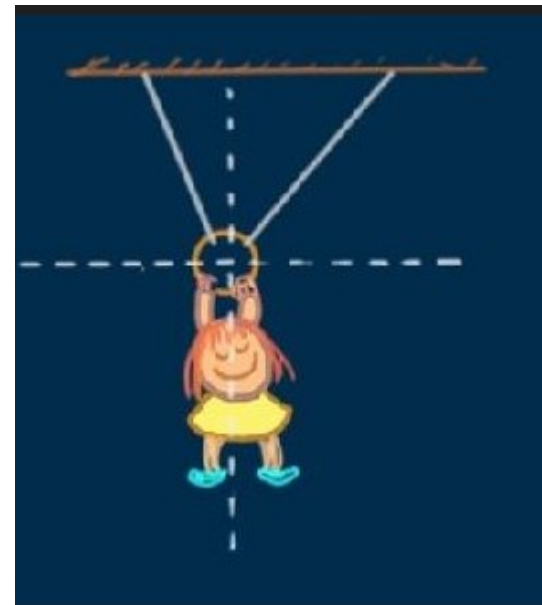
Find the tensions in the 3rd picture. The angle between string is 120

2nd picture. Trace resultant. Angle of T2 with vertical is 30 degrees. Find T1, T2

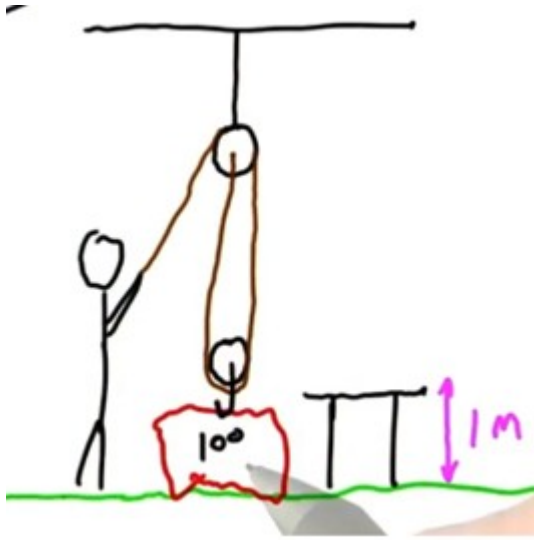
Last picture just trace the tensions. Use parallelogram method.



Just trace the tensions. Use a parallelogram

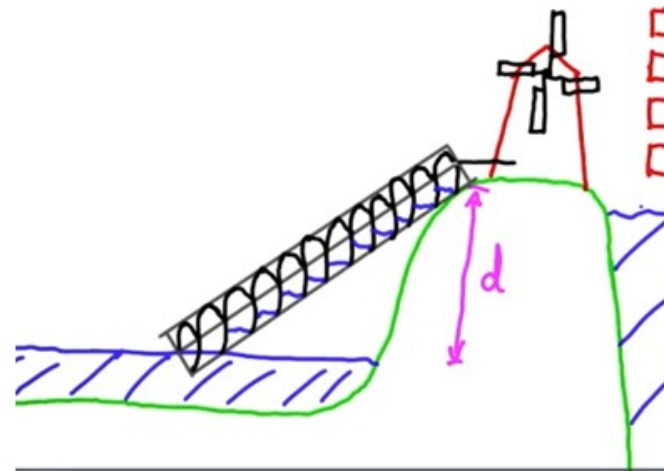


Machines $\rightarrow$ small effort $\times$ large distance = large load $\times$ small distance

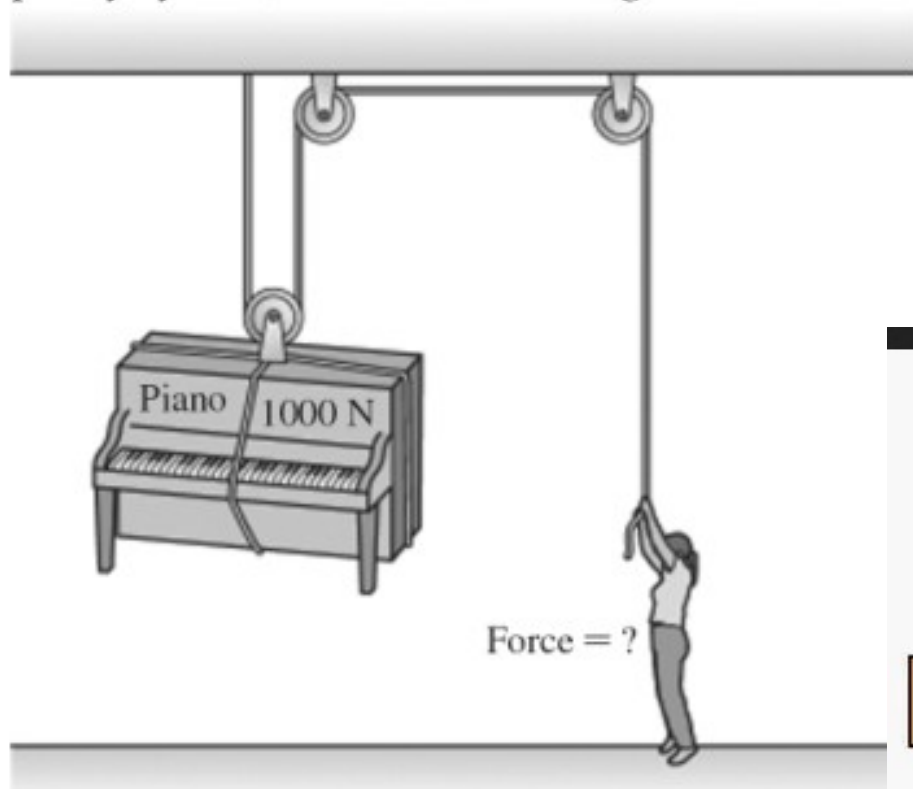


$$F = 500\text{ N}$$
$$d = 2\text{ m}$$

Angle = 30 degrees

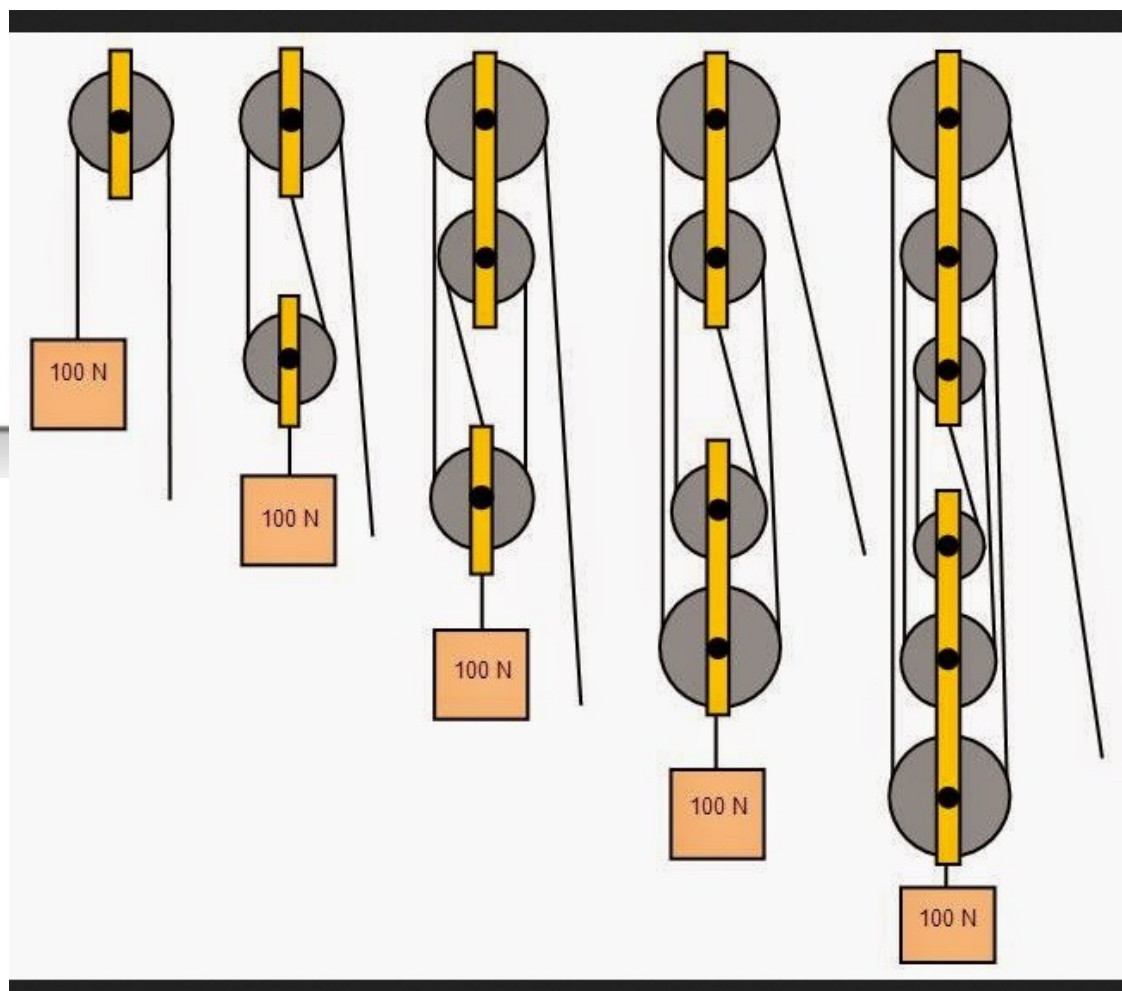


- 3) A piano mover raises a 1000-N piano at a constant speed using a very light rope in a frictionless pulley system, as shown in the figure. With what force is the mover pulling down on the rope?

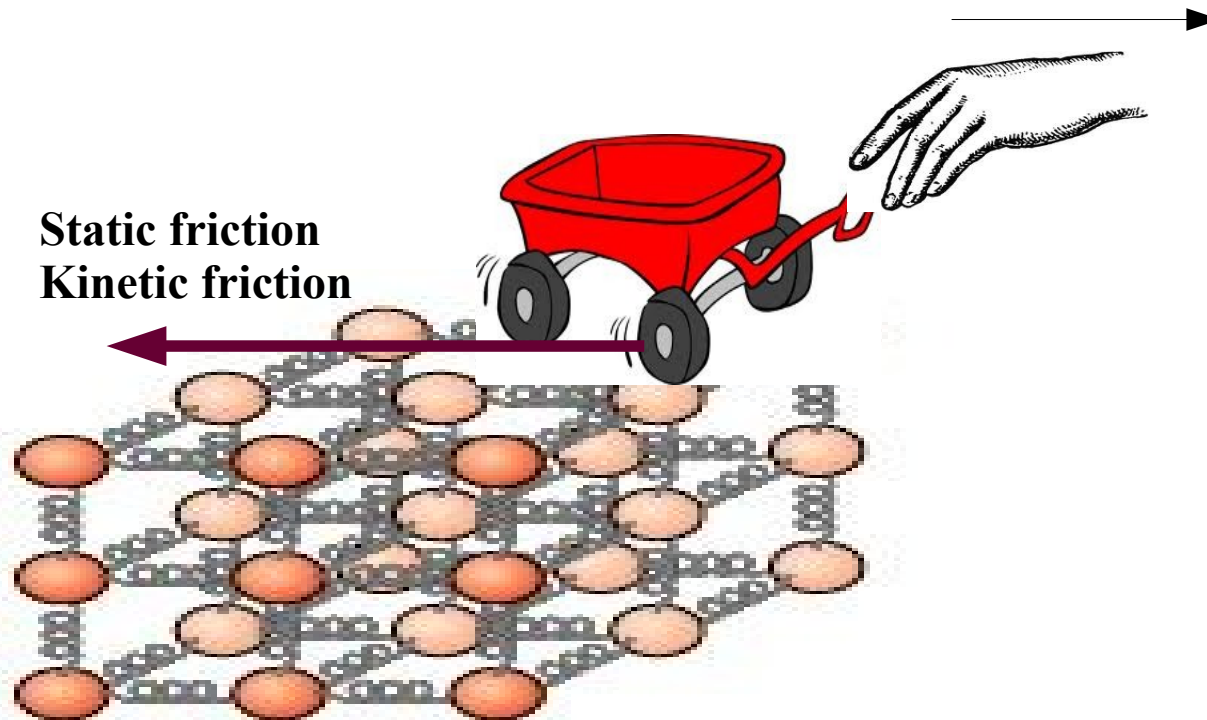


Do one of the problems of the assignments,

See the folder.

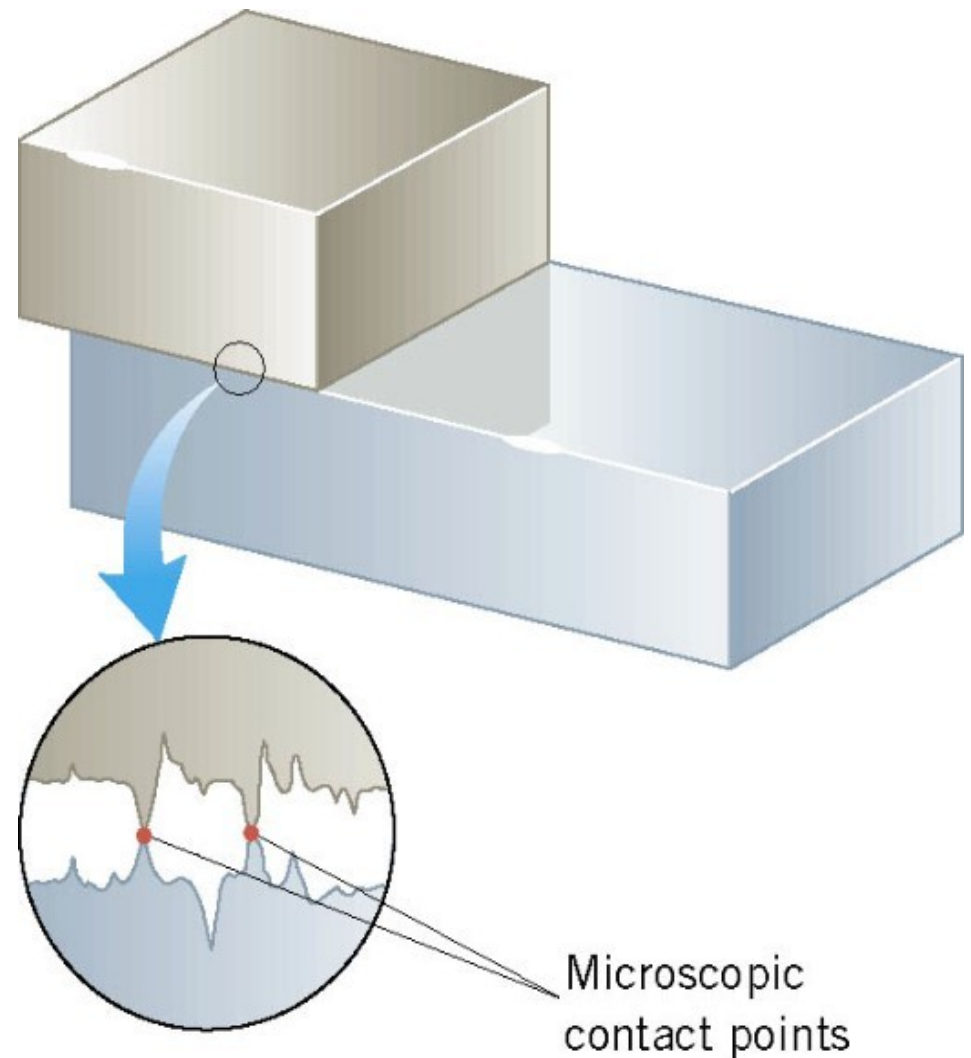


Frictional forces



<http://phet.colorado.edu/en/simulation/forces-and-motion-basics>

When an object is in contact with a surface there is a force acting on that object. The component of this force that is parallel to the surface is called the ***frictional force***.

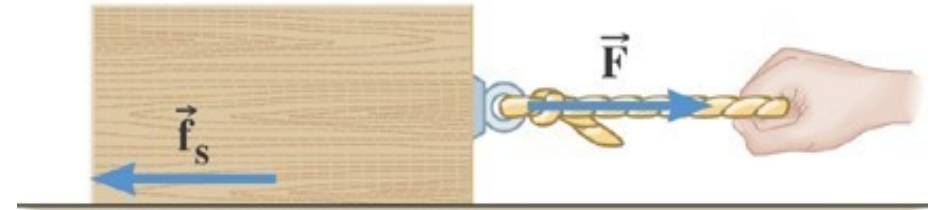


When the two surfaces are not sliding across one another the friction is called ***static friction***.



No movement

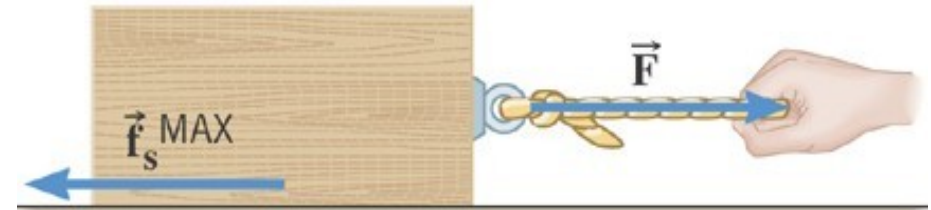
(a)



No movement

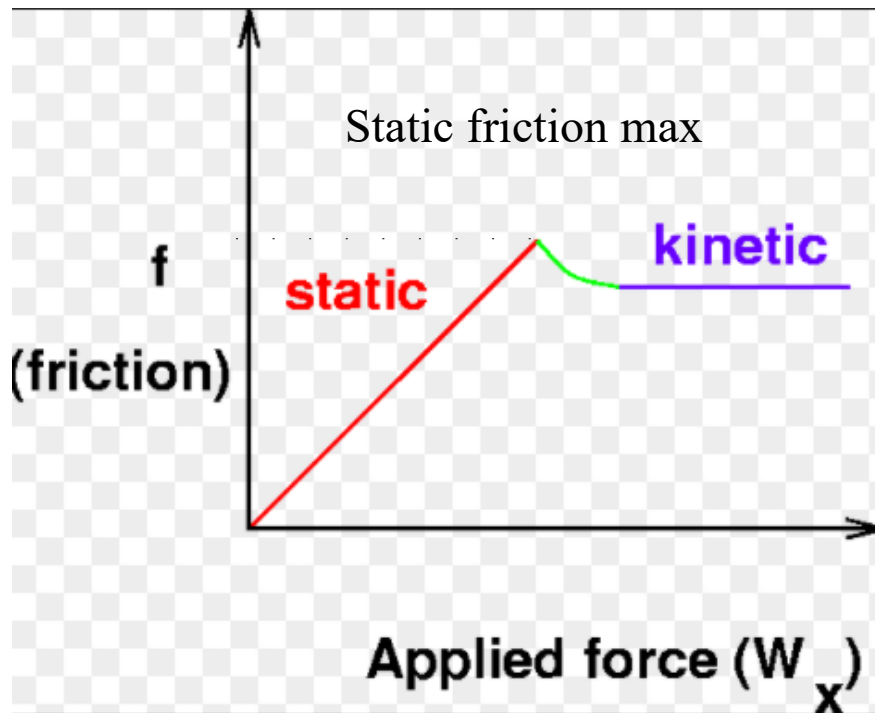
(b)

Static friction max = $\mu \times \text{Normal}$

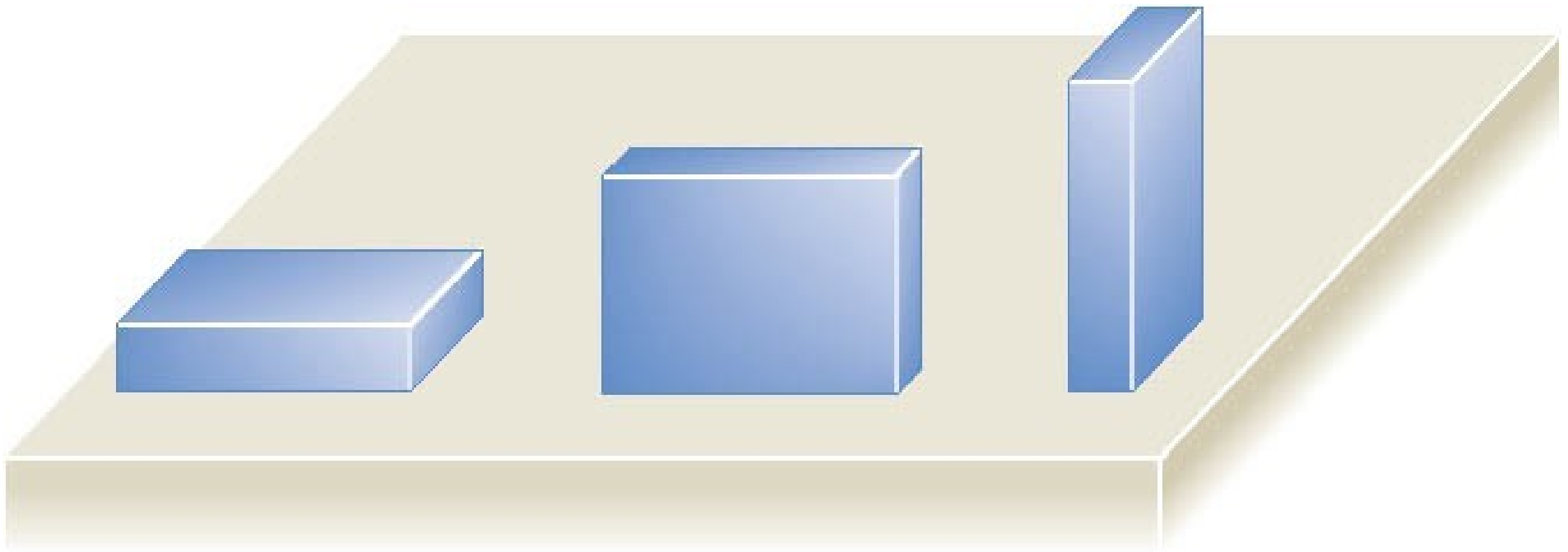


When movement just begins

(c)



Note that the magnitude of the frictional force does not depend on the contact area of the surfaces.



Static friction opposes the *impending* relative motion between two objects.

Kinetic friction opposes the relative sliding motion motions that actually does occur.

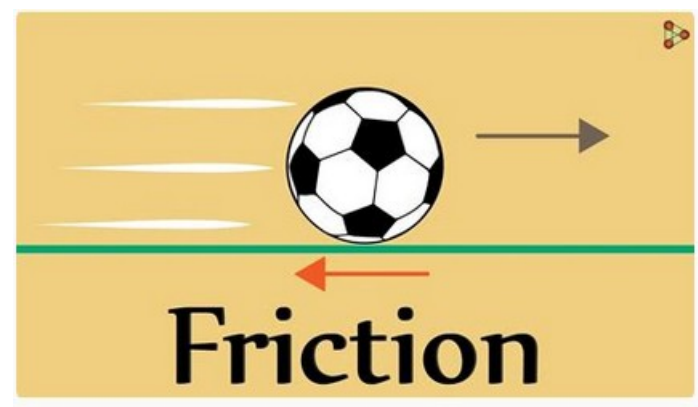
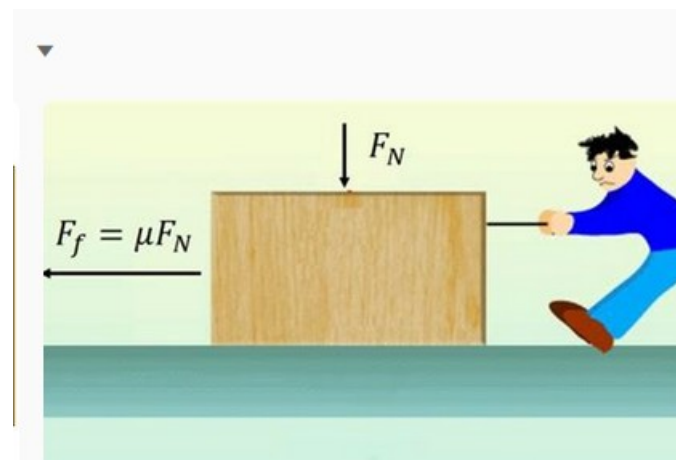
$$f_k = \mu_k F_N$$



$0 < \mu < 1$ is called the coefficient of kinetic friction.

Approximate Coefficients of Friction

	Kinetic	Static
Rubber on concrete (dry)	0.68	0.90
Rubber on concrete (wet)	0.58	
Rubber on asphalt (dry)	0.67	0.85
Rubber on asphalt (wet)	0.53	
Rubber on ice	0.15	
Waxed ski on snow	0.05	0.14
Wood on wood	0.30	0.42
Steel on steel	0.57	0.74
Copper on steel	0.36	0.53
Teflon on Teflon	0.04	



Physics of climbing

→ smearing/edging = use static friction.

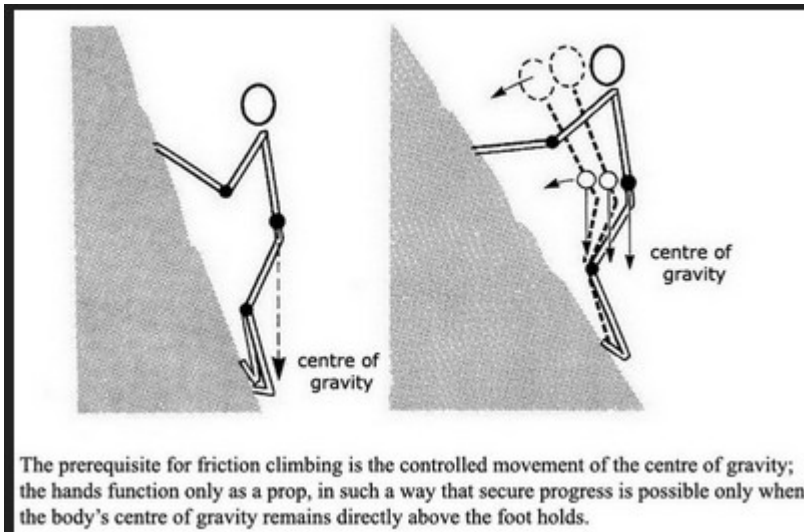
Friction = μ weight . To increase friction, increase μ (special rubber shoes)

And increase the weight by standing on one foot . Noise above the foot

So the center of mass is above shoe.

Powder increases the coefficient μ .

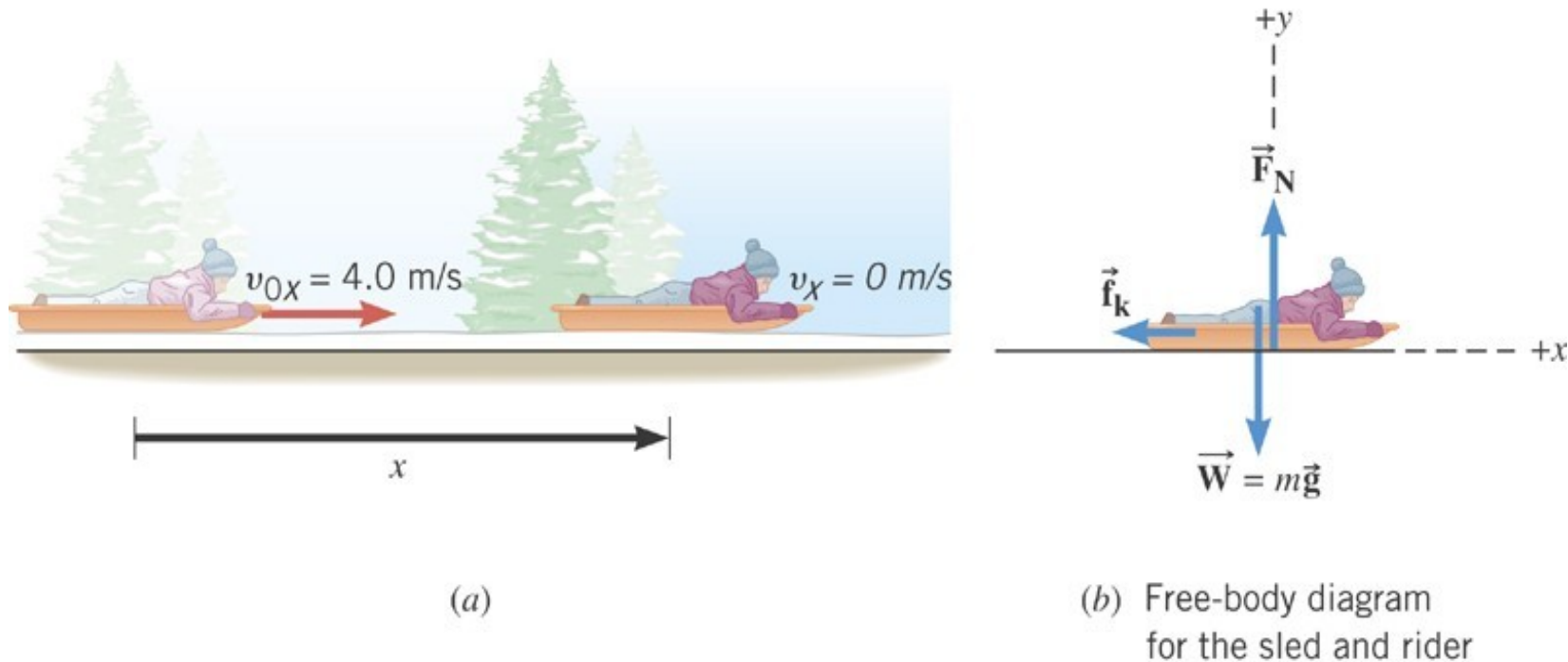
Textbook 5.8



Push on legs first before pulling with Arm (more muscles in legs)

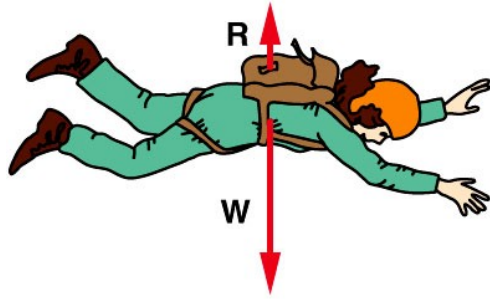


<https://www.youtube.com/watch?v=01vz0g64qQo&t=6s>



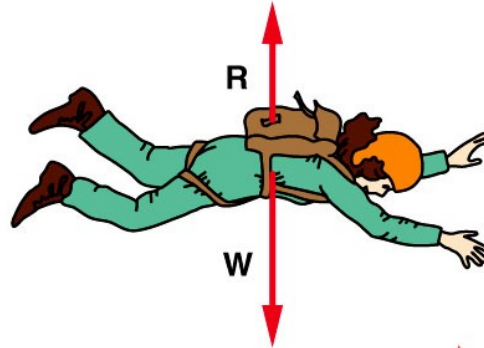
The sled comes to a halt because the kinetic frictional force opposes its motion and causes the sled to slow down.

Suppose the coefficient of kinetic friction is 0.05 and the total mass is 40kg. What is the kinetic frictional force?



$W > R$
 $F_{net} @ \text{down}$
 $0 < a < 10 \text{ m/s/s}$
 Speeding up a @down
 $V @ \text{down}$

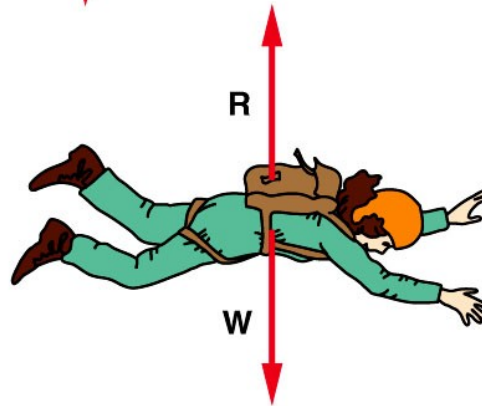
$W > R$ R gets bigger
 $F_{net} @ \text{down}$ smaller than before
 $0 < a < 10 \text{ m/s/s}$ smaller than before
 Speeding up a @down
 $V @ \text{down}$



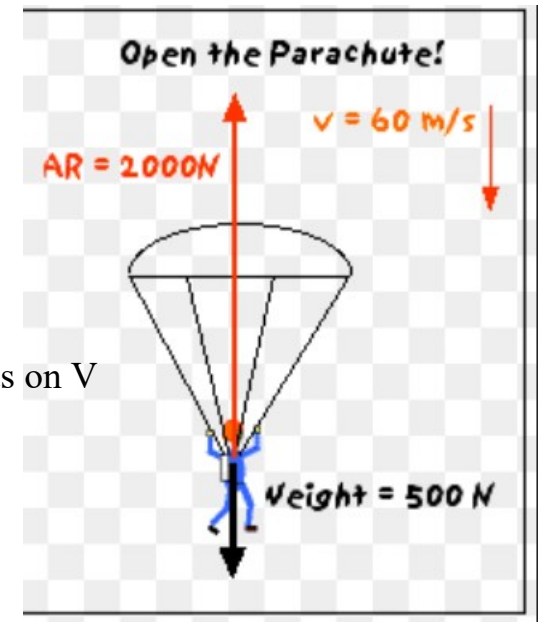
Air resistance depends on the speed²
 Faster $\rightarrow$ more drag

$R = W$
 $F_{net} = 0$
 $a = 0$ constant velocity
 Terminal speed is reached
 $V @ \text{down}$ constant

Do one of the problems of the assignments,



$R > W$ but will decrease since depends on V
 $F_{net} @ \text{up}$
 $0 < a < 10 \text{ m/s/s}$ but @ up
 Slowing down
 $V @ \text{down}$



AIR RESISTANCE /And terminal speed /See exploration of physical see trajectory

Reference: The physics of everyday phenomena: a conceptual introduction to physics by W.
 Thomas Griffith, McGraw-Hill



She opens her parachute. $\text{Drag} > \text{weight}$
 Now the net force is @ up so the acceleration is @ up
 While she is moving @ down.
 So she is slowing down and is still moving @ down.
 The acceleration is negative.
 Drag is increasing



$\text{Drag} = \text{weight}$
 Net force is 0
 Acceleration is 0
 She is going down at a constant speed.
 (terminal speed)
 She is about to touch ground.

TERMINAL SPEEDS :

(from 2007 edition book from same author,
Richard A. Muller)

Food dropped from airplane (3000m)

Example: Afghanistan 2002.

9mph

A person: **74mph**

(see 1000 ways to die, you can survive)

With parachute = **13mph**

otherwise about 540 mph without air resistance

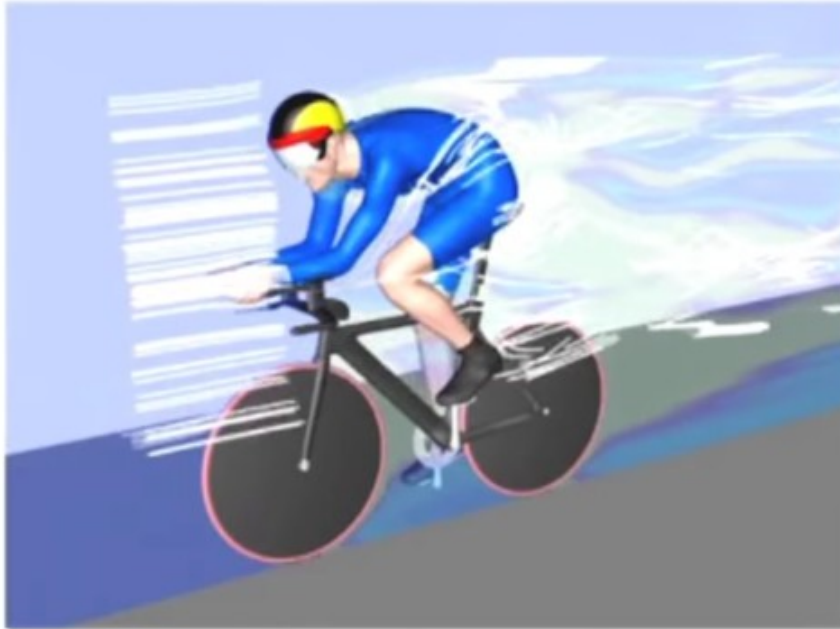
You can use the equation $v^2 = 2 d g$ to compute

This free-fall speed (from m/s to mph multiply by 2.2)



Physics of sport:
Drag is another type of friction

Drag balances the thrust → no acceleration



$$F = \frac{1}{2} \rho v^2 C_d A$$

Aim = Small F

Drafting

small v, low ρ

Design

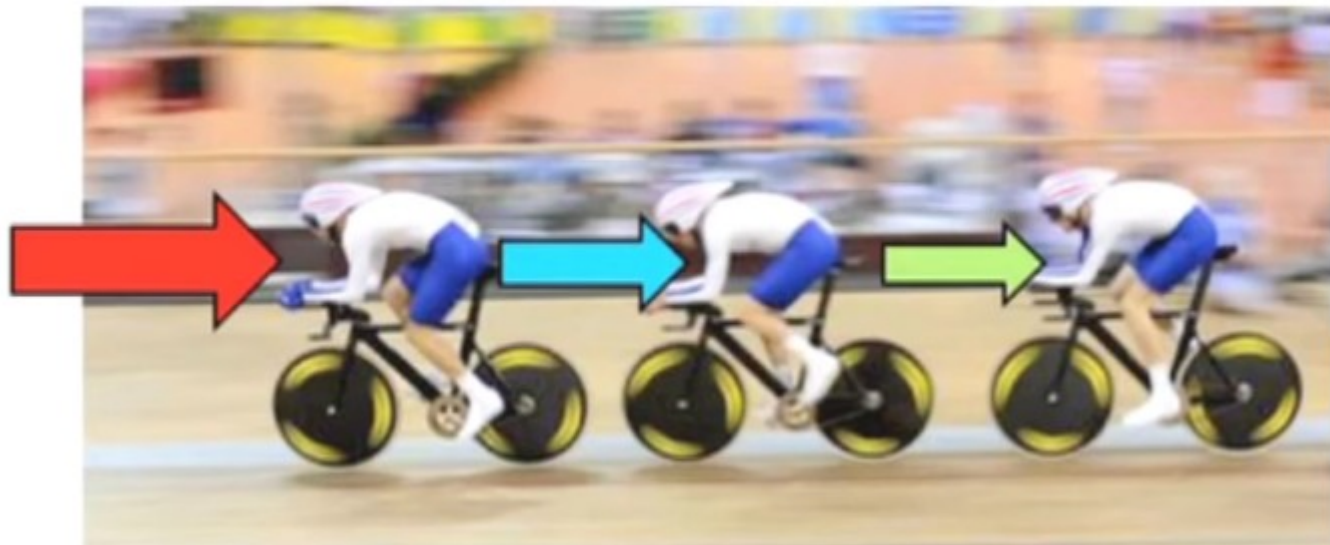
optimisation

Small area, A

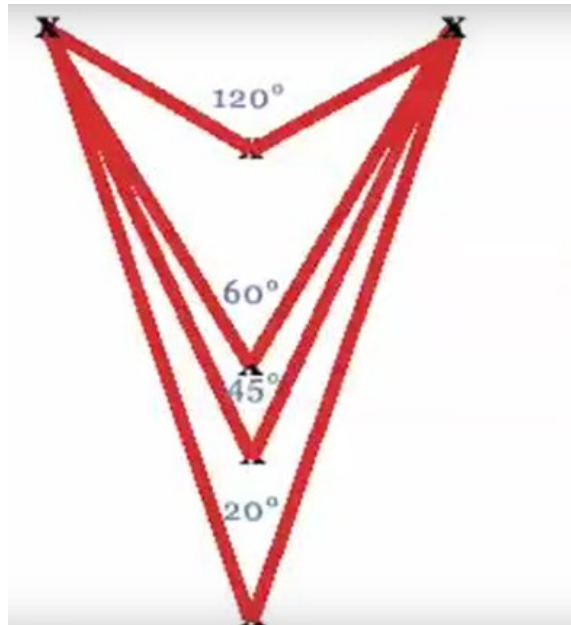
Low drag

coefficient, C_d

Video – Italian
aerodynamics



Static equilibrium Physics of climbing



Smaller the angles between
The cables, larger the weight.
(y-components larger)

Its an equilibrium. You can not have
Horizontal rope supporting a weight.

The max weight is 8000N
Convert to lbs

Draw the forces
Acting on a person
Doing push up on a tree

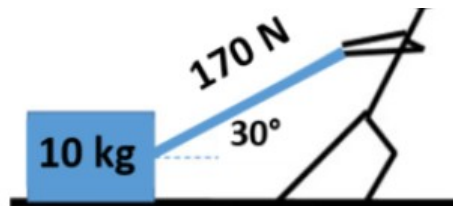


Draw the forces acting on
A sofa being pushed over the
Floor at a constant speed

Draw the forces acting on
A beetle that jumped
Out of a window
And has reached
Terminal speed

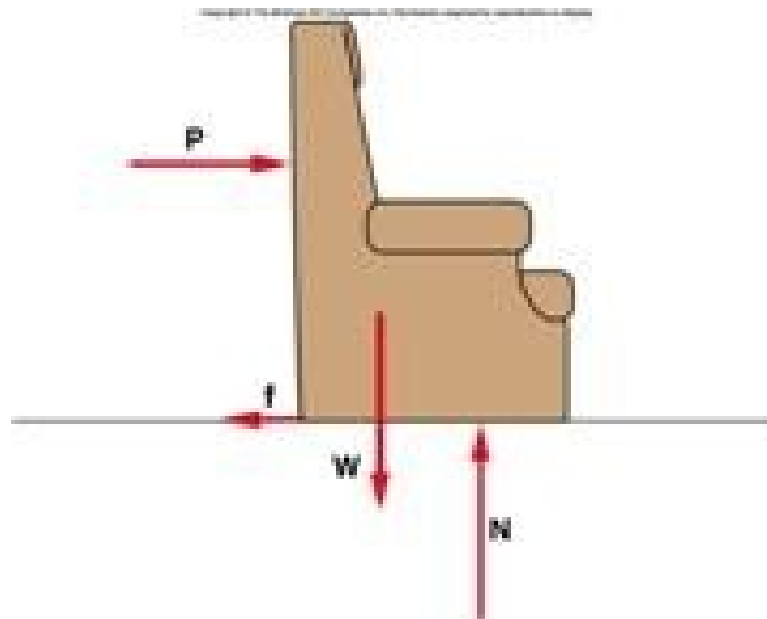
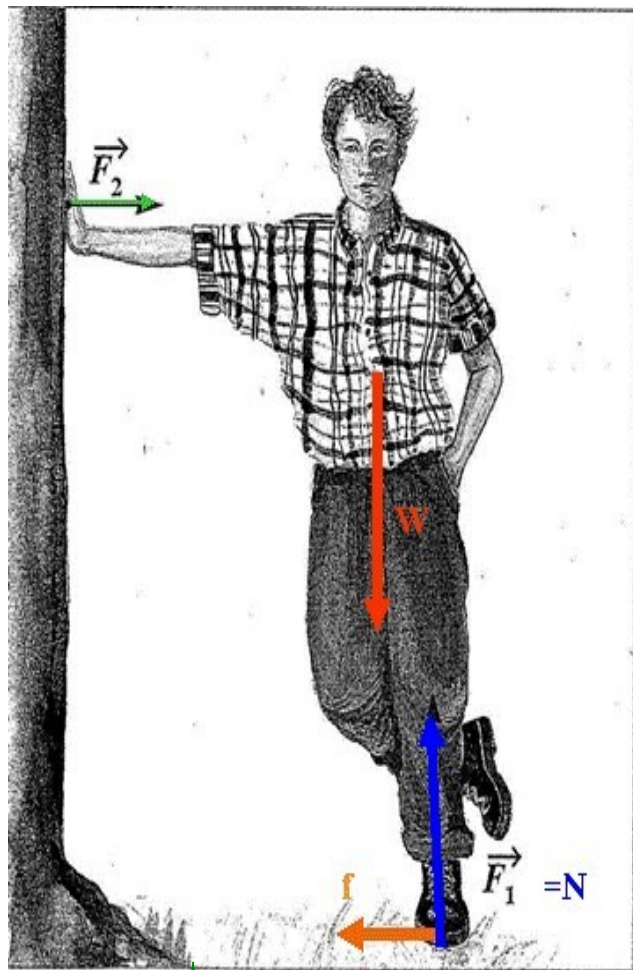


Draw the forces acting on
An airplane moving at a constant
Speed.



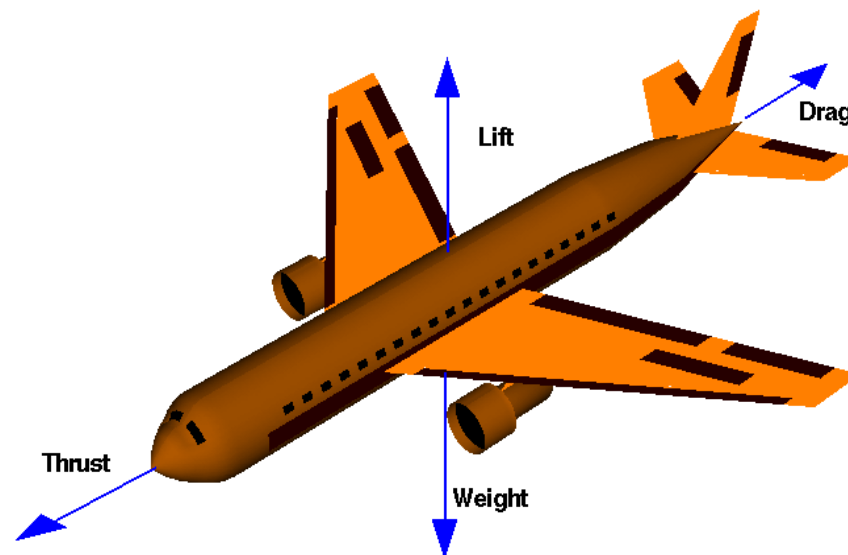
Draw the forces acting on a box being pulled
At a constant speed over the floor.

Draw the forces acting on a box sliding down
A slope



Four Forces on an Airplane

Glenn
Research
Center

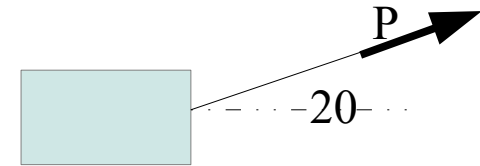


The effect that a force has in any direction can be found by calculating its component in that direction

- 1) A 2.5kg brick is being pulled by a cord (at a constant speed) that makes an angle of 20 degrees with the horizontal and has a 7N (P) of tension in it.

The forces are: tension, weight, normal, friction

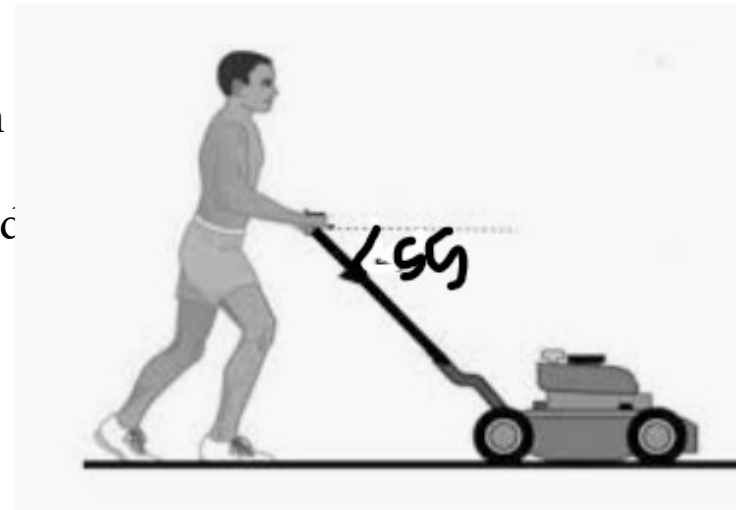
- A) find the weight of the brick
 - B) draw a free-body diagram with the 4 forces
 - C) write the forces in standard notation.
 - D) build a table with the components of the forces.
- Use net force = 0 to find the normal force and the friction.



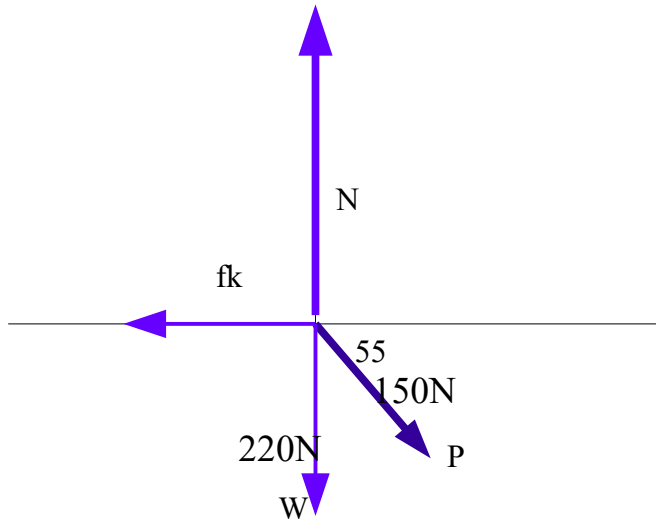
P

- 2) option 1: just draw the forces acting on the lawnmower.
- You are pushing a lawn mower at a constant speed. (so it is an equilibrium)
- The mass of the mower is 22 kg (weight is 220N).
- and is being pushed with a force of 150N (magnitude of P) against friction
- The shaft makes an angle of 55 degrees with ground. See figure.

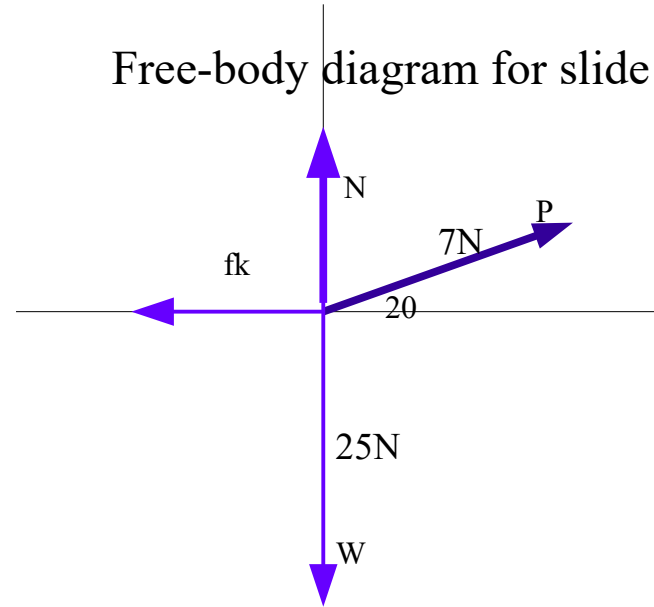
So the forces are the push down along the shaft, the normal from the ground (recoil), the weight and the friction opposing the motion along the ground



Free-body diagram for slide 74 Q2.



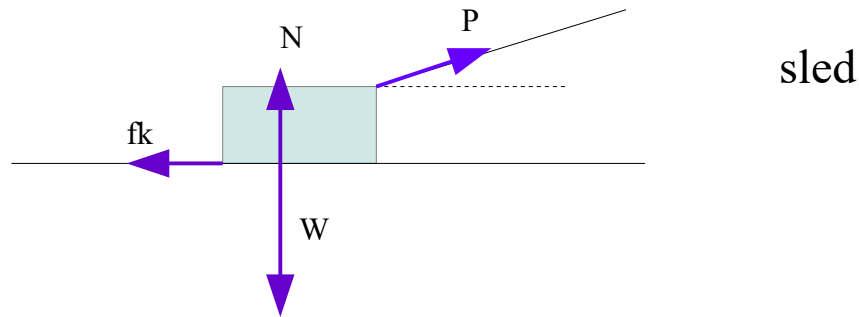
Free-body diagram for slide 74 Q1.



4) A sled is being pulled along a horizontal road at a constant speed by means of a rope that makes an angle of 25 degrees with the horizontal. If the friction between the sled and the snow is 85N, how much is the tension P in the rope ? (free-body diagram please – follow same steps as in 2)

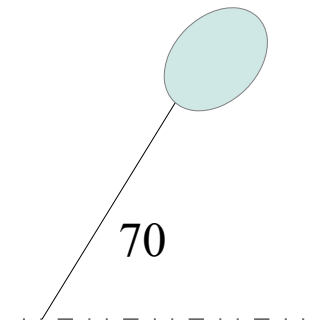
Use $\sum x\text{-components} = 0$ to find P .

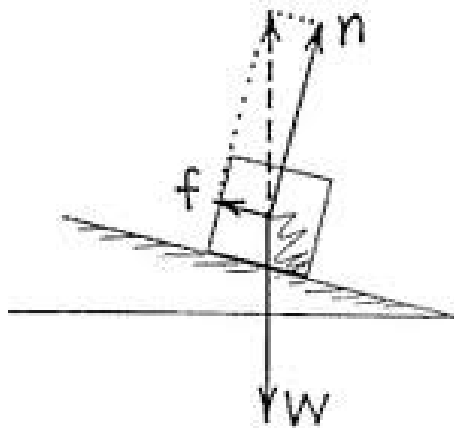
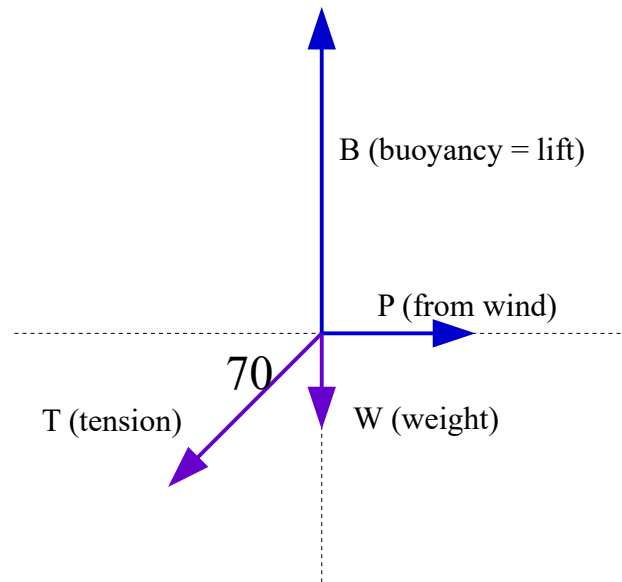




3) A helium balloon is in equilibrium as shown. The balloon weighs $W = 25\text{N}$ and is acted upon by 4 forces. The weight (down), the buoyancy B (up) the tension T in the rope that is holding down the balloon. The force P of the wind pushing the Balloon to the right. The tension T is 16N @ 70 degrees.

- A) draw the forces on the balloon (solution next slide)
- B) draw a free-body diagram
- C) build the table with the components of the forces.
- C) Find the buoyancy and the force exerted by the Wind.





5) see slide 52 for free body diagram

This block is not moving on the inclined plane. If the weight is 10N and the angle of the plane with horizontal 20° :

A) find the frictional force f

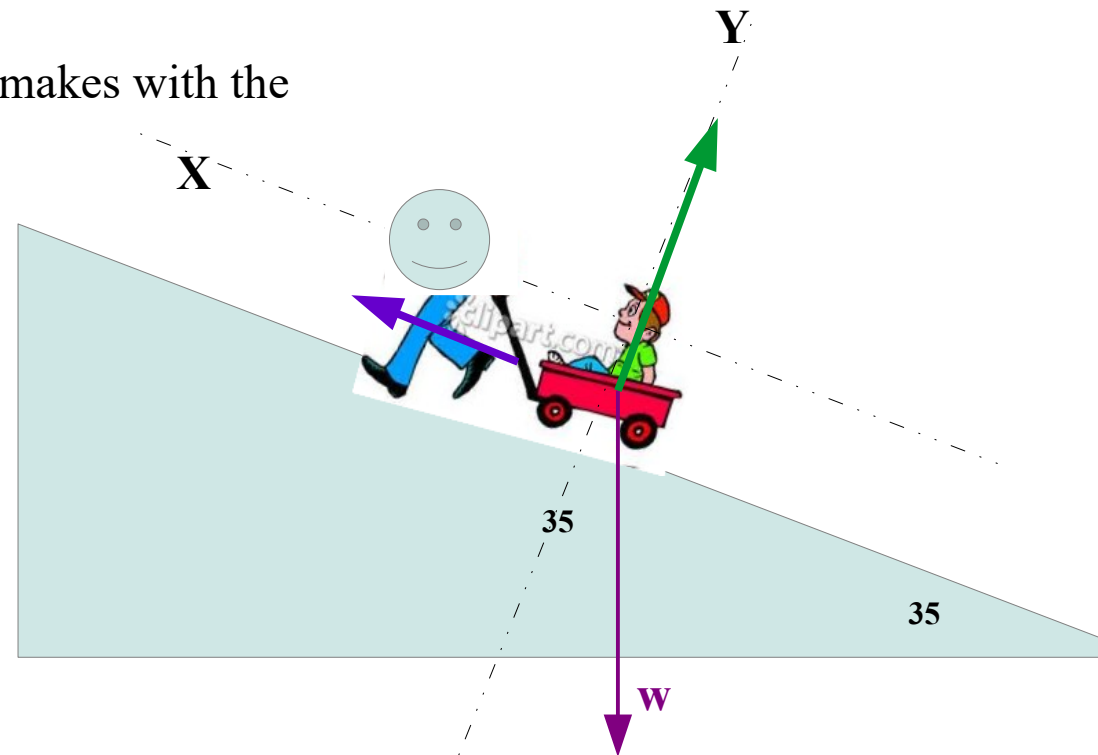
B) find the normal N (recoil force from the ground)

A wagon weighs 40 lb and the angle that the hill makes with the horizontal is 35 degrees. Keep pounds.

The wagon is moving at a constant speed .
That means the acceleration is 0.

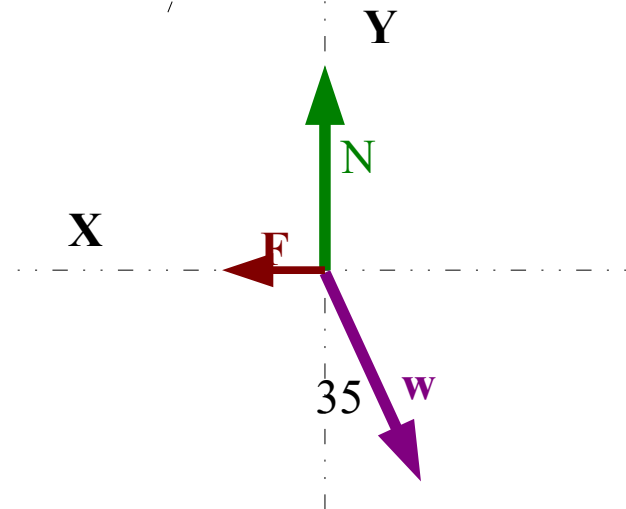
That means the net force
(or sum of the force is 0).

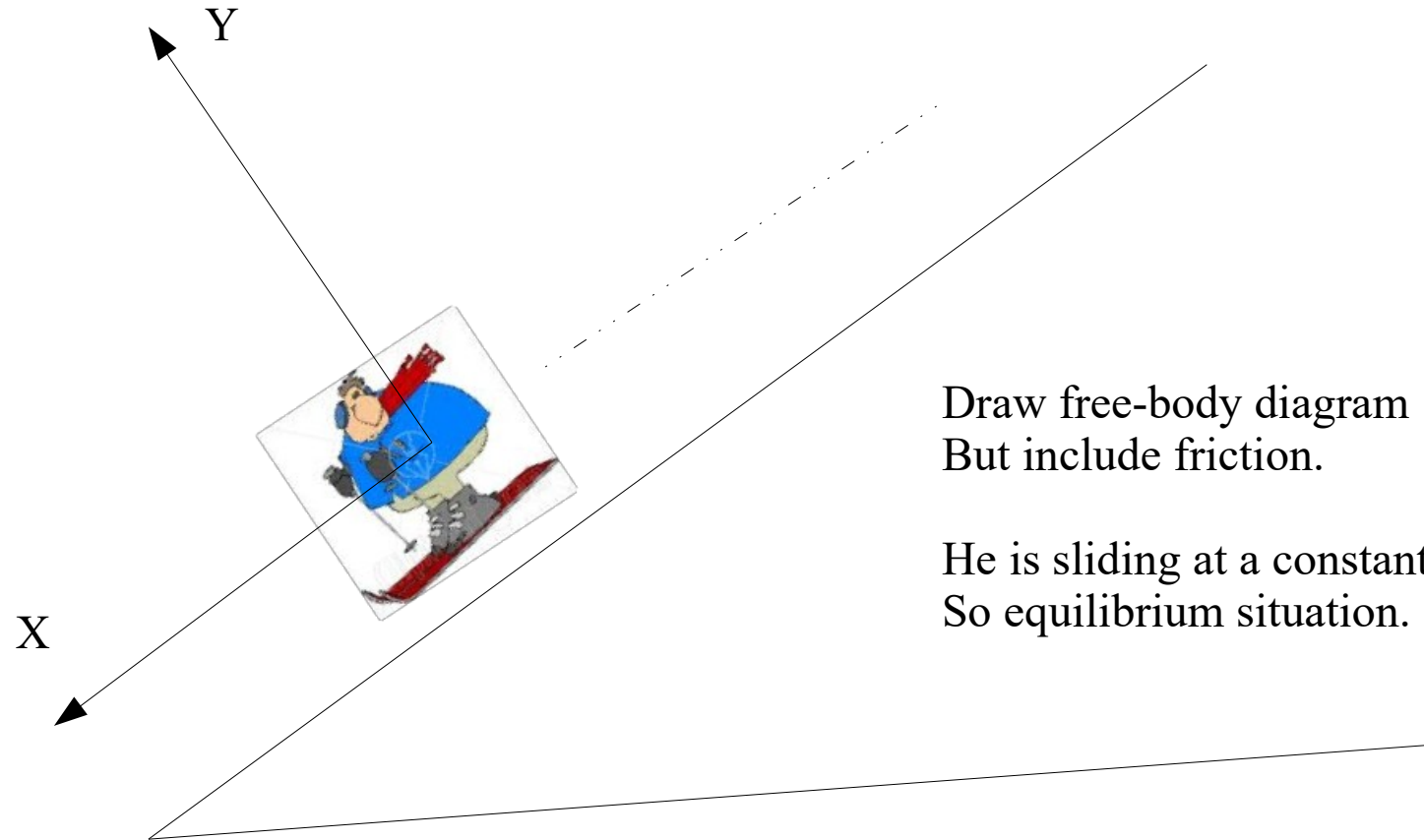
It is an equilibrium situation,
The goal is to find the normal force N.



- 1) Draw a free-body diagram.
- 2) Write each force in standard notation
- 3) Fill the table below and find N

	x-axis	y-axis
N	?	?
F		
mg		
SUM=FNET	0	0

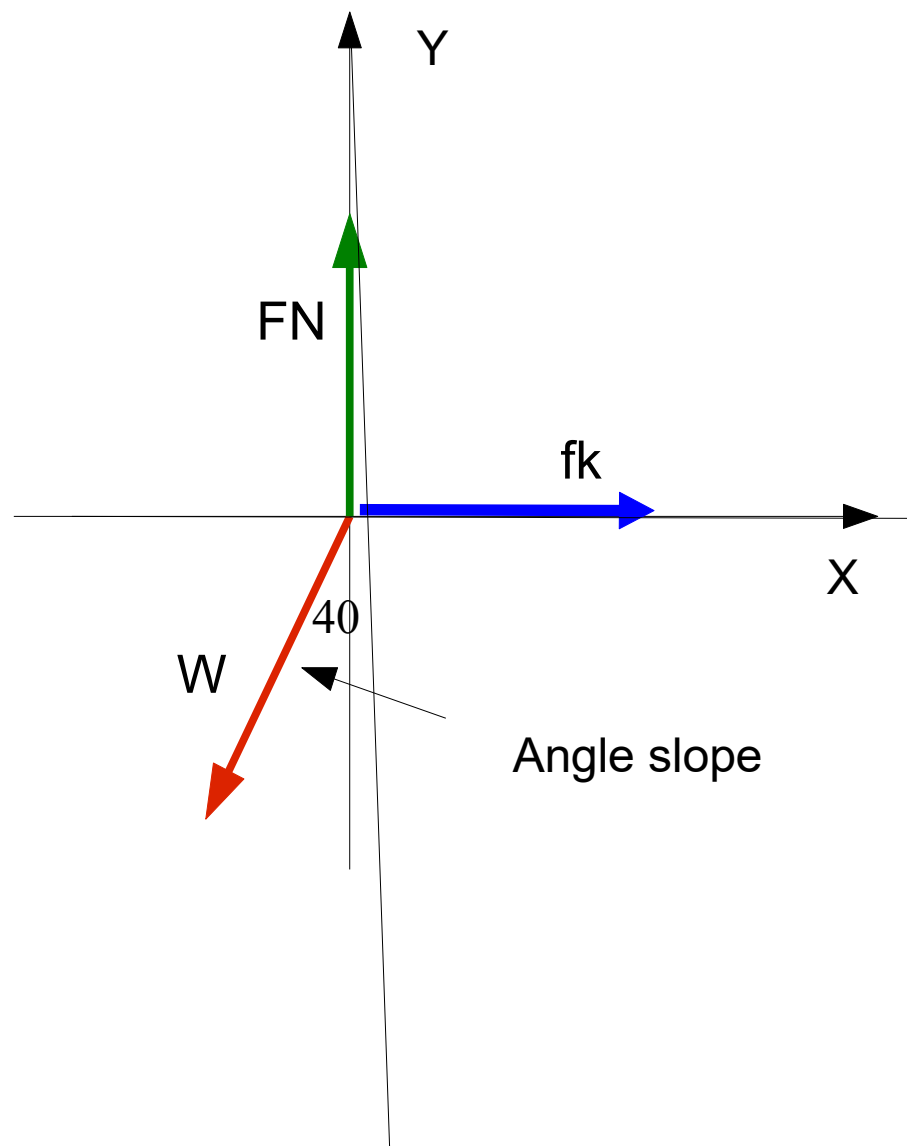
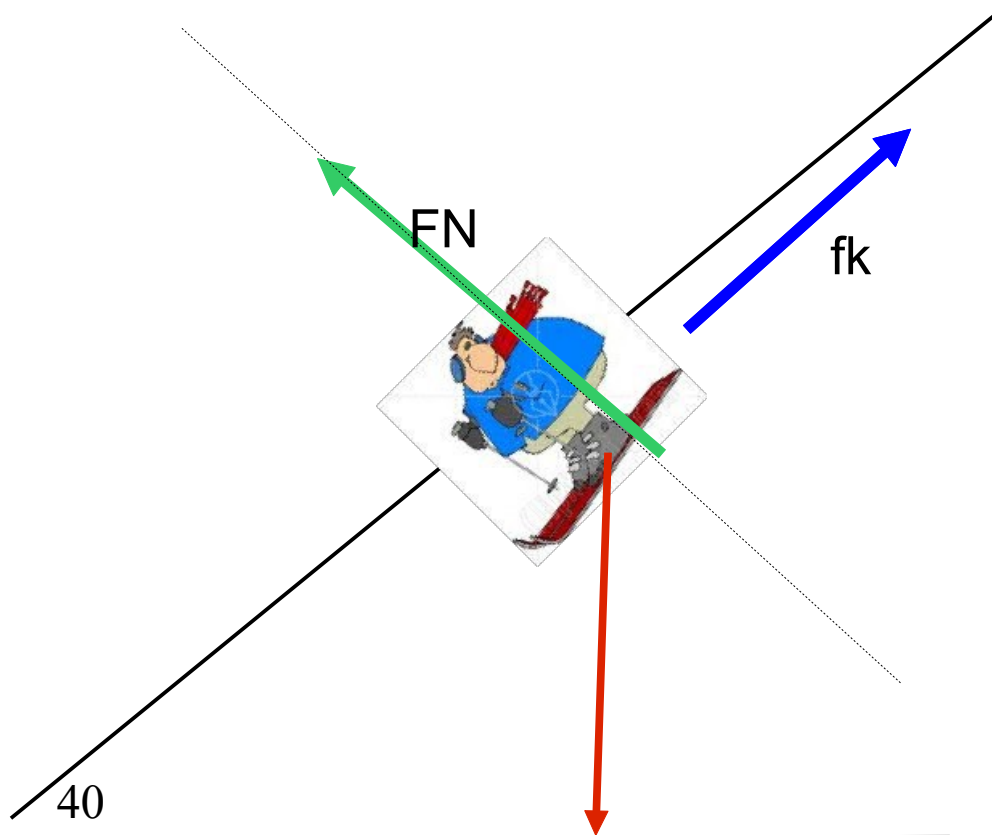




Draw free-body diagram
But include friction.

He is sliding at a constant speed.
So equilibrium situation.

Draw the forces applied to the skier.
Then trace the forces in a free-body diagram



Equilibrium means: @ rest or moving at a constant speed:

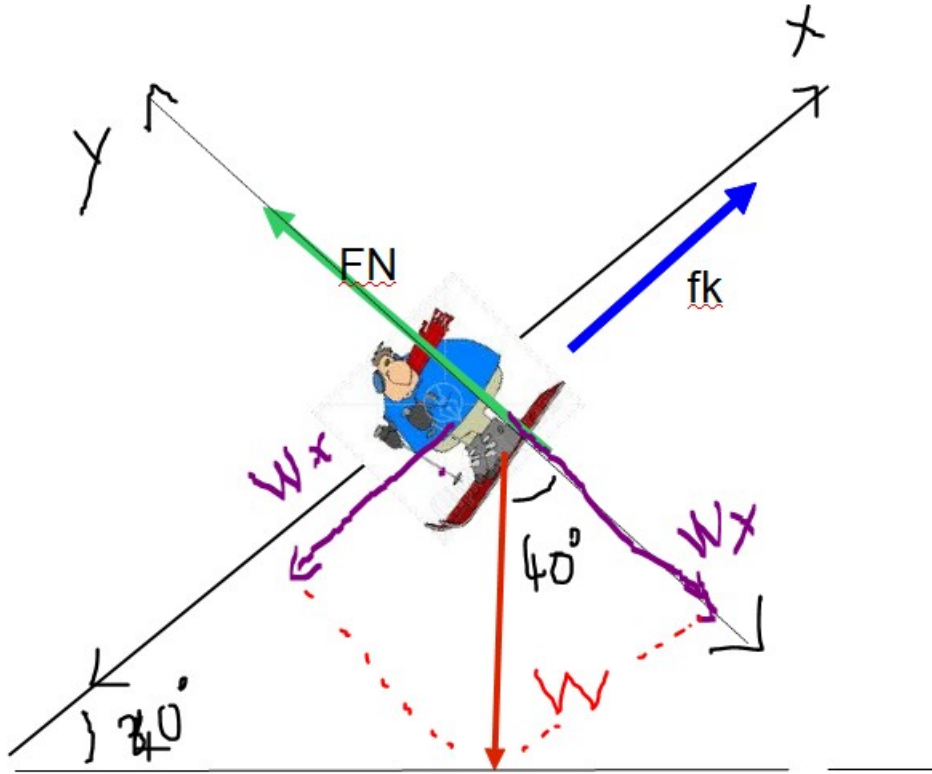
1) sum for forces = 0

→ sum of x-component = 0 → sum (@up components) = sum (@down components)

→ Sum of y-component = 0 → sum (@left components) = sum (@right components)

2) If a force F (ex: N, W, F_{app}) is up or down along the y-axis then its x-component = 0

If a force F is left or right along the x-axis then its y-component = 0



The weight has 2 components:
W_x balances out the friction and makes the Forward motion possible

W_y pushes the snow = normal force

If W_x > friction the skier accelerated. (W_x = m a)

3 forces are acting on the skier:

Weight W, the kinetic friction f_k, the normal force.

W + f_k + N = 0 because its moving at a constant speed.

→ You see that f_k has no y-component f_{ky}=0 (left)
N has no x-component N_x=0 we have (up)

sum(up) = sum (down) means W_y = N

sum(left) = sum(right) means W_x = f_k

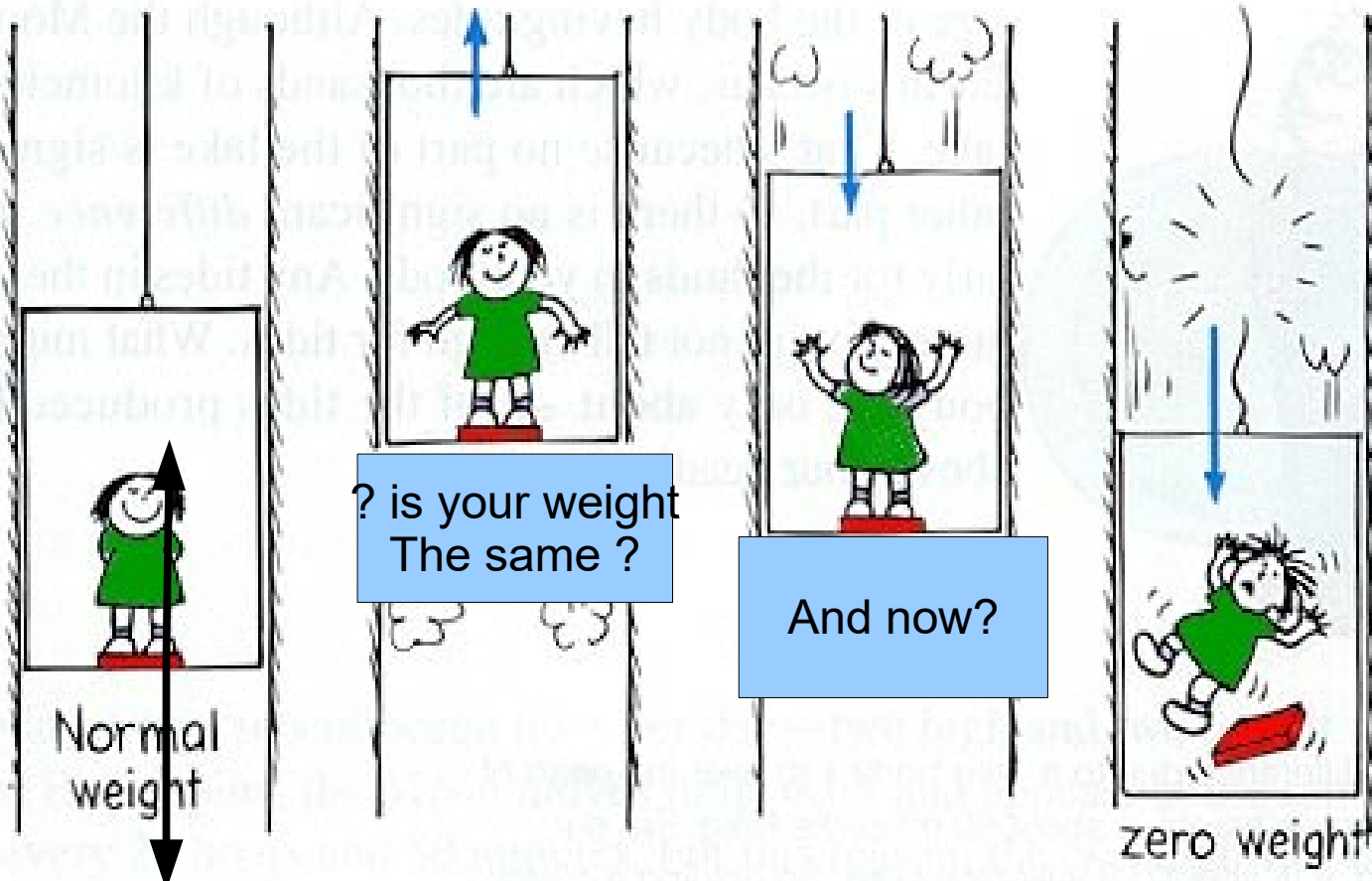
Example: weight is 400N @ 230

	X	Y
w	-257	-306
N	0	N
f _k	f _k	0
R	0	0
	$\sum F_x$	$\sum F_y$

$-257 + f_k = 0 \Rightarrow f_k = 257 \text{ N}$
 $-306 + N = 0 \Rightarrow N = 306 \text{ N}$
 so { Normal = W_y
 Friction = W_x

What is weightlessness?

Gravity is still acting on you but you don't feel gravity anymore because there is no normal force. This is because your framework is falling with you. **You are in free-fall.**



The weight pulls the girl and
The ground pushes the girl up
This is the normal force. You feel
The normal force not the weight..

The weight pulls the girl but
There is no normal force.
So the girl does not "feel" her weight

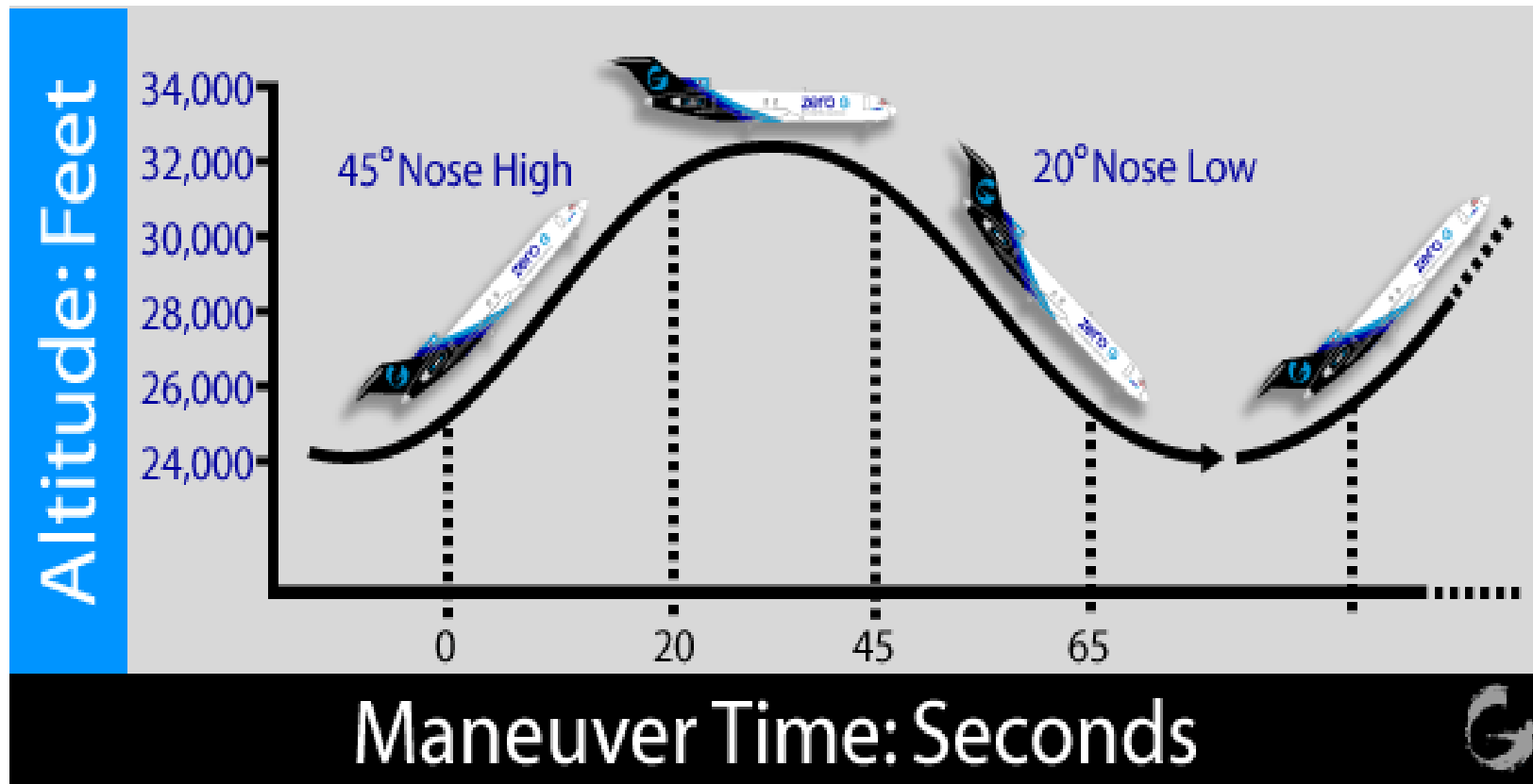
Spinning in zero gravity

http://www.youtube.com/watch?v=7tEkKhMVoS4&feature=player_embedded

http://www.youtube.com/watch?v=H0pGEq7bhLM&feature=player_embedded

Zero gravity

<https://www.youtube.com/watch?v=LWGJA9i18Co>



While going down the plane and its inhabitants are in zero gravity for 10-15 seconds.